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HookDesigner: a Code Generator for Hook-Based Dynamic Binary Instrumentation on Windows

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Abstract

This thesis presents the design and implementation of HookDesigner, a code generator that produces hook-based dynamic binary instrumentation code for Windows from a policy written in a domain-specific language (DSL). The thesis evaluates freely available Windows hooking libraries and selects some of them as HookDesigner compilation targets. The generated hooks support performance measurement, security policy enforcement, and runtime monitoring. Particular attention is given to thread safety and to reliable attachment and detachment from running processes in production settings.

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Chapter 1

Introduction

1.1 Motivation

Many industrial processes are supervised and controlled by software. Operators and security analysts often need to know what this software does at runtime, and sometimes they also need to change how it behaves. For example, they may want to block a process from accessing specific resources, measure how long a critical routine takes, or count how often a function fails.

These applications are often built on very large codebases that have grown over many years, have been changed by different teams, and are poorly documented. For these reasons, even adding a single log line or a new security check can be a slow task: it requires understanding a huge codebase and rebuilding the whole application every time.

This task can be simplified with dynamic binary instrumentation (DBI). At its core, DBI is the technique of inserting additional code into a process while it is running, in order to observe or modify its behavior without needing access to the original source code.

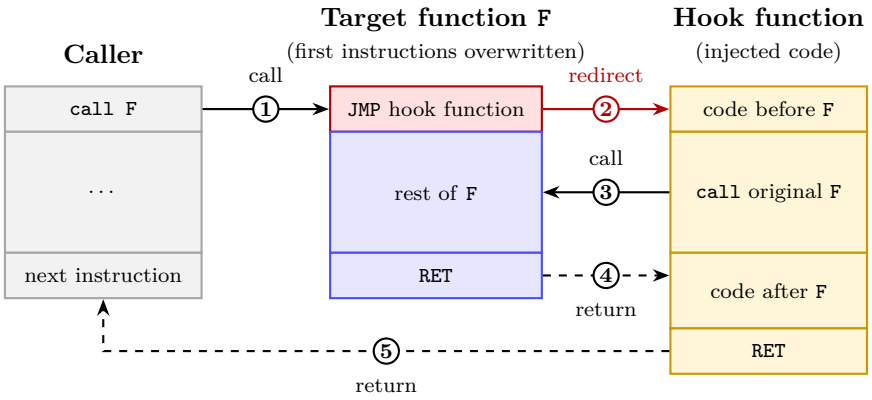
Modern DBI frameworks generally follow one of two architectural paradigms, shown in Figure 1.1:

- *Inline Hooking:* the first instructions of the target function are overwritten in memory with a jump instruction. The jump sends the execution to a custom injected function, called the *hook function*, which runs in place of the target function: it can run code before and after calling the target function, or skip the call, and then it returns to the caller. The jump and the hook function together are called a *hook*.

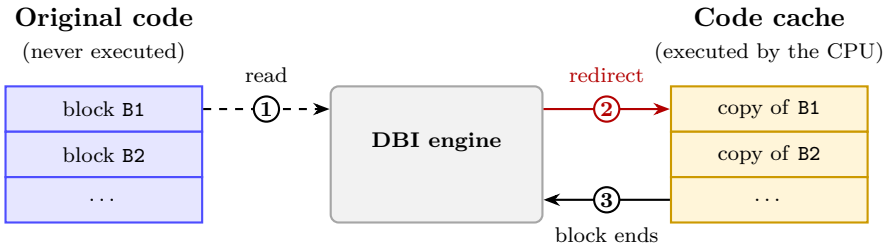
  Seen from the caller, the hook function takes the place of `F`; seen from `F`, it is the caller. The hook function therefore works like a wrapper of `F`: it receives the arguments of

`F`, runs its own code before and after and returns the result of `F` to the caller.

- *Code-Cache Based DBI:* instead of modifying the target code in place, the DBI engine copies the instructions, translates them, inserts the instrumentation code requested by the developer, and executes them in a separate memory area called the *code cache*.



(a) Inline hooking: the code of the target function is patched in memory. Step 3 is simplified: the overwritten instructions of `F` still run before the rest of `F` (Figure 2.1).



(b) Code-cache based DBI: the original code is left intact and only its copies are executed.

Figure 1.1: How the two DBI paradigms redirect the execution of the target program.

Writing the code of a hook by hand is difficult and error-prone. The logic that the developer wants to add, for example a check that blocks the access to a resource, is only a small part of this code. The hook function runs inside the target process, in place of the target function and on the threads of the program, so the rest of its code must make sure that the hook never crashes the program and never changes its behavior in ways the developer did not intend. For example, the hook function must behave towards the caller exactly as the target function would, it must catch its own errors, and it must not end up calling itself again and again. Moreover, the hooks must be attached and removed while the other threads of the program keep running, and some of these threads may be inside a hook function at that moment (Section 2.3.1). None of this code is part of the task, yet much

of it is repeated in every hook function, and a single mistake in it can crash the program that the developer only wanted to observe or change.

A second problem is that instrumentation tasks differ from each other: some only observe the calls to a function, others change their arguments or results, and others block them. Each task needs a different structure of the hook function. For example, to block a call the hook function must skip the target function, to measure how long a call takes it must start and stop a timer right around this call, and if the target function never returns, such as the function that ends the process, no code can run after the call. The safety code described above must therefore be adapted to each structure, and cannot simply be copied from one hook function to another (Section 5.1.5).

A third problem is that each inline hooking framework has its own application programming interface (API), and the frameworks also differ in which of the safety problems above they solve themselves and which ones they leave to the developer (Chapter 3). Moving a hook from one framework to another, for example to use a feature that only the other framework offers, is therefore not a matter of renaming function calls: the developer must also write again the safety code that the new framework does not provide.

These difficulties show a gap between the *what* and the *how* of instrumentation. The *what* is short and easy to state: which functions to intercept and which behavior to apply to each of them, for example to observe, change or block their calls. The *how* is the much longer, low-level code that turns this description into working hooks: code that makes the hooks safe, follows the structure that each behavior needs, and uses the API of one specific framework. With an inline hooking framework alone, the developer writes this code by hand, adapts it for every new task and rewrites part of it for every other framework.

1.2 Research Question and Running Example

The gap between the *what* and the *how* leads to the question that this thesis answers:

> *Can a concrete DBI task be written as a declarative policy, and can the framework-specific instrumentation code be generated from it automatically, for more than one framework, so that the target process' behaviour changes according to what the policy states? And at what runtime cost?*

The work of this thesis tries to answer to the research question through the design and implementation of the *HookDesigner* framework. With this tool, the developer describes the *what* in a *policy* written in a domain-specific language (DSL) and HookDesigner generates the framework-specific *how* (the code) from it.

To keep the following chapters concrete, the thesis follows a single example from start to end, the *running example* selected as the *concrete DBI task* mentioned in the research question.

> A Windows program opens a set of files. The goal is to stop this program from opening some of these files, while all the other files still open normally, and to write a log line for every blocked attempt. The source code is not touched and the target program is not rebuilt: the new behavior is added by using DBI.

1.3 Content of the Thesis

Chapter 2 introduces the foundational concepts of DBI within a Windows environment necessary to understand the proposed work.

Chapter 3 analyze three existing inline hooking engines for Windows, Microsoft Detours, MinHook and Frida-Gum. These engines are the candidate compilation targets of HookDesigner.

Chapter 4 presents the design, architecture, and implementation of HookDesigner, focusing on how it generates and build DBI payload starting from a DSL policy.

Chapter 5 follows this payload into the target process, where an error in its code would crash the target: it shows how the generated code prevent possible errors, and how the payload is injected and later removed while other threads may still be inside the hook functions.

Appendix A describes the code-cache based engines Intel PIN and DynamoRIO, Appendix B explains how a Windows program opens a file, and Appendix C reports the first experiments that I ran with the inline hooking and the code-cache based engines before designing HookDesigner.

Chapter 2

Background

This chapter begins by introducing the foundational concepts of DBI within a Windows environment necessary to understand the proposed work. It first explains how instrumentation code enters a target process through dynamic-link libraries (DLLs) and process injection, and how this code finds the address of a function inside the process (Section 2.1), and then the application binary interface (ABI) rules that this code must follow when it runs in place of a target function (Section 2.2). Then it explains how inline hooking works and which problems it must solve in a running process (Section 2.3), and it briefly presents code-cache based DBI (Section 2.4). Finally, it introduces the logging and metrics libraries that the hooks generated by HookDesigner use to report what they observe at runtime (Section 2.5).

2.1 Dynamic-Link Libraries

The hook function in Figure 1.1a must be in the memory of the target process, or the jump written into `F` could not reach it. In Windows, this code is usually brought into a running process as a *DLL*, through *process injection*. A DLL is a binary module that contains functions, resources, and data that processes can load and use at runtime [Lev00, Ch.10]. A DLL is built like a program: the developer writes its functions in a language such as C or C++, and the compiler and the linker turn them into machine code. The hook function is one of these functions, so it is ordinary compiled code.

2.1.1 Address Space Isolation

In modern operating systems, every running application is allocated an isolated, private virtual address space. This architectural boundary prevents one application from directly reading, writing, or corrupting the code or memory space of another.

A DLL does not break this isolation, because it does not run as a separate process. When a process loads a DLL, the operating system loader *maps* the DLL into the address space of that process: it places the code and data of the DLL file in a free range of addresses of that space [Yos+17]. From then on, the DLL is part of the process, like the code of the program itself: its functions are called by the threads of the process and have the same access to the memory of the process as the program's own code.

2.1.2 DLL Injection

Target processes are unaware of our instrumentation, so the developer must force the target to load the instrumentation DLL from the outside. This is achieved through *DLL Injection* [The26a]. The standard injection technique leverages the Windows API functions `VirtualAllocEx`, `WriteProcessMemory`, and `CreateRemoteThread` targeting `LoadLibrary`. The injector process allocates memory in the remote target, writes the absolute path of the instrumentation DLL, and spawns a remote thread pointing to `LoadLibrary`, which executes within the target process and triggers the operating system (OS) loader to map the DLL.

When the loader maps a DLL, it calls a function of the DLL called its *entry point*, `DllMain`, so that code in the DLL can start to run, for example to install the hooks. While `DllMain` runs, the loader holds a lock of the process, so Microsoft advises doing only simple work inside it [Mic26f]; Section 5.2.1.3 explains why this matters for HookDesigner.

When the instrumentation is no longer needed, the DLL first removes its hooks, writing back the original first bytes of each target function, and then the injector makes the target call `FreeLibrary`, the counterpart of `LoadLibrary`, which unmaps the DLL. The target keeps running without a restart, and its calls no longer go through the hook functions, so they no longer pay their cost.

2.1.3 Modules and Function Addresses

Once the DLL is in the target process, its code can install the hooks. To hook a target function `F`, this code must know the address of `F` in the address space of the target process, because it writes the jump of Figure 1.1a at that address. This subsection explains how

Windows describes where a function is, and why this address is known only while the process runs.

In Windows, every DLL that a process loads are called *modules* [Mic26i]. DLL use the portable executable (PE) format, the format of Windows executable files [Mic26n; Mic26b]. The first address of the range where the loader maps a module is the *base address* of the module. The Windows API identifies a loaded module by a *module handle*, of type `HMODULE`, whose value is exactly this base address [Mic26q]. For example, `GetModuleHandleA` receives the name of a module that the calling process has already loaded and returns its handle; if the name is `NULL`, it returns the handle of the program file [Mic26i].

Inside a module, each function is at a fixed distance from the base address. This distance is the relative virtual address (RVA) of the function: its address in memory minus the base address of the module [Mic26n]. The linker writes in the module file where each part of the module lies relative to the base address, and the loader keeps these distances when it maps the module [Mic26n]. So the address of a function is the base address of its module plus the RVA of the function. For example, a function with RVA `0x1A30` in a module mapped at base address `0x7FF6A0000000` is at address `0x7FF6A0001A30`.

The base address, instead, is not fixed. The module file states a preferred base address, but the loader might not use it [Mic26n]. Moreover, by default the Microsoft linker marks a module so that Windows can map it at a random base address when it is loaded, a feature called address space layout randomization (ASLR) [Mic26b]. If the module of the example above is mapped at another base address, the function is at another address, but its RVA is still `0x1A30`. So the address of a function is known only after its module has been loaded, while its RVA depends only on the module file.

A module can also offer some of its functions to other modules: these functions are called *exported*. The module lists them in its *export table*: the part of this table called the export address table (EAT) holds the RVA of each exported function, and other parts link a public name to it [Mic26n]. For example, the Windows DLL `ntdll.dll` exports `NtOpenFile`, a function that opens a file and that the running example hooks (Appendix B). `GetProcAddress` receives a module handle and the public name of an exported function, and returns the address of the function in the calling process [Mic26j]. This public name can differ from the name that the function has in the source code [Mic26j]. A function that is not exported, such as an internal function of a program, has no entry in the export table, so `GetProcAddress` cannot find it: its address must be computed from the base address of its module and its RVA.

2.2 The Application Binary Interface (ABI)

In inline hooking (Figure 1.1a), the caller executes `call F` and prepares the call for the target function `F`, but the jump sends it to the hook function instead. The caller does not know about this change, so the hook function must behave towards the caller exactly as `F` would: it must find the arguments where `F` would find them, put its result where `F` would put it, and leave the registers and the stack of the thread, together called its *machine state*, as `F` would leave them. These rules are fixed by the ABI.

The ABI defines how arguments are passed to a function, where the return value is placed, who *cleans the stack*, that is, who removes the arguments from the stack after the call (the caller or the called function, the *callee*), and which registers a function must preserve. A set of these rules is called a *calling convention*. When it compiles a function, the compiler follows the specified calling convention: for example, if the function uses a register that it must preserve, the compiler saves that register in the first instructions of the function, the *prologue*, and restores it in the last instructions before the return, the *epilogue*. The compiler applies these rules also to the hook function, which is an ordinary compiled function of the injected DLL (Section 2.1). So the hook function leaves the machine state that the caller expects only if it is declared like `F`: with the same calling convention, the same parameters in the same order and with the same types, and the same return type. The calling convention fixes the rules; the parameters and the return type tell the compiler where, under these rules, each argument and the result are placed. The next two subsections show what goes wrong when one of them differs.

2.2.1 Windows x86 (32-bit) Architecture

In 32-bit environments, multiple calling conventions coexist [Mic26d], requiring alignment between the target function and the hook function defined by the developer. The most common conventions include:

- `__cdecl`: Arguments are pushed onto the stack from right to left, and the *caller* cleans the stack. This is the default convention for C/C++ binaries.
- `__stdcall`: The standard convention for the Windows API. Arguments are pushed right to left, but the *callee* cleans the stack.
- `__fastcall`: The first two arguments are passed in the `ECX` and `EDX` registers, while the rest are placed on the stack. The *callee* must clean the stack.
- `__vectorcall`: An extended convention designed to pass several floating-point arguments in registers. The first two integer-type arguments in the parameter list, from

left to right, are placed in ECX and EDX, and up to six vector arguments in XMM0–XMM5. The *callee* must clean the stack.

Across these 32-bit conventions, registers are typically divided into *caller-saved registers* (EAX, ECX, and EDX), which the caller assumes can be freely overwritten by the called function, and *callee-saved registers* (EBX, ESI, EDI, and EBP), which the called function is contractually required to preserve and restore before returning.

A hook function that uses a different convention from its target breaks what the caller expects. For example, the functions that the running example intercepts (Appendix B) are declared with the macro NTAPI, which on x86 stands for __stdcall, so their callers expect the callee to clean the stack. A hook function declared __cdecl would return without removing the arguments, and the caller would continue with a wrong stack pointer (ESP).

2.2.2 Windows x64 (64-bit) Architecture

Windows x64 defines a single native calling convention [Mic26r]. The compiler accepts __cdecl, __stdcall and __fastcall for compatibility but ignores them; __vectorcall remains in effect on x64 and must be matched exactly when a target uses it. The default convention works as follows:

- Arguments are assigned to positions. Position $i \in \{1, \ldots, 4\}$ is passed in RCX, RDX, R8, R9 if the argument is an integer or a pointer, and in XMM0–XMM3 if it is floating point. In `f(int a, double b, int c, double d)`, a travels in RCX, b in XMM1, c in R8 and d in XMM3.
- Further arguments are pushed on the stack.
- Integer and pointer results are returned in RAX, floating-point results in XMM0.
- *Volatile (caller-saved)*: RAX, RCX, RDX, R8–R11, XMM0–XMM5. *Non-volatile (callee-saved)*: RBX, RBP, RDI, RSI, R12–R15, XMM6–XMM15; a function that uses them must save them in its prologue and restore them before returning.

__vectorcall keeps the positional scheme but extends it: vector-type arguments among the first six are passed by value in XMM0–XMM5, whereas the default convention passes an __m128 by reference and uses XMM0–XMM3 only.

A hook function whose parameters or return type differ from those of its target also breaks what the caller expects, even with the same convention. If the hook function of f above declared b as an int, it would read b from RDX, while the caller put it in XMM1. In the same way, if f returned a double and its hook function returned an int, the hook function would put its result in RAX, while the caller reads it from XMM0.

2.3 Inline Hooking

An inline hook, or simply *hook*, involves three components:

1. *The Target:* the original function whose execution we wish to intercept. The goal is to patch the first 5 to 7 bytes of the target function with an unconditional jump (`JMP`) to redirect execution to the hook function. These bytes are written back when the hooks are removed (Section 2.1.2).

2. *The Hook Function:* the custom function, in the injected DLL, that the jump leads to. This function receives the execution flow, inspects the target function's arguments or return value, performs the desired analysis or policy enforcement, and then dictates whether the execution of the target function should continue or be blocked.

3. *The Trampoline:* in the paradigm of inline hooking, placing an interception branch (`JMP`) inherently corrupts the target function's prologue. To maintain execution continuity, all inline instrumentation frameworks must implement an Out-of-Line Execution Bridge, universally referred to as a *"Trampoline"*. Conceptually, the trampoline is a dynamically allocated, executable memory buffer that must fulfills two architectural mandates: it securely relocates the displaced prologue instructions and unconditionally routes the instruction pointer back to the unpatched remainder of the target function.

Figure 2.1 shows how the three components work together. The caller calls `F` (step 1), and the jump at the start of `F` sends the execution to the hook function (step 2). The hook function runs its *pre-logic*, the code that runs before `F`, for example to read the arguments, and then calls the trampoline (step 3). The trampoline runs the moved instructions and jumps to the rest of `F` (step 4). Since the hook function reached `F` through a `call` to the trampoline, the return address on the stack points into the hook function, so when `F` returns, it returns to the hook function (step 5). The hook function then runs its *post-logic*, the code that runs after `F`, for example to read the return value, and returns to the caller (step 6). To block a call, the hook function skips steps 3 to 5 and returns at once, so `F` never runs.

2.3.1 Challenges in Inline Hooking

An inline hooking engine changes the code of a process while the process is running, and the hook function then runs inside that process, on the threads of the program. This creates different challenges; Chapter 3 shows which of them the existing engines solve, and which ones they leave to the developer of the hook function.

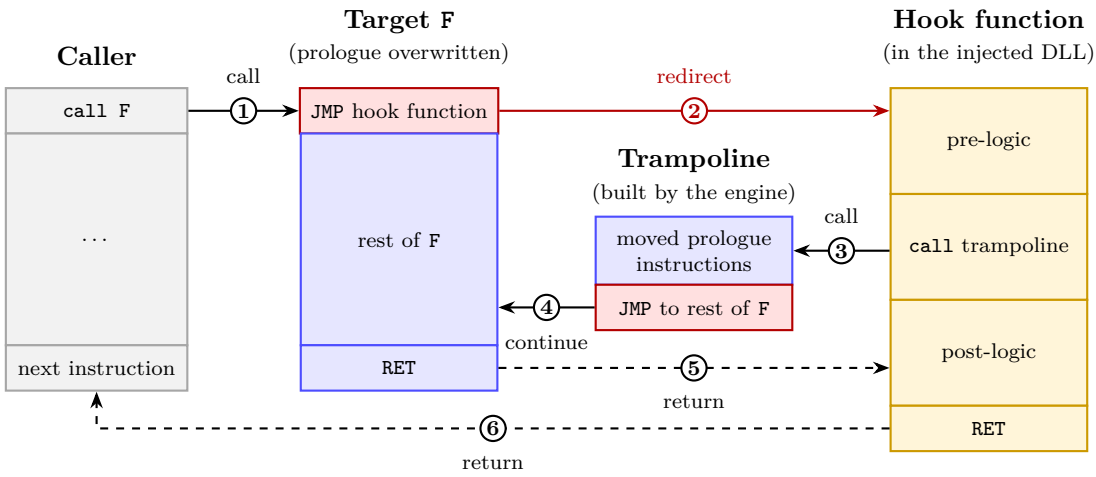


Figure 2.1: The three components of an inline hook: the target `F`, the hook function and the trampoline.

1. *Instruction boundaries:* Instructions do not all have the same length. On x86, the jump written into the prologue takes 5 bytes (1 byte for the opcode and 4 bytes for the relative address). If the engine blindly overwrites exactly 5 bytes, it may cut an instruction in half. For example, if the prologue starts with two 3-byte instructions, the jump overwrites the first one and only part of the second, and leaves one byte of the second instruction behind it. The trampoline would then hold an incomplete instruction and jump back to that leftover byte, so the CPU would run wrong instructions and the process would crash or behave in an unexpected way. The engine must therefore know where each instruction of the prologue ends, and move only whole instructions into the trampoline.

2. *Patching a running process:* the engine writes the jump while the other threads of the process keep running, and one of them may reach the target function at that moment. This can go wrong in two ways:

   - a thread runs the prologue while the jump is only partly written, so the CPU decodes a mix of old and new bytes as an invalid instruction and the process crashes;
   - a thread was paused by the operating system with its instruction pointer (IP) inside the bytes that are being replaced. When it continues, it starts in the middle of the new jump and runs the destination address of the jump as machine code.

3. *Unloading the DLL:* when the hooks are removed, writing the original prologue back stops *new* calls from entering the hook function. However, other threads may still be

inside the hook code that they entered before. A thread is inside the hook code not only while it runs the hook function, but also while it runs the original function that the hook function called through the trampoline: when that function returns, the thread continues in the hook function (step 5 in Figure 2.1). If the DLL is unloaded at that moment, these threads run code at addresses that are no longer mapped, and the process crashes with an Access Violation. The DLL can therefore be unloaded only after every thread has left the hook code. How long this takes has no upper limit when the original target function *blocks*, that is, when it waits for an event before it returns. For example, a thread that calls `recv` waits until data arrives on a network connection, and a thread that calls `WaitForSingleObject` with no timeout waits until another thread signals the object. If the target function blocks, or calls a function that blocks, a thread may stay inside the hook code for minutes or hours, and during all this time the DLL cannot be unloaded.

4. *Re-entrant calls:* the hook function itself may call the target function, directly or through a library function that calls it internally. Such a call passes through the jump again and enters the hook function a second time, on the same thread, before the first execution of the hook function has finished: the hook function is *re-entered*.

   For example, suppose that the hook function of the running example (Section 1.2) reads the list of the files to block from a configuration file the first time it runs. Opening the configuration file is itself a file open, so it reaches the target function and enters the hook function again; the list is still not loaded, so this second execution opens the configuration file again, and so on. Each nested call takes a new frame on the stack of the thread, so the stack soon runs out, and the process crashes with a stack overflow.

   The *re-entrancy* can also come back in an *indirect* way: if the hook function of a function A calls another target function B, and the hook function of B calls A, the two hook functions call each other with no end.

2.4 Code-Cache Based DBI

Inline hooking works at function level: it patches the entry of selected functions and leaves the rest of the program untouched. Code-cache-based DBI engines, such as Intel PIN [Luk+05] and DynamoRIO [BGA03], work instead at instruction level: the original code of the application is never executed directly by the CPU. A *dispatcher* passes each new block of code to a just-in-time (JIT) compiler, which translates it, inserts the instrumentation code requested by the developer, and stores the result in a *code cache* (Figure 1.1b). The CPU then runs the translated block from the cache, and blocks that are already translated are reused.

This approach allows a very detailed analysis, such as logging every conditional branch taken by a program. However, every instruction must pass through the JIT compiler, even when it is not instrumented, so the runtime overhead is much higher than with inline hooking.

HookDesigner supports inline hooking engines (Section 4.1.3), and the integration for code-cache based DBI is left to a future extension of this work.

Appendix A describes the architecture of this paradigm in detail, compares it with inline hooking, and presents the APIs of Intel PIN and DynamoRIO.

2.5 Logging and Metrics

Often the goal of inline hooking is intercepting a function in order to monitor it, and what it observes must then reach the analyst. The data that a running program reports about its own execution is called *telemetry*. Usually the two most common forms are *logs*, such as a specific message written each time a file is opened, and *metrics*, such as how often a call fails or how long a routine takes. HookDesigner generates the code for both (Section 4.4) on top of two existing libraries: *Spdlog* [Mel26] for logs and *Prometheus* [The26b; Mp26] for metrics. This section explains how these libraries work.

2.5.1 Logging with Spdlog

A *log* is a text message with a related *severity level*. Spdlog uses six levels, from the least to the most severe: trace, debug, info, warning, error and critical.

A *logger* is the object that receives the messages: it discards those below a minimum level and passes the others to one or more *sinks*, the objects that write messages to a destination. Spdlog provides a sink that writes to a file (`basic_file_sink_mt`) and one that sends each message as a network packet, this is the most important difference with most of the logging engines available on Windows that include only file sink. Industrial deployments frequently aggregate telemetry across distributed nodes into monitoring systems (such as Promtail, Grafana Loki), and with spdlog's built-in we can do this without writing any additional code.

A *synchronous* logger runs the sinks in the thread that logs the message. Writing to a disk or to the network are usually slow operations, and the thread waits until the write is done. In a hook function, this wait would be added to every intercepted call that logs a message.

An *asynchronous* logger separates the two tasks with a producer–consumer scheme. The thread that logs (the producer) only formats the message and copies it into a queue in

memory; one or more *background threads* (the consumers) are created to take the messages from the queue and pass them to the sinks. The disk and network writes therefore happen on the background threads without burden the worker thread of the process.

In spdlog, `spdlog::init_thread_pool(q_size, thread_count)` creates the queue and the background threads, and a `spdlog::async_logger` sends its messages to them. The queue is a circular buffer with a fixed number of slots, `q_size`.

A mutex protects the queue, so threads that log at the same time take turns, but each thread holds it only for the time needed to copy one message.

When messages arrive faster than the consumer threads can write them, the queue fills up. The *overflow policy* of the logger then decides what happens to a new message:

- `block`, the default, makes the producer wait until a slot is free;
- `overrun_oldest` overwrites the oldest message in the queue;
- `discard_new` drops the new message.

Only with the last two policies a producer never wait for the sinks. Section 4.4.4 describes the queue size, sinks and policy that HookDesigner uses.

2.5.2 Metrics with Prometheus

A *metric* is a number that summarizes many events, for example the number of calls to a function or the current length of a queue. A log keeps one message per event: to know how many calls failed in the last minute, a tool must store and parse every message. A metric gives this answer directly, and its size does not grow with the number of events.

Prometheus is an open-source monitoring system [The26b]. It follows a *pull model*: the monitored program does not send its metrics, but publishes their current values on an endpoint that answers HTTP requests, and the Prometheus server reads this endpoint at regular intervals. Each read is called a *scrape*. The server stores the scraped values as *time series*, that is, sequences of values ordered in time. HookDesigner uses three of the metric types of Prometheus:

- a *counter* is a value that can only increase, such as the number of calls or of errors;
- a *gauge* is a value that can go up and down, such as the number of items in a queue;
- a *histogram* counts observations by size. A list of increasing upper bounds divides the possible values into *buckets*, and the histogram counts how many observations

fall into each bucket, together with the number and the sum of all observations. It shows how a quantity, such as the duration of a call, is distributed, and not only its average.

A metric can carry *dynamic labels*: pairs of a key-value that describe the runtime context of an observation. Each combination of label values is a separate time series. For example, in the running example (Section 1.2), a counter of file opens with the label `verdict` produces the series `file_opens_total{verdict="allowed"}` and `file_opens_total{verdict="denied"}`. The set of series that share a metric name is called a *metric family*.

Dynamic Labels have a cost. Every new combination of label values creates a new time series, which both the program and the Prometheus server must keep. If a dynamic label takes its values from a large or unbounded set, such as user IDs or memory addresses, the number of series grows without limit. This problem is called *cardinality explosion*, and the Prometheus naming guidelines[1] warn against such labels.

HookDesigner uses prometheus-cpp [Mp26], a C++ client library for Prometheus. A `Registry` holds the metric families of the program, and `BuildCounter()`, `BuildGauge()` and `BuildHistogram()` create a family and register it in a registry. `Family::Add(labels)` returns the metric of one combination of label values, and creates it the first time that combination appears. An `Exposer` runs a small hypertext transfer protocol (HTTP) server, inside the program, and at each scrape it serves the families of the registries attached to it (on the path `/metrics` by default).

Calling `Family::Add` also has a cost at runtime. The labels are a `std::map` of strings, so building them allocates memory on the heap. `Add` then creates a new metric object, also on the heap, takes a mutex that protects the family, and looks the labels up in a hash table. If the labels are already there, it discards the new object and returns the existing metric. When several threads call `Add` on the same family at the same time, they must wait for each other on the mutex. This waiting is called *lock contention*. Section 4.4.5 shows how HookDesigner uses these objects.

2.6 Summary

This chapter introduced the concepts used in the rest of the thesis. The ABI fixes how a function receives its arguments and which registers it must preserve, and therefore how a hook function must be declared. DLL injection places the instrumentation code inside the address space of the target process, where any error in that code affects the target

[1] https://prometheus.io/docs/practices/naming/

itself. Inline hooking redirects single functions through a jump, a hook function and a trampoline, and in a running process it must deal with several challenges. Code-cache based DBI rewrites every instruction instead, at a much higher runtime cost. A hook function reports what it observes through logs and metrics: an asynchronous logger such as spdlog moves the disk and network writes off the thread of the target, and Prometheus collects counters, gauges and histograms, split by labels, from an HTTP endpoint inside the program.

Chapter 3

State of the Art

Chapter 2 described inline hooking paradigm in general terms and the problems that every inline hooking engine must solve in a running process. This chapter presents three existing inline hooking engines for Windows, Microsoft Detours, MinHook and Frida-Gum, and shows how each of them handles those problems. These engines are the candidate compilation targets of HookDesigner.

3.1 Microsoft Detours and MinHook

Microsoft Detours [HB99] is the well-known standard for inline hooking on Windows x86/x64. Detours calls the hook function the *detour function*; this thesis keeps the name *hook function* for all the engines. The jump that Detours writes into the prologue of the target leads directly to the hook function, without executing intermediate assembly code as made in Frida (Section 3.2); this thesis calls this model *direct-jump hooking*.

MinHook [Kag26] is a smaller open-source library that follows the same direct-jump model and puts the same requirements on the hook function.

Since no engine code runs before it, the hook function finds the machine state exactly as the caller prepared it for the target, so it must follow the same ABI rules as the target. The hook function is ordinary compiled code, so the compiler applies these rules: its prologue and epilogue preserve the non-volatile registers. This holds only if the developer gives the hook function exactly the convention, parameters and return type of the target, as explained in Section 2.2.

The listings of this chapter hook the running example (Section 1.2). The function they intercept is `NtOpenFile`, the function of `ntdll.dll` that opens a file; Appendix B explains

this function and its parameters.

The hook function calls the original function through a pointer, which the developer declares as in Code 3.1. The type of this pointer repeats the convention (`NTAPI`), the parameters and the return type (`NTSTATUS`) of `NtOpenFile`, so that a call through the pointer follows the same ABI rules as a call to the target. In Detours, the pointer `TrueNtOpenFile` starts with the address of `NtOpenFile`, which tells Detours which function to hook; when the hook is applied, Detours replaces this address with the address of the trampoline.

Code 3.1: Detours and MinHook target function pointer declarations.

```
// Strict function pointer definition required by Detours
static NTSTATUS (NTAPI * TrueNtOpenFile)(PHANDLE, ACCESS_MASK, POBJECT_ATTRIBUTES,
  PIO_STATUS_BLOCK, ULONG, ULONG) = NtOpenFile;

// Strict function pointer definition required by MinHook
typedef NTSTATUS (NTAPI * NTOPENFILE)(PHANDLE, ACCESS_MASK, POBJECT_ATTRIBUTES,
  PIO_STATUS_BLOCK, ULONG, ULONG);
NTOPENFILE fpNtOpenFile = NULL;
```

Code 3.2 shows the structure of the hook function for the running example, `HookedNtOpenFile`. In the pre-logic phase, the hook function reads the name of the file from `ObjectAttributes` and decides whether the program may open it. For a protected file, it returns the error `STATUS_ACCESS_DENIED` at once, so the original function never runs and the file is not opened. Otherwise, it calls the trampoline: the original function opens the file, and control returns to the hook function (post-logic), which sees the result before returning it to the program.

Code 3.2: Structure of the hook function for Detours and MinHook.

```
// Detours (same for MinHook)
NTSTATUS NTAPI HookedNtOpenFile(PHANDLE FileHandle, ACCESS_MASK DesiredAccess,
  POBJECT_ATTRIBUTES ObjectAttributes, PIO_STATUS_BLOCK IoStatusBlock,
  ULONG ShareAccess, ULONG OpenOptions)
{
  // PRE-LOGIC: read the file name from ObjectAttributes
  if (/* the file is protected */)
    return STATUS_ACCESS_DENIED;   // the original function never runs

  // Call to the trampoline
  NTSTATUS status = TrueNtOpenFile(FileHandle, DesiredAccess, ObjectAttributes,
    IoStatusBlock, ShareAccess, OpenOptions);

  // POST-LOGIC: for example, write a log line with the file name and the status

  return status;
}
```

3.2 Frida-Gum

Frida-Gum [Fri26] is the underlying low-level C library of the Frida instrumentation toolkit, representing a modern, multi-architecture approach to Inline Hooking.

At the core of Frida-Gum is a centralized execution *dispatcher*, a C function of the engine that automatically handle the entire hook execution flow.

When a target function is called, the jump in its prologue does not lead to the hook function, but to the *entry thunk*, a short piece of assembly code that Frida-Gum generates at run time used for triggering the dispatcher. The dispatcher is compiled C code, so it may overwrite the volatile registers (Section 2.2), which still hold the arguments that the caller prepared for the target. For this reason, the entry thunk first saves all the registers of the thread into a `GumCpuContext` structure, and only then calls the dispatcher.

The Interceptor API of Frida-Gum offers two modes: *Invocation Listener Mode* and *Function Replacement Mode*.

In Invocation Listener Mode, the developer writes two callbacks, `on_enter` and `on_leave`, which run before and after the original function. The callbacks do not need an ABI-compatible signature (Section 2.2): they receive a `GumInvocationContext` handle and read or change arguments and the return value by index (`gum_invocation_context_get_nth_argument`). However, on Windows x64 this index-based access reads only general-purpose registers (`RCX`, `RDX`, `R8`, `R9`), so it cannot read floating-point arguments, which are passed in `XMM0`–`XMM3` (Section 2.2.2). For this reason, HookDesigner uses only Function Replacement Mode (Section 4.1.3).

3.2.1 Function Replacement Mode

To overcome the limitations of index-based parameter extraction, Frida-Gum exposes *Function Replacement Mode* via `gum_interceptor_replace` and `gum_interceptor_replace_fast`.

1. **Standard Replacement (`gum_interceptor_replace`):** Execution passes through Frida-Gum's `enter_thunk`, where `GumCpuContext` captures the CPU state and triggers the dispatcher. However, instead of invoking high-level callback listeners and then jumping to the original function, the dispatcher redirects execution to the hook function, which Frida-Gum calls the *replacement function* (the `replacement_function` argument of `gum_interceptor_replace`). It is an ordinary C/C++ function, as with Detours or MinHook. Before jumping to it, the entry thunk restores all the registers saved in `GumCpuContext`, so the hook function finds the arguments exactly where the caller placed them. The thunk and the dispatcher therefore do

not free the developer from the ABI: as with Detours, the hook function must have exactly the convention, parameters and return type of the target (Section 2.2). In return, the compiler reads every argument from the right register, including the floating-point ones.

2. **Fast Replacement (`gum_interceptor_replace_fast`):** Emits a direct assembly `JMP` instruction from the target prologue straight to the hook function, bypassing Frida-Gum's dispatcher creation entirely, yielding an execution flow identical to traditional frameworks like Detours and MinHook.

In Function Replacement mode, the hook function of the running example has the same signature and the same structure as the one for Detours in Code 3.2: the pre-logic, the call to the original function through the trampoline pointer `TrueNtOpenFile`, and the post-logic. The code that differs is the code that registers the hook, which Section 3.5 shows for both engines.

3.3 Hook Execution Flow

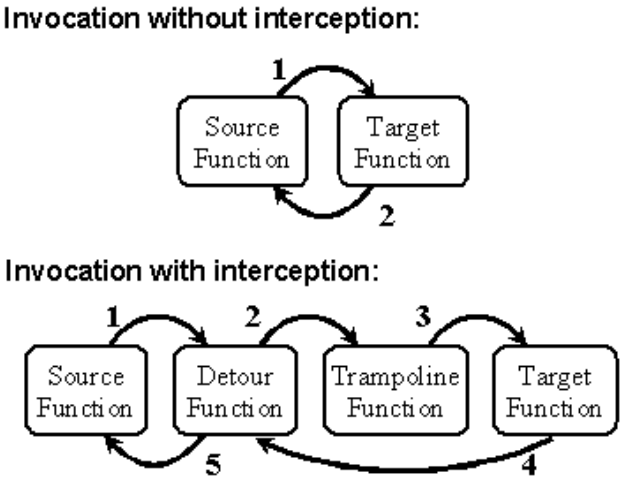
While all the presented frameworks utilize inline prologue patching, the role of the engine in the execution flow of a hook differs:

- **Passive Dispatch (Detours, MinHook, Frida Fast Replacement):** Overwrites the target prologue with a direct assembly jump (`JMP`) to the hook function. The engine acts purely as a passive observer, delegating all execution routing and state management entirely to the hook function written by the developer.
- **Hybrid Dispatch (Frida Standard Replacement Mode):** Combines active engine-dispatcher with user-controlled execution routing. The engine intercepts execution via its dispatcher, but then yields execution control to the hook function, which has the same structure seen in passive dispatch frameworks.

To visually illustrate these operational differences, Figure 3.1 contrasts the direct, user-managed routing (Figure 3.1a) with the dispatcher-mediated execution path (Figure 3.1b).

3.3.1 Advantages of Engine-Mediated Dispatch

Frida-Gum employs a framework dispatcher in Standard Replacement mode to deliver useful capabilities that traditional frameworks do not automatically provide:



(a) Detours/MinHook/Frida Fast Replacement Execution Flow (Passive)

Host Process: Invocation of `TargetFunction()`

JMP

Entry: `enter_thunk` and Dispatcher

- **CPU State Preservation:** Saves all registers into `GumCpuContext`
- **Thread Tracking:** Increments the active invocation counter in `GumFunctionContext`
- **Re-entrancy Check:** Runs the original function directly if the newest stack entry of the thread is for the same target; otherwise pushes a new entry
- **CPU State Restoration:** Restores the registers from `GumCpuContext` and jumps to the hook function

Hook Function: `HookedTargetFunction`

- **Hook Logic:** Runs its pre-logic, calls the original function through the trampoline and runs its post-logic

Exit: `leave_thunk` and Dispatcher

- **CPU State Preservation:** Saves all registers into `GumCpuContext`
- **Thread Tracking Completion:** Decrements the active invocation counter
- **Re-entrancy Tracking Completion:** Removes the entry of this call from the stack of the thread
- **CPU State Restoration:** Restores the registers from `GumCpuContext` and returns to the caller

RET

Host Process: Control Returned to Caller

(b) Frida-Gum Standard Replacement Execution Flow (Hybrid)

Figure 3.1: A comparison of the execution flows of a hook.

1. **Built-in Thread Tracking for Safe Ejection (`gum_interceptor_flush`):** To unload the DLL safely (Section 2.3.1), the instrumentation code must know when every thread has left the hook function. Under passive dispatch (Detours, MinHook) and Frida's fast replacement mode, the engine does not count threads. Frida-Gum's dispatcher, instead, keeps the counter itself inside `GumFunctionContext`: it increments it in `enter_thunk` and decrements it in `leave_thunk` (Figure 3.1b), so a thread is counted from its entry until it returns to its caller, the call to the original function included. The instrumentation detachment logic can call `gum_interceptor_flush()` in a loop; it returns `TRUE` only when no thread is still inside the hooks, and proceed with a safe detachment.

2. **Built-in Protection Against Re-entrancy:** Under passive dispatch (Detours, MinHook) and Frida's fast replacement mode (`gum_interceptor_replace_fast`), the engine does nothing against re-entrant calls (Section 2.3.1): every call to the target function passes through the jump and reaches the hook function, and only the developer can stop the loop. Frida-Gum's dispatcher, instead, keeps for each thread a stack with one entry for each call that is running a hook function; the entry is removed only when the call returns to its caller Figure 3.1b. When a target function is called, the dispatcher looks only at the newest entry: if it belongs to the same function, the dispatcher skips the hook function and runs the original function directly through the trampoline [Fri26]. This stops the *direct form* of re-entrancy, where the hook function calls its own target. It does not stop the indirect form: if the hook function of a function A calls another target function B, the newest entry belongs to A, so the hook function of B runs; if this one calls A, the newest entry belongs to B, so the hook function of A runs again, and the loop goes on.

   The check has a second limit: since the entry stays on the stack until the call returns, it stays there also while the original function runs. If the original function calls itself (recursive function), as a reentrant function does, the nested call finds the entry as the newest one and is skipped too, so the hook function sees only the outermost call.

3.4 Instruction Boundaries and Disassemblers

To avoid cutting an instruction in half (Section 2.3.1), every inline hooking engine needs a disassembler that finds the length of each instruction in the prologue of the target function. A disassembler that computes only this length, without analyzing what the instruction does, is called a length disassembler engine (LDE). Before writing the jump, the engine decodes the prologue one instruction at a time. It then copies whole instructions into the trampoline until it has moved at least as many bytes as the jump needs, and fills the remaining bytes in the original function with `NOP` instructions. The engines differ in the

disassembler they use, and this choice changes the size and the dependencies of the engine:

- *Microsoft Detours* relies on its own internal decoder optimized specifically for standard compiler prologues.
- *MinHook* utilizes a highly optimized, custom-built Hacker Disassembler Engine (HDE32/HDE64) to calculate instruction lengths, rather than fully analyze instruction semantics.
- *Frida-Gum* must handle multiple architectures instruction sets. To accomplish this, it statically links against Capstone, a comprehensive and heavy multi-architecture disassembly framework, significantly increasing the size of the resulting DLL.

3.5 Patching a Running Process

Section 2.3.1 showed that the engine writes the jump while the other threads of the process keep running, so a thread may run a half-written prologue, or continue inside the bytes that are being replaced. This section presents the techniques that prevent these dangers, and shows which of them each engine performs itself and which it leaves to the developer.

Engines patch a running process following one of two models. In the *pessimistic* model, the engine suspends the other threads of the process before it writes to the target, so that no thread runs while the prologue is being patched. Suspending the threads is not enough on its own: a suspended thread may have its IP inside the bytes that are being replaced. For this reason, the engine reads the registers of each suspended thread (`GetThreadContext`). If the IP of a thread falls inside the replaced bytes, the engine moves it to the matching copied instruction in the trampoline. When the hook is removed, the engine does the opposite: it moves a thread that is running the copied instructions in the trampoline back to the matching instruction of the restored prologue. In the *non-blocking* model, the engine does not suspend threads and does not read their IP: it makes the code pages writable and overwrites the prologue while the other threads keep running, so it prevents neither of the dangers while patching described in Section 2.3.1. It accepts them because the chance that it happens is very small: the engine overwrites only the first few bytes of the function, a thread runs them very quickly, and they are only a small part of the code of the function, so a thread would have to be running them at the very moment of the write.

Frameworks tailored natively for Windows, such as *Microsoft Detours* and *MinHook*, follow the pessimistic model, but they differ in which threads are suspended.

Microsoft Detours groups hook attachments/detachments into a *transaction*, a set of tasks that the engine applies together, and lets the developer manage it explic-

itly [Mic26e]. The developer begins a transaction (`DetourTransactionBegin`), enlists every thread to be inspected via `DetourUpdateThread`, queues the hook attachments or detachments (`DetourAttach`, `DetourDetach`), and commits the transaction (`DetourTransactionCommit`). `DetourUpdateThread` suspends the enlisted thread at once, and `DetourAttach` already builds the trampoline, but the jump is written into the target only during the commit. The commit then relocates the instruction pointers of the suspended threads, flushes the CPU instruction cache, and resumes the threads. The transaction is *all-or-nothing*: if any queued operation has failed, the commit cancels the whole transaction and no hook is applied. The developer can also cancel it explicitly with `DetourTransactionAbort`, which frees the trampolines and resumes the suspended threads, in case of a fault before reaching the commit.

Code 3.3 shows a transaction that attaches the `NtOpenFile` hook of Code 3.2. Detours has no function that finds the threads of a process, so the developer must collect their handles before the transaction begins (`otherThreads` in the listing). The list leaves out the thread that runs the transaction: Detours does not suspend it, because a thread that suspends itself would never reach the commit.

Code 3.3: Attaching the `NtOpenFile` hook of Code 3.2 in a Detours transaction.

```
DetourTransactionBegin();

// Enlist every other thread of the process: each one is suspended at once
for (HANDLE hThread : otherThreads)
  DetourUpdateThread(hThread);

// Queue the hook: the trampoline is built now, the jump is not written yet
DetourAttach(&(PVOID&)TrueNtOpenFile, (PVOID)HookedNtOpenFile);

// Write the jump, move the instruction pointers, flush the cache and resume
// the threads; if a queued operation failed, cancel the transaction instead
DetourTransactionCommit();
```

MinHook adopts the same underlying mechanism but hides the transaction boundaries from its public API. Internally, when `MH_EnableHook` or `MH_DisableHook` is invoked, the library enumerates all the threads of the process, suspends them, performs instruction-pointer relocation, applies the patch, and resumes the threads.

Frida-Gum, in contrast, follows the non-blocking patching model on Windows [Fri26]. Unlike in Detours, the transaction is not automatically all-or-nothing: its transactional API (`gum_interceptor_begin_transaction` and `gum_interceptor_end_transaction`) is only a boundary, the hooks registered with `gum_interceptor_replace` or reverted with `gum_interceptor_revert` between the two calls are applied together when the transaction ends. Each `gum_interceptor_replace` returns its own error at once, and the hooks that succeeded are still applied when the transaction ends. There is also no call to cancel a

transaction; to undo staged hooks, the developer must revert them one by one. When the transaction ends, Frida-Gum makes the pages that hold the target prologues readable, writable and executable, and writes the jump into them with ordinary memory writes, one part at a time, while the other threads keep running. It does not enlist threads and does not inspect or relocate their IP, so the write is not atomic for a thread that runs the prologue at the same moment.

Code 3.4 registers the same `NtOpenFile` hook with Frida-Gum, and can be compared with the Detours transaction of Code 3.3.

Code 3.4: Registering the `NtOpenFile` hook in a Frida-Gum transaction.

```
gum_interceptor_begin_transaction(interceptor);

GumReplaceReturn r = gum_interceptor_replace(interceptor,
  targetNtOpenFile,
  (gpointer)HookedNtOpenFile,
  NULL,
  (gpointer*)&TrueNtOpenFile);

if (r != GUM_REPLACE_OK) {
  // the error is returned at once: revert the hooks staged so far, one by one
}

// Write the jumps while the other threads keep running
gum_interceptor_end_transaction(interceptor);
```

3.6 Comparison of the Engines

To conclude the discussion on inline hooking, Table 3.1 provides a summary comparison of the three analyzed frameworks: Microsoft Detours, MinHook, and Frida-Gum.

Appendix C reports the preliminary experiments that I ran with these engines before designing HookDesigner.

Feature	Detours	MinHook	Frida-Gum
Engine Role	Passive	Passive	Hybrid (`replace`) / Passive (`replace_fast`)
Disassembler	Proprietary LDE	HDE32 / HDE64	Capstone (full semantic decoder)
Patching Concurrency	explicit thread enlistment & IP relocation	automated thread suspension & IP relocation	Non-blocking patching (plain writes while threads run, no IP relocation)
Thread-Safe Ejection	Manual tracking (counter written by the developer)	Manual tracking (counter written by the developer)	Built-in tracking (`gum_interceptor_flush`)
Re-entrancy Protection	None (left to the developer)	None (left to the developer)	Direct form only (`replace`); none (`replace_fast`)
Architectures	x86, x64	x86, x64	x86, x64, ARM, ARM64, MIPS

Table 3.1: Comparison of the analyzed inline hooking frameworks.

Chapter 4

From Policy to Instrumentation DLL

This chapter presents the design, architecture, and implementation of HookDesigner: a DBI code generator for Windows, introduced in Chapter 1.

All the examples of this chapter come from the running example of Section 1.2, the goal is stopping a process from opening some files. Chapter 3 presented DBI code examples written by hand; this chapter shows how HookDesigner generates it from a DSL policy.

4.1 Design Principles and Architecture

Section 1.1 described the gap between the *what* of an instrumentation task (which functions to intercept and which behavior to apply to each of them) and the *how* (the low-level, engine-specific code that implements it). HookDesigner tries to close this gap leveraging a *multi-target transpilation* process.

In computer science, a *transpiler* (or source-to-source compiler) is a program that translates an input specification or program authored in one high-level language into equivalent source code in another programming language. HookDesigner uses a transpiler that automatically translates high-level, policy declarations (the what of an instrumentation task) directly into the corresponding engine-specific production-ready C++ code. This code is mainly designed to run safely inside a multi-threaded target process.

Trust model: Policy authors are trusted, for this reason DSL expressions—compiled into instrumentation DLL, which runs with the privileges of the target—are not sanitized.

4.1.1 Design Principles

- **Engine-Independent Transpilation** As analyzed in State of the Art (Chapter 3), inline hooking engines expose different APIs and put different constraints on the instrumentation code. HookDesigner keeps the parts that depend on each engine separate from the rest of the transpiler. In this way, it generates the code for every selected engine from the same policy, and a new engine can be added without changing the core of the transpiler (Section 4.3).

- **Crash Safety** As analyzed in Section 2.1.1, the instrumentation payload is mapped directly into the target application's private address space. Consequently, any runtime error will crash the entire host process. For this reason, HookDesigner automatically adds safety mechanisms to every hook function it generates, so that an error inside the hook does not crash the target process (Section 5.1).

- **Thread Safety** Hooking a running, multi-threaded process introduces synchronization issues both when the hooks are installed or removed and when the DLL is unloaded (as analyzed in Section 2.3.1). HookDesigner automatically generates the code that installs and removes the hooks safely, and makes sure that the DLL is unloaded only when no thread is still running inside a hook function (Section 5.2).

- **High-Performance Telemetry** Logs and metrics (Section 2.5) show what the hooks observe, but the code that produces them runs inside the hook functions, on the threads of the target, so it must slow down the target as little as possible. HookDesigner generates this code automatically from the policy, using spdlog for logs and Prometheus for metrics (Section 4.4).

- **Separation of Policy from Code** The user describes an instrumentation task in two separate files. A declarative YAML policy states *which* functions are intercepted and *what* each hook does, for example logging the call or blocking it. The C++ code needed by this behavior, such as the code that decides whether a call is blocked, is written in a separate C++ *logic file* (Section 4.2). Since the logic file is ordinary C++, the user can write it with the usual development tools, and the same logic file can be reused with different policies and engines.

4.1.2 The Seven-Stage Pipeline

HookDesigner handles a DBI request through a pipeline of seven stages. The pipeline is split into two execution contexts: an *offline* context, which runs on the machine of the developer (stages 1–4), and an *online* context, which runs inside the target process, on the machine where the compiled DLL is deployed (stages 5–7).

- **Stage 1 – Input:** The user provides the two input files introduced in Section 4.1.1: the DSL policy and the C++ logic file (Section 4.2).
- **Stage 2 – Parse & Validate:** The transpiler reads the policy, checks that it is valid, and checks that it agrees with the logic file (Section 4.2).
- **Stage 3 – Generate:** From the validated policy, the transpiler generates the complete C++ source code of the hook DLL, one source file for each selected engine (Section 4.3).
- **Stage 4 – Compile:** Each generated source file is compiled (Section 4.3.6) into a Windows DLL (Section 2.1).
- **Stage 5 – Inject:** An injector program loads the DLL into the target process (Section 2.1.2) and remotely attaches its hooks. The target can be a process that is already running, or a process that the injector starts with the DLL already loaded (Sections 5.2.1 and 5.2.2).
- **Stage 6 – Execute:** The calls to the target functions now enter the hook functions, which apply the behaviors stated in the policy: they run the code of the logic file, can deny the call, and write logs and metrics (Sections 4.4 and 5.1.5).
- **Stage 7 – Eject:** When the instrumentation is no longer needed, the injector removes the hooks and unloads the DLL only when this is safe, as required by the *Thread Safety* principle (Section 5.2.3).

Thanks to this context split, the machine of the target needs only the compiled DLL and the injector binaries: the policy and the external dependencies that the transpiler uses in the offline stages stay on the machine of the developer. This matters in industrial environments, where the machine of the target is often a restricted system on which installing new programs, such as interpreters or developer tools, is not allowed.

4.1.3 Selection of Compilation Targets

The engine analysis in State of the Art (Chapter 3) and the preliminary experiments in Appendix C directly guided the choice of instrumentation engines supported by HookDesigner.

- **Microsoft Detours:** Microsoft Detours was selected because, despite its minimalistic model (analyzed in Section 3.1), it represents the gold standard for DBI in Windows x86 and x64 environments. By delegating execution routing directly to the

hook function without intermediate layers (as illustrated in Figure 3.1a), Detours achieves near-zero runtime overhead, prioritizing execution speed over framework complexity.

- **Frida-Gum:** Frida-Gum was chosen as a complementary engine to provide modern dynamic interception capabilities. Only Function Replacement mode is used, because the Invocation Listener cannot read floating-point arguments on Windows x64 (Sections 3.2 and C.2). This allows developers to manually handle floating-point parameter extraction bypassing Frida default indexing, while retaining some of Frida's features not included in Detours/MinHook (presented in Section 3.3.1).

MinHook was not selected because it uses the same direct-jump model as Detours (Section 3.1) and it is also less documented. Its main practical difference, the automatic suspension of all threads (Section 3.5), is already provided for Detours by HookDesigner.

4.2 Declarative Policy Specification and Logic File Integration

This section analyzes the two input artifacts processed by HookDesigner in Stage 1 of its pipeline (Section 4.1.2): the declarative policy file, and the C++ user logic file.

4.2.1 The Core YAML Policy DSL

The policy DSL is structured as a human-readable document expressed in YAML: a human-readable data serialization language—meaning, a format used to convert complex data structures (like objects, lists, or key-value pairs) into a plain-text format that humans can easily read.

The policy contains two top-level blocks: `configure`, which establishes global setting for the instrumentation, and `hooks`, which specifies the list of target functions to intercept along with their runtime address resolution strategies and the optional code to run before and after them, the pre-logic and the post-logic (Section 2.3).

4.2.1.1 The `configure` Block

The `configure` section declares target process parameters, the relative path to the companion C++ logic file, the instrumentation paradigm and the correspondent target engines, as summarized in Table 4.1.

Field	Type	Req.	Default	Description
`process`	string	Yes	—	Target executable filename (e.g., `filedeny_target.exe`).
`logic_file`	string	Yes	—	Relative path (from the policy file) to the C++ logic source file.
`engines`	string / list	No	`"detours"`	Instrumentation engine(s) to generate code for. (e.g., `["detours", "frida"]`).
`paradigm`	string	No	*inferred*	Hooking paradigm (e.g., `inline_hook`).
`telemetry`	object	No	`{}`	Global Telemetry settings (detailed in Section 4.4)

Table 4.1: Schema definition for the top-level `configure` block.

- `engines` accepts either a single engine name or a list of target engines. When multiple engines are declared, the transpiler generates separate, engine-specific translation C++ files in a single invocation, enabling multi-engine deployments from a single policy definition.
- `paradigm` (in the current version of HookDesigner must be inline_hook) explicitly declares the instrumentation paradigm. If specified, transpiler validates that all `engines` belong to this paradigm. If omitted, it is automatically inferred from the configuration registered in `engines.yaml` of the specified `configure.engines`.

4.2.1.2 The `hooks` List

The `hooks` section defines a sequence of hook configuration objects. Each entry specifies a target function to intercept, the module (within `configure.process`), the runtime address resolution strategy to locate its address within the specified module, and the behaviors to apply to the target function (pre-logic, post-logic, call denial, logging and metrics), as detailed in Table 4.2.

- **Address Resolution Strategies:**
  - *Export Strategy (`strategy: export`):* Resolves the function entry address dynamically via `GetProcAddress` (Section 2.1.3). When hooking C++ member functions or overloaded APIs where compiler name mangling alters the exported symbol name, the optional `symbol_name` field allows specifying the decorated symbol.

Field	Type	Req.	Default	Description
`function_name`	string	Yes	—	Target function name.
`module`	string	No	`""`	Target DLL module or `configure.process`.
`is_noreturn`	bool	No	`false`	Flag for non-returning functions.
`address_resolution`	object	No	`{strategy: "export"}`	Strategy for runtime function address lookup.
`behaviors`	object	No	`{}`	Sub-policies for metrics, logging, and pre/post-logic.
Address Resolution Configuration (`address_resolution`)				
`strategy`	string	Yes	—	Resolution method: `"export"` or `"offset"`.
`symbol_name`	string	No	`function_name`	Exported decorated symbol for `GetProcAddress`.
`rva_offset`	string / int	If offset	—	Hexadecimal RVA of the function (e.g., `"0x1A30"`).
Behavioral Sub-Policies Configuration (`behaviors`)				
`security_policy`	object	No	`{}`	Names of the pre-logic and post-logic functions, and optional call denial.
`log_rules`	list	No	`[]`	Logging rules (detailed in Section 4.4).
`performance_metrics`	object	No	`{}`	Target latency and Prometheus metrics (detailed in Section 4.4)
Call Denial Configuration (`security_policy.on_deny`)				
`return_value`	string	If non-void	—	C++ expression returned to the caller of a denied call (Section 4.2.1.2.1).
`last_error`	string	No	`null`	C++ `DWORD` expression set as the caller's last-error code.

Table 4.2: Schema definition for the `hooks` list entries

  - *Offset Strategy* (`strategy: offset`): For unexported functions, HookDesigner computes their address at runtime by adding the RVA given in `rva_offset` to the base address of the specified `module`, or of the program file if `module` is empty (Section 2.1.3).

- **Non-Returning Functions (`is_noreturn`):** this flag is automatically inferred for standard non-returning APIs (such as `RaiseException`, `ExitProcess`, and `ExitThread`). Because these functions never return to the caller, the transpiler leverages this flag during semantic validation to prevent policy authors from declaring a post-logic or non-compatible telemetry options (Section 4.4.3) and for correctly selecting the hook function structure designed for non-returning functions (Section 5.1.5.2).

- **Security_policy**: is a dictionary that names the pre-logic and the post-logic of the hook, two functions implemented in the C++ logic file (Section 4.2.2.1). Optionally, an `on_deny` entry lets the pre-logic deny the original target call (Section 4.2.1.2.1).

4.2.1.2.1 Call Denial The pre-logic and the post-logic can inspect and change arguments and return values, but the original function is always executed. Access-control policies, such as preventing a process from opening a protected file, need a different behavior: the original function must not run, and the call must fail as if the operating system had refused it.

HookDesigner supports this through the optional `on_deny` entry of `security_policy`. When `on_deny` is declared, the pre-logic becomes a decision point: it must return `bool`, where `true` allows the call and `false` denies it. The entry has two fields (Table 4.2):

- `return_value`: a C++ expression of the target's return type
- `last_error`: an optional C++ `DWORD` expression set as the caller's last-error code (Section 5.1.4)

A denied call behaves as follows:

- the original function is not called;
- the caller receives `return_value`;
- the caller's last-error code is set to `last_error` if declared, otherwise it is left as it was before the call;
- the post-logic is skipped, since there is no original result to process;

- telemetry still runs and sees `return_value` as the result of the call, so denied calls can be logged and counted through the `__hook_denied` wildcard (Section 4.4.2.1).

During validation, the transpiler rejects an `on_deny` entry that is attached to a non-returning function, that lacks a `return_value` for a non-void function or declares one for a void function.

Code 4.1 shows the hook of the running example on `NtOpenFile`. The pre-logic `CheckFileAllowed` receives the `ObjectAttributes` argument of `NtOpenFile`, which holds the name of the file (Appendix B), and returns its verdict. A denied call returns `STATUS_ACCESS_DENIED`, the `NTSTATUS` error for a refused access. The hook declares no `last_error`, because native functions report their errors only through the returned `NTSTATUS`.

Code 4.1: YAML policy snippet using call denial on `NtOpenFile`.

```
- module: ntdll.dll
  function_name: NtOpenFile
  address_resolution: { strategy: export }
  behaviors:
    security_policy:
      pre_hook: CheckFileAllowed   # bool CheckFileAllowed(POBJECT_ATTRIBUTES
    ObjectAttributes); true = allow
      on_deny:
        return_value: "STATUS_ACCESS_DENIED"
```

4.2.2 Logic File Integration

A core design principle of HookDesigner is the separation between declarative policy and imperative logic to inject, as presented in Section 4.1.1.

For guiding the transpiler in parsing the logic file, it must present a specific structure built using tag markers:

1. **Signatures Block** (`// @signatures-start` ... `// @signatures-end`): This block defines the exact parameter types, parameter names and order, optional calling conventions, and return types of the functions targeted by the policy.
2. **Callback Implementation Regions** (`// @<CallbackName>-start`): Each callback, a pre-logic or a post-logic, is prefixed with a dedicated tag matching the name referenced in the YAML policy's `security_policy` dictionary (`pre_hook` or `post_hook`).

The logic file will be included directly via `#include` in the transpiler generated C++ file, so it can be used also as a standard helper file.

4.2.2.1 Clang Parsing

Parsing C++ function declarations with regular expressions is fragile, because of complex types, nested pointers and calling conventions. For this reason, Stage 2 of HookDesigner uses Clang [LLV26], an open-source C and C++ compiler. Like most compilers, Clang first reads the source code and turns it into an abstract syntax tree (AST): a tree in which each node is an element of the program, such as a function declaration or one of its parameters, and stores its name and its type.

The transpiler uses Clang to build the AST of the logic file, using the `-std=c++17` standard, and traverses its nodes to perform the following tasks:

1. **Extracting Signatures:** The transpiler locates the *Signatures Block* and iterates over all the function declarations. For every signatures node within this range, it extracts:
   - The *target function name* and *return type*
   - The *calling convention* (Section 2.2) by scanning the token preceding the function identifier, looking for calling convention keywords (`__stdcall`, `__cdecl`, `__fastcall`, etc.).
   - The *parameter list*, capturing parameter names and Clang type handles (`_clang_type`).
2. **Extracting Callback Parameters:** The transpiler scans for `// @<CallbackName>-start` tag patterns and filters the Clang `FUNCTION_DECL` nodes, extracting the parameter names for cross-validation against target function prototypes, and the callback return type.

During Stage 2, the transpiler enforces the following semantic constraints:

1. **Signature Bindings:** Every `function_name` specified in the YAML policy must have a corresponding function declaration inside the Signatures Block.
2. **Callback Resolution:** Every callback identifier declared under `security_policy` (`pre_hook` - `post_hook`) must have a matching `// @<CallbackName>-start` implementation in the logic file.
3. **Parameter Matching:** Each callback parameter must match the name of a parameter in the target function's signature specified in the *Signatures Block*. Type checking and reference binding is left to the C++ compiler, since we could have arguments passed as reference in the callbacks.

4. **Return Value Overriding:** For a post-logic attached to a non-void function, an optional reference parameter named `__hook_result`, of the target's return type, can be declared. It holds the value returned by the original function, so the post-logic can inspect it or override the value returned to the caller.

5. **Deny Verdict:** If the hook declares `on_deny`, its pre-logic must return `bool`: this value is the verdict that allows or denies the call (Section 4.2.1.2.1).

4.2.2.2 Callback Reusability

In some cases, sharing a pre-logic or a post-logic across multiple distinct target functions could be useful for saving time and avoiding code duplication. Because HookDesigner binds callback arguments by matching parameter names against the target function's parameters (delegating type checking and reference binding to the C++ compiler), one callback can serve every target function whose parameters have the same names.

Codes 4.2 and 4.3 show this on an extension of the running example, a *read-only mode*: instead of denying a protected file, the hook lets the program open it, but removes the rights to change it from the requested access. `NtOpenFile` and `NtCreateFile` both call their access mask `DesiredAccess` and their file name `ObjectAttributes`, so a single pre-logic, `ForceReadOnly`, serves both functions.

Code 4.2: C++ user logic file illustrating callback reusability

```
// @signatures-start
NTSTATUS NTAPI NtCreateFile(PHANDLE FileHandle, ACCESS_MASK DesiredAccess,
    POBJECT_ATTRIBUTES ObjectAttributes, PIO_STATUS_BLOCK IoStatusBlock,
    PLARGE_INTEGER AllocationSize, ULONG FileAttributes, ULONG ShareAccess,
    ULONG CreateDisposition, ULONG CreateOptions, PVOID EaBuffer, ULONG EaLength);

NTSTATUS NTAPI NtOpenFile(PHANDLE FileHandle, ACCESS_MASK DesiredAccess,
    POBJECT_ATTRIBUTES ObjectAttributes, PIO_STATUS_BLOCK IoStatusBlock,
    ULONG ShareAccess, ULONG OpenOptions);
// @signatures-end


// IsReadOnlyFile: true if the file is on the read-only list

// @ForceReadOnly-start
static void ForceReadOnly(ACCESS_MASK& DesiredAccess, POBJECT_ATTRIBUTES
    ObjectAttributes) {
    if (!ObjectAttributes || !IsReadOnlyFile(ObjectAttributes)) return;
    if (DesiredAccess & (GENERIC_ALL | MAXIMUM_ALLOWED)) DesiredAccess |= GENERIC_READ;
      // keep the read rights
    DesiredAccess &= ~kWriteRights;
}
```

Code 4.3 attaches the same callback to both hooks.

Code 4.3: YAML policy snippet attaching a shared callback across multiple hooks.

```
- module: ntdll.dll
  function_name: NtOpenFile
  behaviors: { security_policy: { pre_hook: ForceReadOnly } }
- module: ntdll.dll
  function_name: NtCreateFile
  behaviors: { security_policy: { pre_hook: ForceReadOnly } }
```

To summarize the integration between declarative policy configuration and callback definitions, Code 4.4 presents a complete YAML policy for another extension of the running example. The hook on `NtOpenFile` combines the pre-logic verdict of call denial with a post-logic.

Code 4.4: Example of core HookDesigner YAML policy (extension of the running example).

```
configure:
  paradigm: inline_hook
  engines: [detours, frida]
  process: "filedeny_target.exe"
  logic_file: "logic.cpp"

hooks:
  - module: ntdll.dll
    function_name: NtOpenFile
    address_resolution: { strategy: export }
    behaviors:
      security_policy:
        pre_hook: CheckFileAllowed
        post_hook: CheckOpenedFile
        on_deny:
          return_value: "STATUS_ACCESS_DENIED"
```

The corresponding companion `logic_file` (`logic.cpp`) is shown in Code 4.5. It holds the signatures block and the two tagged callbacks. The post-logic `CheckOpenedFile` runs only when the pre-logic allowed the call, after the original function. The pre-logic can judge only the name that the caller passed, but once the file is open, `GetFinalPathNameByHandleW` can return its *final path*, for example the target of a symbolic link [Mic26g]. If the final path names a protected file, the post-logic closes the handle and changes `__hook_result` to `STATUS_ACCESS_DENIED`.

Code 4.5: C++ logic file (`logic.cpp`) illustrating signature declarations, tag annotations, a pre-logic verdict and a post-logic that overrides the return value (simplified).

```
#include <windows.h>
#include <winternl.h>
```

```
#include "../hd_paths.h"     // CopyObjectName
#define STATUS_ACCESS_DENIED ((NTSTATUS)0xC0000022L)

// @signatures-start
NTSTATUS NTAPI NtOpenFile(PHANDLE FileHandle, ACCESS_MASK DesiredAccess,
    POBJECT_ATTRIBUTES ObjectAttributes, PIO_STATUS_BLOCK IoStatusBlock,
    ULONG ShareAccess, ULONG OpenOptions);
void WINAPI ExitProcess(UINT uExitCode);
// @signatures-end

// IsDeniedName: true if the last component of a path is in the deny list

// Pre-logic of NtOpenFile: true = allow
// @CheckFileAllowed-start
static bool CheckFileAllowed(POBJECT_ATTRIBUTES ObjectAttributes) {
    /* checks the name of the file with IsDeniedName */
}

// Post-logic of NtOpenFile: runs only for allowed calls
// @CheckOpenedFile-start
static void CheckOpenedFile(PHANDLE FileHandle, NTSTATUS& __hook_result) {
    if (__hook_result < 0 || !FileHandle || !*FileHandle) return;
    wchar_t path[MAX_PATH];
    const DWORD length = GetFinalPathNameByHandleW(*FileHandle, path, MAX_PATH,
                                                   FILE_NAME_NORMALIZED |
    VOLUME_NAME_DOS);
    if (length == 0 || length >= MAX_PATH) return;   // name not available: keep
    __hook_result
    if (!IsDeniedName(CopyString(path))) return;
    CloseHandle(*FileHandle);
    *FileHandle = nullptr;
    __hook_result = STATUS_ACCESS_DENIED;
}
```

After signature extraction and semantic validation, for each declared hook, the transpiler matches its `function_name` against the extracted signatures, extending the hook's in-memory representation with its C++ return type, parameter list, and calling convention. In Stage 3, this merged data structure is passed to the template engine that generates the C++ code (Section 4.3).

4.3 C++ Code Generation and DLL Compilation

Once Stage 2 (Section 4.1.2) has validated the declarative policy and extracted the data within the logic file, Stage 3 and Stage 4 translate the in-memory policy representation into compilable C++ source code and build the final instrumentation DLL.

Stage 3 generates the C++ code with a *template engine*. The program gives the template engine a template and a set of named values, the *render context* (Section 4.3.2.2), and the engine *renders* the template. HookDesigner uses Jinja2 [Pal24], a well-known template engine for Python. The templates of HookDesigner are C++ files that also contain Jinja2 code (Section 4.3.2).

To realize the foundational principle of *engine-independent transpilation* established in Section 4.1.1, this phase must overcome a specific challenge: inline hooking engines expose fundamentally different low-level interfaces as examined in State of the Art (Chapter 3). Hard-coding engine-specific boilerplate into the transpiler would couple the transpilation process to specific frameworks, preventing the addition of new instrumentation engines without modifying the transpiler core.

To resolve this challenge, HookDesigner organizes code generation process through two different layers:

1. An *external engine registry* (`engines.yaml`) that parameterises engine-specific template selection, cast expressions, and build arguments (Section 4.3.1).
2. A *modular template hierarchy* that factors shared instrumentation logic into engine-independent partial templates while delegating specific features to engine templates (Section 4.3.2).

4.3.1 Engine Registry

The `engines.yaml` registry is a declarative configuration file that contains all engine-specific parameters and build requirements. The transpiler loads this registry at startup via `load_engines()` and uses it to drive subsequent template rendering and compilation decisions. Crucially, the registry decouples the transpiler's core code generation logic from individual hooking backends: supporting an additional hooking engine requires only supplying its template and defining its parameters in `engines.yaml`, requiring zero modifications to the transpiler logic.

4.3.1.1 Registry Schema

Engine entries are organized under a *paradigm* top-level key that groups engines by their hooking model (e.g., `inline_hook`). This classification enables HookDesigner to support heterogeneous instrumentation paradigms (such as code-cache-based DBI engines like DynamoRIO or Intel PIN, Section 2.4). Table 4.3 documents the configuration schema for engines operating under the `inline_hook` paradigm, which mandates four core parameters alongside an optional list of auxiliary system libraries.

Field	Type	Req.	Role / Description
`template`	string	Yes	Filename of the engine's root template located in `templates/` (e.g., `detours_template.cpp`).
`cast_format`	format string	Yes	Python format string for function-pointer casting during dynamic symbol resolution (e.g., `"(True{name}_t)"` or `"(gpointer)"`).
`include_path`	string (path)	Yes	Header include directory for the hooking engine; supports `${VAR}` environment variable expansion.
`lib_path`	string (path)	Yes	Static library (`.lib`) file path for the hooking engine; supports `${VAR}` environment variable expansion.
`extra_libs`	list of strings	No	Auxiliary Windows system libraries

Table 4.3: Schema definition for engine entries registered in `engines.yaml`.

Code 4.6 lists the registry configuration supporting Microsoft Detours and Frida-Gum. The `cast_format` field illustrates how the registry handles low-level engine differences without introducing conditional branching into the template files. By encoding these conventions in `cast_format`, the address resolution handler (`handle_address_resolution`, detailed in Section 4.3.3) emits the appropriate C++ cast expression directly in the transpilation process.

Code 4.6: Registry definitions for Detours and Frida-Gum in `engines.yaml`.

```yaml
inline_hook:
  detours:
    template: "detours_template.cpp"
    cast_format: "(True{name}_t)"
    include_path: "${DETOURS_INC}"
    lib_path: "${DETOURS_LIB}"
    extra_libs:
      - "User32.lib"
      - "Advapi32.lib"

  frida:
    template: "frida_template.cpp"
    cast_format: "(gpointer)"
    include_path: "${FRIDA_INC}"
    lib_path: "${FRIDA_LIB}"
    extra_libs:
      - "User32.lib"
      - "Advapi32.lib"
      - "Ole32.lib"
```

```
    - "Shell32.lib"
```

4.3.2 Jinja2 Template Architecture

Jinja2 marks its own code inside a template with tags [Pal24]. The templates of HookDesigner use three kinds of tags:

- `{{ x }}` is replaced by the value of `x`, for example by the name of a target function.
- `{% for %}` ... `{% endfor %}` repeats the text between the two tags once for each element of a list, for example once for each hook of the policy, and `{% if %}` ... `{% endif %}` keeps this text only when a condition is true.
- `{% include %}` inserts another template file at that point; the inserted file is rendered with the same render context, and it can also use the variable of the `for` loop that contains the tag.

In the listings of this chapter, all the text outside these tags is the C++ code that goes into the generated file.

A naive approach to multi-target code generation would maintain monolithic, duplicated template files for each supported engine. However, with the framework we have selected, the majority of generated code is engine-independent. Duplicating this code across engines would introduce maintenance issues.

4.3.2.1 Two-Layer Hierarchy

To maximize code reuse while preserving engine autonomy, in HookDesigner templates are organized into a two-layer hierarchy:

1. **Engine-Specific Templates** (`detours_template.cpp`, `frida_template.cpp`): The files that define the translation skeleton. Each template defines module-level includes, global synchronisation primitives, function-pointer typedefs, and the exported `InitializeHooks` and `DeinitializeHooks` hooks lifecycle routines (Section 4.3.5).
2. **Shared Partial Templates** (`_*.cpp`): Reusable partial templates located in `templates/`. The engine templates insert them with `{% include %}`, so the Jinja2 code inside a partial template is rendered with the same render context as the engine template.

The two lifecycle functions belong to the engine-specific templates because their bodies cannot be shared across engines: as explained in Section 3.5, guaranteeing thread safety during hook attachment and detachment requires engine-specific transactional models.

4.3.2.2 Render Context

The render context that the transpiler passes to both engine-specific templates has four entries:

- `hooks`: the list of hooks extracted in Stage 2, containing the hook policy data merged with the data derived from logic file (Section 4.2).
- `logic_file_rel`: The normalized relative filesystem path from the output C++ file to the companion user logic file (Section 4.2.2), ensuring that the generated `#include` directive resolves correctly across arbitrary build directories.
- `telemetry`: The telemetry configuration dictionary (or `None`), for managing the conditional rendering of logging and metrics C++ code (Section 4.4.1).
- `has_execution_time`: `true` if at least one hook enables `execution_time` (Section 4.4.1). It decides whether the template declares the timer frequency `g_PerfFrequency` and initializes it with `QueryPerformanceFrequency` in `InitializeHooks`.

4.3.3 Target Address Resolution

Before template rendering begins, the transpiler resolves the virtual address lookup expression for every declared hook and stores it in `hook["address_resolution_cpp"]`, this will be used by the hooks activation routine (`InitializeHooks`, Section 4.3.5.1). To support both public exported APIs and proprietary functions, HookDesigner provides two resolution strategies:

- **Export Resolution:** For functions exposed in a module's EAT, the transpiler generates dynamic lookup via `GetProcAddress`, supporting custom or compiler-mangled symbol names where required.
- **Offset Resolution:** In proprietary software where internal subroutines are unexported, the transpiler generates code that adds the RVA to the base address returned by `GetModuleHandleA`; the result stays correct under ASLR, because the RVA does not depend on the base address (Section 2.1.3).

To maintain engine independence, the resulting pointer expression is formatted according to the active engine's `cast_format` (e.g., typed function pointers for Microsoft Detours vs. `gpointer` for Frida-Gum, as defined in Table 4.3), ensuring that templates remain decoupled from engine-specific type casting conventions.

4.3.4 Hook Function Generation

For every function declared in the policy, HookDesigner generates a hook function named `Hooked<FunctionName>`. As established in Section 2.2, to preserve stack alignment and register integrity across interception boundaries, the hook function must match the target function's ABI specifications exactly. On 64-bit Windows the compiler ignores the classic qualifiers (`__stdcall`, `__cdecl`, `__fastcall`), but `__vectorcall` remains in effect (Section 2.2.2); on 32-bit targets, every qualifier changes how arguments are passed. For this reason, the hook function must keep the calling convention qualifier of the target.

The transpiler achieves strict fidelity by rendering the function signature directly from the metadata extracted from the logic file in Stage 2 (Section 4.2.2.1), as illustrated in the template definition in Code 4.7:

Code 4.7: Hook function signature rendering template.

```
{{ hook.return_type }} {% if hook.calling_convention %}{{ hook.calling_convention }}
  {% endif %}Hooked{{ hook.function_name }}({%- for param in hook.parameters -%}{{
  param.type }} {{ param.name }}{%- if not loop.last -%}, {%- endif -%}{%- endfor
  -%})
```

While the classic inline hook function structure (Section 2.3) follows a basic three-step pattern (*pre-logic* → *trampoline dispatch* → *post-logic*), deploying such a function into a live, multi-threaded process could be dangerous: an unhandled exception inside the hook function crashes the host application, reentrant API calls cause stack overflow, in-flight threads crash during DLL unloading.

For this reason in HookDesigner, the body of each hook function (`_hook_body.cpp`) extends this basic pattern with automated telemetry collection (Section 4.4) and runtime safety defenses (Section 5.1). The call to the trampoline through the pointer `True<F>` splits this body into two parts. We call *pre-trampoline code* all the code that the hook function runs before this call, and *post-trampoline code* all the code that it runs after the trampoline returns. The pre-trampoline code includes the call to the pre-logic, and the post-trampoline code includes the call to the post-logic and the telemetry code. The complete structure of a generated hook function is shown in detail in Section 5.1.5.

4.3.5 Hook Lifecycle Function Generation

In addition to individual hook functions (Section 4.3.4), the generated code must provide the functions that install and remove the hooks of the instrumentation DLL. HookDesigner generates two exported functions with standard C linkage: `InitializeHooks`, which resolves the targets and installs the hooks, and `DeinitializeHooks`, which removes them before the DLL is unloaded.

The hooks could also be installed in `DllMain`, which the loader calls when it maps the DLL (Section 2.1.2). HookDesigner does not do so, because `DllMain` runs while the loader holds a lock of the process, and the work of these two functions is not simple: when telemetry is enabled, they start the background threads of the logger and of the metrics server (Section 4.4), and they stop them again, waiting for them to end, when the hooks are removed or when the installation fails. A thread that waits for other threads while the loader holds this lock can block the process forever (Section 5.2.1.3). For this reason, the `DllMain` of the generated DLL installs no hooks, and the injector calls the two functions itself, each through a new remote thread that runs outside `DllMain`: `InitializeHooks` after `LoadLibrary` has mapped the DLL, and `DeinitializeHooks` before `FreeLibrary` unmaps it.

4.3.5.1 Transactional Attachment and Detachment

Once the target functions are located, the template performs batch hook registration using the engine's transactional API (Section 3.5).

Under Frida-Gum a hook that the engine refuses to install reports such a failure only through its return value, so the template turns any result other than `GUM_REPLACE_OK` into a C++ exception, which starts the rollback (Code 4.9).

Depending on the active hooking engine, the template emits the corresponding transactional attachment for declared hooks, as illustrated in Code 4.8 for Microsoft Detours and Code 4.9 for Frida-Gum.

In both listings only the `catch` handlers are left as comments; their content is described in Section 5.1.1.2.

Code 4.8: Simplified Jinja2 template logic for Microsoft Detours hook lifecycle functions.

```
extern "C" __declspec(dllexport) DWORD WINAPI InitializeHooks(LPVOID lpParam) {
    try {
        // Locate Functions
        {% for hook in hooks %}
        True{{ hook.function_name }} = {{ hook.address_resolution_cpp }};
        {% endfor %}
```

```
        // Transactional Attachment
        std::vector<HANDLE> threads = EnlistAllThreads();
        DetourTransactionBegin();
        for (HANDLE hThread : threads) {
            DetourUpdateThread(hThread);
        }
        {% for hook in hooks %}
        if (True{{ hook.function_name }}) {
            DetourAttach(&(PVOID&)True{{ hook.function_name }}, (PVOID)Hooked{{ hook.
    function_name }});
        }
        {% endfor %}
        DetourTransactionCommit();

        for (HANDLE hThread : threads) {
            CloseHandle(hThread);
        }

    } catch (const std::exception& ex) {
        //diagnostic and rollback mechanism...
    }
}

extern "C" __declspec(dllexport) DWORD WINAPI DeinitializeHooks(LPVOID lpParam) {
    try {
        // Transactional Detachment
        std::vector<HANDLE> threads = EnlistAllThreads();
        DetourTransactionBegin();
        for (HANDLE hThread : threads) {
            DetourUpdateThread(hThread);
        }
        {% for hook in hooks %}
        if (True{{ hook.function_name }}) {
            DetourDetach(&(PVOID&)True{{ hook.function_name }}, (PVOID)Hooked{{ hook.
    function_name }});
        }
        {% endfor %}
        DetourTransactionCommit();

        for (HANDLE hThread : threads) {
            CloseHandle(hThread);
        }

    } catch (const std::exception& ex) {
        //diagnostic and rollback mechanism...
    }
}
```

Code 4.9: Simplified Jinja2 template logic for Frida-Gum hook lifecycle functions.

```
extern "C" __declspec(dllexport) DWORD WINAPI InitializeHooks(LPVOID lpParam) {
    try {
        gum_init_embedded();
        interceptor = gum_interceptor_obtain();

        // Locate Functions
        {% for hook in hooks %}
        target_{{ hook.function_name }} = {{ hook.address_resolution_cpp }};
        {% endfor %}

        // Transactional Replacement
        gum_interceptor_begin_transaction(interceptor);
        {% for hook in hooks %}
        if (target_{{ hook.function_name }}) {
            GumReplaceReturn r = gum_interceptor_replace(interceptor,
                                    target_{{ hook.function_name }},
                                    (gpointer)Hooked{{ hook.function_name }},
                                    nullptr,
                                    (gpointer*)&True{{ hook.function_name }});
            // Gum reports failures only through the return code: throwing starts the
    rollback
            if (r != GUM_REPLACE_OK) {
                throw std::runtime_error(fmt::format("Frida failed to replace function
    '{}'. Error code: {}", "{{ hook.function_name }}", (int)r));
            }
        }
        {% endfor %}
        gum_interceptor_end_transaction(interceptor);

    } catch (const std::exception& ex) {
        //diagnostic and rollback mechanism...
    }
}

extern "C" __declspec(dllexport) DWORD WINAPI DeinitializeHooks(LPVOID lpParam) {
    try {
        // Transactional Detachment
        gum_interceptor_begin_transaction(interceptor);
        {% for hook in hooks %}
        if (target_{{ hook.function_name }} != nullptr) {
            gum_interceptor_revert(interceptor, target_{{ hook.function_name }});
        }
        {% endfor %}
        gum_interceptor_end_transaction(interceptor);

        g_object_unref(interceptor);
        gum_deinit_embedded();
```

```
    } catch (const std::exception& ex) {
        //diagnostic only, no rollback...
    }
}
```

While Codes 4.8 and 4.9 illustrate the simplified Jinja2 template loops used to generate batch attachment and detachment, deploying these routines into a live, multi-threaded process requires additional safeguards to coordinate thread synchronization and handle telemetry subsystems. The complete definition of `InitializeHooks` and `DeinitializeHooks` is detailed in Section 5.2.

4.3.6 DLL Compilation

The generated C++ is compiled into a 64-bit (or 32-bit) Windows DLL by `compile_dll()`. The transpiler locates the Microsoft Visual C++ (MSVC) compiler by invoking Microsoft's `vcvars64.bat` environment script through a spawned subprocess, eliminating the requirement for manual configuration. To assemble the complete build command, `compile_dll()` combines the compiler flags in Table 4.4 with the engine-specific settings retrieved from `engines.yaml` (Section 4.3.1).

Table 4.4 documents the specific compiler flags applied to `cl.exe` and their justifications.

Flag	Category	Behavior
`/LD`	Linker Mode	Instructs MSVC to produce a DLL
`/EHa`	Exception Model	Enables asynchronous structured exception handling (SEH) translation. (Section 5.1.1).
`/O2`	Optimization	Minimizes Interception Overhead on APIs called at high frequencies
`/utf-8`	Source Encoding	Enforces UTF-8 character encoding across source files, ensuring cross-platform consistency.
`/std:c++17`	Language Standard	Enables C++17 language constructs
`rt-link`	CRT Linkage	{/MT,/MD}, default /MT. Links the C runtime library (CRT) statically (`/MT`, `libcmt.lib`) or dynamically (`/MD`).

Table 4.4: MSVC compiler flags utilized in `compile_dll()` and their behavior

4.3.6.0.1 DLL portability A key design constraint for HookDesigner is reliability in restricted industrial supervisory control and data acquisition (SCADA) environment.

If the instrumentation DLL were dynamically linked against the universal CRT (`/MD`), loading the payload onto a target machine lacking the exact corresponding redistributable build would trigger a loader failure, silently failing injection. By enforcing static CRT linkage (`/MT`) and statically linking the hooking engine (Detours or Frida-Gum) alongside the telemetry libraries (`spdlog` and `prometheus-cpp`), the transpiler outputs a completely self-contained binary. The resulting DLL carries zero dependencies beyond core Windows system libraries, guaranteeing reliable injection into restricted industrial environments.

4.4 HookDesigner Telemetry

In industrial SCADA environment, critical functions frequently execute thousands of times per second across concurrent worker threads. When instrumenting such processes, where performances are essential, telemetry cannot come at the expense of host responsiveness. A naive instrumentation strategy that performs heavy tasks (such as executing blocking disk writes, or querying network endpoints directly within the hook function) introduces latency spikes and thread contention, destabilizing the monitored system.

To overcome these constraints and fulfill the *High-Performance Telemetry* design principle introduced in Section 4.1.1, HookDesigner implements the following solutions:

1. **Asynchronous Logging (Section 4.4.4):** A non-blocking logging mechanism powered by `spdlog`, utilizing a circular buffer that decouples hot-path hook execution from background I/O draining.
2. **Real-Time Performance Metrics (Section 4.4.5):** A native Prometheus metrics subsystem based on `prometheus-cpp`, exposed through an embedded HTTP server.

Section 2.5 explains how these two libraries work; this section describes how HookDesigner configures them and generates the code that uses them.

Before generating code, the framework processes telemetry policy rules during transpilation: Section 4.4.1 defines the declarative YAML schema and static validation constraints.

4.4.1 Telemetry Policy Specification

HookDesigner defines telemetry behaviors declaratively within the policy document introduced in Section 4.2.1. Telemetry configuration is split into two distinct components:

process-wide parameters declared in the top-level `configure.telemetry` block, and hook-specific behavioral policies declared under each hook entry's `behaviors` mapping.

4.4.1.1 Global Telemetry Configuration

The top-level `configure.telemetry` mapping establishes the general settings for the logging framework and the metric server endpoint. Table 4.5 outlines the schema definitions for this block.

Field	Type	Req.	Default	Description
`logger`	string	No	`null`	Logging engine identifier (currently supports `"spdlog"`).
`logger_config`	object	No	`{}`	if `logger` not `null`, log configuration
`metrics`	string	No	`null`	Metrics engine identifier (currently supports `"prometheus"`).
`metrics_config`	object	No	`{}`	if `metrics` not `null`, HTTP exposer configuration.
Logger Settings (`logger_config`)				
`type`	string	No	`"file"`	Log destination: `"file"` (local disk) or `"udp"` (remote Syslog).
`host`	string	No	`"127.0.0.1"`	Target host IPv4 address only if `type: "udp"`.
`port`	integer	No	`1514`	Target UDP listening port only if `type: "udp"`.
`file`	string	No	`"hook_log.txt"`	Destination file path only if `type: "file"`.
`level`	string	No	`"info"`	Minimum severity filter (`trace`, `debug`, `info`, `warn`, `error`, `critical`).
Metrics Exposer Endpoint (`metrics_config`)				
`endpoint`	string	No	`"127.0.0.1:8080"`	Address and port bound by the embedded Prometheus HTTP server.

Table 4.5: Schema definition for the top-level `configure.telemetry` block.

4.4.1.2 Hook-Level Telemetry Behaviors

At the granularity of individual hooks, the `behaviors.performance_metrics/log_rules` defines logging and performance metrics rules to inject in the `hook.target_function`. Hook-level telemetry is configured across two sub-mappings:

- `log_rules`: A list of conditional rule objects specifying log severity, triggering conditions, and emitted message formats.
- `performance_metrics`: flag for execution latency tracking and Prometheus canonical metrics configuration.

Within the metrics configuration, an optional `label` key-value mapping can be declared. In Prometheus, labels (Sections 2.5.2 and 4.4.5.2) allow a single metric family to partition its observations dynamically according to runtime execution context (e.g., classifying a function invocation counter by return status or error code).

Table 4.6 documents the schema for these behavioral sub-policies.

4.4.1.3 Logging Rule Declaration Modes

In DBI, logging formats can vary widely across different inspection environments. During exploratory reverse engineering, an analyst often requires rapid, call-graph tracing log to understand execution control flow. Conversely, during an incident response, logs should be able to capture precise parameter states and outcome codes formatted to match external ingestion tools.

To reconcile these different requirements, HookDesigner supports different declaration modes within `behaviors.log_rules`, illustrated in Code 4.10:

1. **Default / Empty Rule (`- {}`):** Injects a function invocation log using default metadata (`name: "rule_1"`, `level: "info"`, `condition: "true"`), only for checking if the target_function is called.
2. **Default Rule with Arguments:** The policy author can specify target function parameters to include in the log; the transpiler validates argument identifiers against the target signature and generates a structured, pipe-delimited key-value format string automatically.
3. **Custom Format String:** Parses tokens matching `{placeholder}` (supporting format specifiers like `:x` or `:.2f`) and emits a pre-compiled positional `fmtlib` template representing the custom log.

Field	Type	Req.	Default	Description
Performance Metrics Policy (`behaviors.performance_metrics`)				
`execution_time`	bool	No	`false`	Enables `target_function __hook_latency` measurement
`counters`	list	No	`[]`	Counter metric definitions
`gauges`	list	No	`[]`	Gauge metric definitions
`histograms`	list	No	`[]`	Histogram metric definitions
Metric Instrument Schema (Counters, Gauges, Histograms)				
`name`	string	Yes	—	Unique metric identifier within the hook function.
`description`	string	No	`""`	Human-readable help string exported to the Prometheus registry.
`condition`	string	No	`"true"`	Boolean expression indicating when the observation occurs.
`value`	string	Gauges/Hists	—	C++ numerical expression to record (e.g., `__hook_latency`, parameter).
`buckets`	list	No (Hists)	*default*	Custom strictly increasing positive bucket upper bounds only for histograms.
`label`	dict	No	`{}`	Dynamic key-value label mapping for dimensional partitioning.
Logging Rule Schema (`behaviors.log_rules`)				
`name`	string	No	`"rule_N"`	Unique rule name identifier within the hook
`level`	string	No	`"info"`	Log severity level (`trace`, `debug`, `info`, `warn`, `error`, `critical`).
`condition`	string	No	`"true"`	Boolean expression evaluated at runtime to trigger logging.
`arguments`	list	No	`[]`	Parameter identifiers captured for auto-generated format strings.
`format`	string	No	`null`	Custom message format string supporting positional format specifiers.

Table 4.6: Schema definition for hook-level logs and metrics

4. **Multi-Rule:** Emits independent sequential C++ `if` blocks for generating logs evaluated against return codes or runtime arguments.

5. **No Logging (`log_rules: []`):** Completely eliminates all logging code generation.

Code 4.10 shows the modes on an extension of the running example that records the file activity of the program. `NtOpenFile` uses the empty rule. `NtCreateFile` declares two rules (mode 4): the first lists two arguments (mode 2), and the second uses a custom format on failed calls (mode 3); `__hook_result < 0` selects the failures, because a negative `NTSTATUS` is a warning or an error (Section B.2). `NtReadFile`, the native function that reads data from a file, is called too often to log each call, so it only counts its calls (mode 5).

Code 4.10: Demonstration of the logging rule declaration modes (file-activity extension of the running example).

```
hooks:
  # Mode 1: Empty Rule
  - module: ntdll.dll
    function_name: NtOpenFile
    behaviors:
      log_rules:
        - {}  # Emits: "[HOOK NtOpenFile] [rule_1] called"

  # Mode 4: Multi-rule
  - module: ntdll.dll
    function_name: NtCreateFile
    behaviors:
      log_rules:
        # Mode 2: Default Rule with Arguments
        - name: "create_args"
          level: "debug"
          arguments:
            - input_DesiredAccess
            - input_CreateDisposition
          # Emits: "[HOOK NtCreateFile] [create_args] | input_DesiredAccess=1179785 |
    input_CreateDisposition=1"
        # Mode 3: Custom format string with format specifiers (:x)
        - name: "create_failed"
          condition: "__hook_result < 0"
          level: "warn"
          format: "NtCreateFile failed: name={CopyObjectName(input_ObjectAttributes)}
    status=0x{(ULONG)__hook_result:x}"
          # Emits: "NtCreateFile failed: name=\??\C:\data\a.txt status=0xc0000034"

  # Mode 5: No logging
  - module: ntdll.dll
    function_name: NtReadFile
    behaviors:
```

```
      log_rules: []  # Emits: nothing
      performance_metrics:
        counters:
          - name: "reads_total"
```

The custom format of the second rule calls `CopyObjectName`, a helper function that the logic file of the running example includes from a small header, to read the file name from `ObjectAttributes` (Section 5.1.2.3).

Code 4.11 shows the telemetry of another extension of the running example. The hook on `NtOpenFile` denies files as in the running example, sends its log over UDP to a log collector listening on port 1514, and exposes three Prometheus metrics (Section 2.5):

- `opens_total`, a counter of all the calls, with a dynamic label `verdict` that is `"denied"` or `"allowed"`;
- `failed_total`, a counter without labels of the allowed calls that failed, that is, that returned a negative `NTSTATUS` (Section B.2);
- `latency_ms`, a histogram of the duration of the allowed calls. The condition `!__hook_denied` leaves out the denied calls, whose duration is 0 because the original function does not run (Section 4.4.2.1).

Code 4.11: Example YAML with telemetry configuration (metrics extension of the running example).

```
configure:
  process: "filedeny_target.exe"
  logic_file: "logic.cpp"
  telemetry:
    logger: "spdlog"
    logger_config:
      type: "udp"
      host: "127.0.0.1"
      port: 1514
      level: "info"
    metrics: "prometheus"
    metrics_config:
      endpoint: "0.0.0.0:8085"

hooks:
  - module: ntdll.dll
    function_name: NtOpenFile
    address_resolution:
      strategy: export
    behaviors:
```

```
    performance_metrics:
      counters:
        - name: "opens_total"
          description: "Calls to NtOpenFile, by verdict of the policy"
          label:
            verdict: '__hook_denied ? "denied" : "allowed"'
        - name: "failed_total"
          description: "Allowed calls to NtOpenFile that failed"
          condition: "!__hook_denied && __hook_result < 0"
      histograms:
        - name: "latency_ms"
          description: "Duration of the allowed calls to NtOpenFile, in milliseconds"
          condition: "!__hook_denied"
          value: "__hook_latency"
          buckets: [0.01, 0.05, 0.1, 0.5, 1.0, 5.0]
    log_rules:
      - name: "denied"
        condition: "__hook_denied"
        level: "warn"
        format: "denied open: name={CopyObjectName(input_ObjectAttributes)} tid={
__hook_tid}"
    security_policy:
      pre_hook: CheckFileAllowed
      on_deny:
        return_value: "STATUS_ACCESS_DENIED"
```

4.4.2 Static Telemetry Expression Translation

Inside a hook function, evaluating telemetry expressions—such as conditional triggers, custom format strings, and dynamic metric values—through runtime interpretation would be inefficient. While conventional frameworks rely on embedded interpreters or runtime AST evaluators—introducing C-structure marshaling costs, heap allocations, and garbage collection jitter per each hook invocation—HookDesigner adopts a *static expression translation* approach across Stage 2 and Stage 3 of the pipeline (Section 4.1.2). The transpiler analyzes telemetry expressions ahead of time and translates them directly into native C++ instructions. The core translation and demand-driven parameter capture registration procedure is formalized in Algorithm 1.

4.4.2.1 Telemetry Wildcards

As formalized in lines 1–16 of Algorithm 1, domain-specific wildcards in telemetry expressions are resolved directly into concrete C++ constructs:

Algorithm 1 Static telemetry expression translation (`translate_telemetry_expr`)

Require: telemetry expression e, target hook metadata h
Ensure: translated C++ expression e, updated capture sets and tracking flags on h

```
1: replace every __hook_tid in e with GetCurrentThreadId()
2: if __hook_latency occurs in e then
3:     if h.is_noreturn then
4:         throw semantic error
5:     end if
6:     mark h as requiring execution-time tracking
7:     replace __hook_latency with elapsed_ms_<h.function_name>
8: end if
9: if __hook_result occurs in e then
10:    if h.return_type is void or h.is_noreturn then
11:        throw semantic error
12:    end if
13: end if
14: if __hook_denied occurs in e and h declares no on_deny then
15:    throw semantic error
16: end if
17: for each token input_<p> or output_<p> in e do
18:    if p ∉ h.parameters or (token is output_ and h.is_noreturn) then
19:        throw semantic error
20:    end if
21:    register p in h.input_vars_needed (or h.output_vars_needed)
22: end for
23: return e
```

- `__hook_tid` (line 1) rewrites to a Win32 `GetCurrentThreadId()` call.
- `__hook_latency` (lines 2–8) automatically forces execution time tracking on target hook h and rewrites the symbol to `elapsed_ms_<FunctionName>`, statically rejecting non-returning routines.
- `__hook_result` (lines 9–13) maps to the target routine's return value, rejecting functions that are `void` or non-returning.
- `__hook_denied` (lines 14–16) is the `bool` verdict of call denial (Section 4.2.1.2.1), `true` for a denied call. It can be used in log and metric conditions, formats, arguments and values, but only in hooks that declare `on_deny`. A denied call does not execute the original function, so it reports `__hook_latency` as 0: latency metrics on these hooks should use the condition `!__hook_denied`, to keep these zeros out of the

distribution.

4.4.2.2 Parameter Capturing

A requirement in inline hooking is capturing function parameter values accurately across the execution of a hook function (Section 4.3.4). Target functions, the pre-logic or the post-logic may mutate parameter buffers or pointer values in-place during execution. Consequently, a telemetry rule evaluated *after* the original function returns may observe altered parameter rather than the values originally supplied by the caller.

To provide correct visibility, HookDesigner establishes an explicit way for targeting parameter from the policy file:

- `input_<param>`: Represents the parameter captured upon hook entry, *before* the pre-logic or the original function runs.
- `output_<param>`: Represents the parameter captured *after* the original function returns and after the post-logic runs.

Crucially, argument capturing is *demand-driven*. By scanning telemetry expressions with regular expressions (`_VAR_RE`), Algorithm 1 (lines 17–22) identifies every referenced parameter, validates it against the formal signature of h (rejecting post-invocation captures on non-returning routines), and registers it into the corresponding capture set (`input_vars_needed` or `output_vars_needed`). If an argument is never referenced in telemetry, no capture code is emitted for it.

For every needed parameter, the transpiler invokes `analyze_type()` on its Clang type handle (`_clang_type`, preserved from Stage 2, Section 4.2.2.1), to automatically generate the safe C++ code for saving the parameter's value according to its type, as detailed in Section 5.1.2:

- **C-Style Strings:** The transpiler generates safe string capture via `input_<param> = CopyString(<param>)` helper.
- **Counted Strings:** Some Windows functions pass text as a pointer to a `UNICODE_STRING` structure. For these parameters the transpiler generates `input_<param> = CopyUnicodeString(<param>)`.
- **Primitive Pointers:** The transpiler generates safe pointer dereference chains via `input_<param> = ReadPointer(<param>)` helper.
- **Primitive Values:** Value types are captured directly by value assignment.

4.4.2.2.1 Member Field Access A design trade-off arises when policy rules reference struct members (e.g., `{input_ObjectAttributes->RootDirectory}` in the running example, or `{input_pStruct.value}`). When member operators (`->` or `.`) are detected, the transpiler identifies the base symbol (`ObjectAttributes`) to ensure the base pointer is captured in `input_vars_needed`. However, it deliberately bypasses complex reentrant Clang struct-layout inspection on the accessed field, emitting the raw member path directly into the generated C++ code. This grants policy authors full expressiveness to inspect arbitrary proprietary structures, while deferring a safe access to the member value directly to the policy authors as detailed in Section 5.1.2.

4.4.2.3 Custom Log Format Handling

When processing a custom log format (e.g., `format: "access=0x{input_DesiredAccess:x} status=0x{(ULONG)__hook_result:x}"`), naive instrumentation engines format the message by parsing placeholders dynamically at runtime, introducing non-deterministic latency. Because HookDesigner's logging engine (`spdlog`, Section 4.4.4) relies on the `fmt` library (`fmtlib`, Section 2.5.1) for message formatting, the transpiler rewrites each declarative template into a native `fmtlib` positional format string during code generation, and the C++ compiler then parses this string when it builds the DLL, so no string parsing is left for runtime.

As illustrated in Figure 4.1 for the log rule of the read-only extension of the running example (Code 4.2), the transpiler scans the format string for tokens matching `{placeholder[:specifier]}`. The rule logs the name of the file, the access mask before and after the pre-logic changed it, the result and the duration of the call. For each detected token:

1. The inner expression is resolved into a native C++ expression via Algorithm 1 and appended to the `fmt` argument list.
2. The entire token itself in the template string is rewritten into an empty or specifier positional token (e.g., `{}` or `{:x}`), generating a positional format string.

The template places the positional format string as input to `spdlog::<level>`.

At runtime, the hook function only converts the arguments to text. spdlog receives a plain string, not a format string, so it does not parse the message again. This is why the template does not pass the format string and the arguments to spdlog directly (Code 4.12): spdlog would then parse the string at runtime, each time the rule logs a message.

Because telemetry frequently evaluates function outcome codes (`__hook_result`), elapsed execution latency, and mutated output buffers (`output_`), HookDesigner positions the

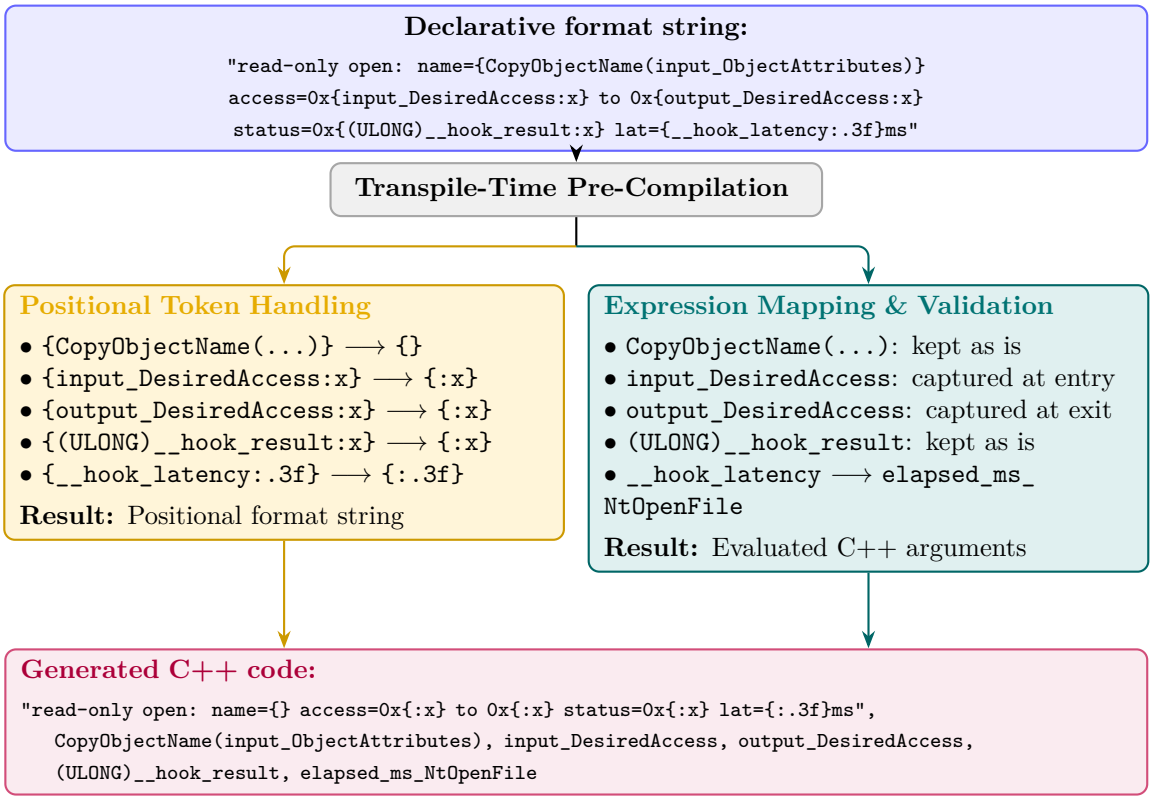


Figure 4.1: Transpile-time format string pre-compilation

telemetry evaluation block (`_telemetry_body.cpp`) in the post-trampoline code, after the post-logic and the output capture.

Code 4.12 shows the code that the transpiler generates inside `_hook_body.cpp` and `_telemetry_body.cpp` for the hook on `NtOpenFile` of the read-only extension, whose log rule is the one of Figure 4.1. The rule references only `DesiredAccess` and `ObjectAttributes`, so only these two parameters are captured at entry, and only `DesiredAccess` at exit. `ObjectAttributes` points to a structure, so it is copied as a plain pointer (Section 5.1.2). The rule writes a line only when the pre-logic has changed the requested access, that is, when `input_DesiredAccess` and `output_DesiredAccess` differ.

Code 4.12: Generated C++ code inside `_hook_body.cpp` and `_telemetry_body.cpp` for `NtOpenFile` (read-only extension of the running example)

```
// Entry Parameter Capturing
ACCESS_MASK input_DesiredAccess = DesiredAccess;
POBJECT_ATTRIBUTES input_ObjectAttributes = ObjectAttributes;

// Pre-logic execution
ForceReadOnly(DesiredAccess, ObjectAttributes);
```

```
// Target function latency measurement
LARGE_INTEGER start_time_NtOpenFile;
LARGE_INTEGER end_time_NtOpenFile;
QueryPerformanceCounter(&start_time_NtOpenFile);
NTSTATUS __hook_result = TrueNtOpenFile(FileHandle, DesiredAccess, ObjectAttributes,
                                        IoStatusBlock, ShareAccess, OpenOptions);
QueryPerformanceCounter(&end_time_NtOpenFile);
double elapsed_ms_NtOpenFile = (g_PerfFrequency.QuadPart > 0)
    ? ((double)(end_time_NtOpenFile.QuadPart - start_time_NtOpenFile.QuadPart) * 1000.0
       / (double)g_PerfFrequency.QuadPart)
    : 0.0;

// Exit Parameter Capturing
ACCESS_MASK output_DesiredAccess = DesiredAccess;

// Telemetry Evaluation
if (input_DesiredAccess != output_DesiredAccess) {
    fmt::memory_buffer ss_readonly;
    fmt::format_to(std::back_inserter(ss_readonly),
                   FMT_COMPILE("read-only open: name={} access=0x{:x} to 0x{:x} status
    =0x{:x} lat={:.3f}ms"),
                   CopyObjectName(input_ObjectAttributes),
                   input_DesiredAccess,
                   output_DesiredAccess,
                   (ULONG)__hook_result,
                   elapsed_ms_NtOpenFile);
    spdlog::info(spdlog::string_view_t(ss_readonly.data(), ss_readonly.size()));
}

return __hook_result;
```

4.4.3 Static Semantic Validation

During Stage 2 (`validate_policy`), HookDesigner performs semantic analysis across declared telemetry rules to catch possible errors offline before DLL compilation. Beyond standard syntactic checks (such as verifying unique rule identifiers and the mutual exclusivity of `arguments` and `format`), the transpiler enforces semantic constraints:

1. **Non-Returning Functions:** For target routines classified as non-returning (`is_noreturn: true`), execution never returns after trampoline invocation. The validator statically rejects any telemetry specification on these targets that references post-invocation properties (including `execution_time: true`, `__hook_latency`, `output_<param>`, or `__hook_result`), preventing the generation of dead code.

2. **Custom Histogram Buckets Monotonicity:** Prometheus registry initialization fails at runtime if histogram bucket upper bounds are unordered or non-positive. The transpiler checks custom `buckets` arrays at compile time to ensure strict positive monotonicity, preventing silent runtime metric registration.

3. **Bounded Label Count:** Every new combination of label values creates a new time series, and labels with many possible values cause *cardinality explosion* (Section 2.5.2). The number of combinations also grows with the number of labels of a metric. HookDesigner reduces this risk by strictly restricting metric definitions to at most a single dynamic label, redirecting high-cardinality inspection to structured log events.

4.4.4 Asynchronous Logging

As explained in Section 2.5.1, a synchronous logger would add the time of every disk or network write to the intercepted call. In an industrial process where intercepted APIs are invoked at high frequencies across multiple threads, this would stall the worker threads and degrade the performance of the software under observation.

4.4.4.1 The Producer-Consumer Scheme

HookDesigner prevents this possible overhead by integrating asynchronous logging powered by `spdlog`. During hook initialization in `_telemetry_init.cpp`, the framework preallocates an asynchronous thread pool:

```
spdlog::init_thread_pool(65536, 1);
```

This creates the spdlog queue (Section 2.5.1) with 65,536 slots instead of the default 8192, and a consumer thread that empties it, as modeled in Figure 4.2.

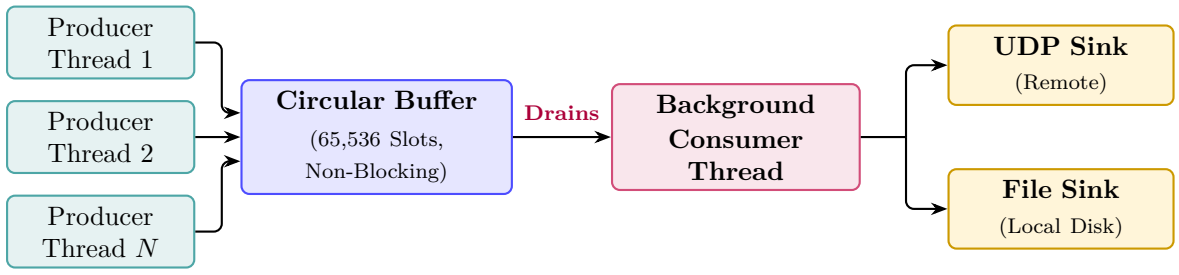


Figure 4.2: Architecture of the asynchronous structured logging subsystem

When a hook function triggers a log rule, the thread of the target only formats the message and copies it into the queue; the disk or network write happens on the worker thread.

4.4.4.2 Buffer Saturation Policy

Under burst load, the threads that run the hook functions may produce messages faster than the worker thread can write them, and the queue fills up. With the default `block` policy (Section 2.5.1), the thread of the target would then wait for a free slot. With a pre-allocated capacity of 65,536 (2^{16}) entries, the circular buffer is intentionally oversized, easily handling standard industrial traffic burst without dropping a single event. Saturating a queue of this size is unlikely under normal operating conditions; it represents extreme scenario—such as complete physical disk stalls or total network interface disconnection. In the rare event that saturation does occur, HookDesigner avoids this by configuring the asynchronous logger with the `overrun_oldest` policy:

```
spdlog::async_overflow_policy::overrun_oldest
```

Under queue saturation, if the 65,536-entry queue fills completely, the logger drops the oldest unwritten messages to make space for incoming events. This guarantees that application worker threads are *never* blocked by logs operations, following the minimal-overhead telemetry design principle established in Section 4.1.

4.4.4.3 Network and Storage Sinks

The logging subsystem supports two sinks (Section 2.5.1):

- **UDP Sink (`logger_config.type: "udp"`)**: HookDesigner integrates a non-blocking UDP sink using `spdlog::sinks::udp_sink_mt`. Utilizing connectionless UDP transport provides an advantage: if the remote logging server crashes or network connectivity is down, the operating system immediately discards the packets. The payload never blocks attempting to re-establish transmission control protocol (TCP) handshakes, isolating the host process from external network failures.
- **Local File Sink (`logger_config.type: "file"`):** HookDesigner instantiates also a asynchronous basic file sink (`spdlog::sinks::basic_file_sink_mt`). File flushing is explicitly triggered only on error conditions (`spdlog::flush_on(spdlog::level::err)`) to minimize disk write amplification during a possible host process crash.

4.4.5 Performance Metrics Engine

Logs describe single events, while quantities such as call rates, error ratios and latency distributions are better served by metrics (Section 2.5.2). HookDesigner exposes metrics

through `prometheus-cpp`: the hook DLL hosts the embedded HTTP server of the library, from which a Prometheus server scrapes the current values.

4.4.5.1 Registration and Metric Family Hierarchy

During the initialization of the payload (`InitializeHooks`), the framework allocates a global metric registry and starts an embedded HTTP exposer on the configured socket address, here the endpoint of Code 4.11 (simplified):

```
g_PromRegistry = std::make_shared<prometheus::Registry>();
g_PromExposer = std::make_unique<prometheus::Exposer>("0.0.0.0:8085");
g_PromExposer->RegisterCollectable(g_PromRegistry);
```

For every counter, gauge and histogram (Section 2.5.2) declared under `hook.behaviors.performance_metrics`, HookDesigner generates a typed global pointer to its metric family (`g_PromFamily_<Function>_<Metric>`), created with `BuildCounter`, `BuildGauge` or `BuildHistogram` and registered in the central registry.

4.4.5.2 Dynamic Labels

A HookDesigner metric can declare one dynamic label (Section 2.5.2) whose value is a C++ expression evaluated inside the hook function (Table 4.6), such as the `verdict` label of Code 4.11, which distinguishes denied calls from allowed ones. The value is known only when the hook function runs, so on each observation the hook function must find the time series of that value with `Family::Add`. Each call allocates memory and takes the mutex of the family (Section 2.5.2). When several threads of the program call the target function at the same time, they wait for each other on this mutex, and the wait is added to every intercepted call.

4.4.5.2.1 The Thread-Local Cache HookDesigner avoids this cost with a *Caching strategy* generated into `_telemetry_body.cpp`.

Behind this optimization is *temporal label locality*: within processes, worker threads frequently execute batches of operations that share identical label values over extended time windows (e.g., in the running example, a thread that opens many files in a row, all with the same verdict).

To exploit this locality, the transpiler generates two thread-local variables for every labeled metric, accessible:

1. `thread_local uint64_t tl_last_hash`: Stores the 64-bit hash of the last observed label value on that specific thread.
2. `thread_local MetricType* tl_last_ptr`: Stores a direct pointer to the underlying Prometheus metric instance.

Code 4.13 illustrates the caching logic generated by the transpiler for the `opens_total` counter of Code 4.11.

Code 4.13: thread-local caching mechanism

```
if (g_PromFamily_NtOpenFile_opens_total) {
    // Evaluate label string and compute 64-bit hash
    std::string cur_val = ToLabelString(__hook_denied ? "denied" : "allowed");
    uint64_t cur_hash = std::hash<std::string>{}(cur_val);

    // Thread-local storage variables
    thread_local uint64_t tl_last_hash_opens_total = 0;
    thread_local prometheus::Counter* tl_last_ptr_opens_total = nullptr;

    // Fast-Path Check: bypass mutex if label matches cached state
    if (tl_last_ptr_opens_total == nullptr || tl_last_hash_opens_total != cur_hash) {
        // SLOW PATH: Mutex acquired only on start or label transition
        tl_last_ptr_opens_total = &g_PromFamily_NtOpenFile_opens_total->Add({ {"verdict
    ", cur_val} });
        tl_last_hash_opens_total = cur_hash;
    }

    // FAST PATH: Lock-free atomic increment
    tl_last_ptr_opens_total->Increment();
}
```

4.4.5.3 Direct Binding for Label-less Metrics

For metrics defined without dynamic labels, HookDesigner applies a compile-time optimization. Because the metric dimensionality is static, resolving the metric instance dynamically at runtime is entirely redundant. During initialization (`_telemetry_init.cpp`), the transpiler generatess code that registers the metric family and immediately adds the base instance, storing the result in a static global pointer, as for the `failed_total` counter of Code 4.11:

```
g_PromMetric_NtOpenFile_failed_total = &g_PromFamily_NtOpenFile_failed_total->Add({ {"
    function", "NtOpenFile"} });
```

Inside the hook function (`_telemetry_body.cpp`), the observation bypasses even the thread-local hash check, updating the metric directly through the global pointer:

```
if (g_PromMetric_NtOpenFile_failed_total) {
    if (!__hook_denied && __hook_result < 0) {
        g_PromMetric_NtOpenFile_failed_total->Increment();
    }
}
```

This delivers the best possible execution efficiency, reducing metric recording overhead to a single atomic instruction.

To demonstrate how these heterogeneous telemetry features operate cohesively within an intercepted function, Code 4.14 illustrates the complete telemetry evaluation block of the hook function, generated with `_telemetry_body.cpp` for the policy of Code 4.11. In this example we can see spdlog logging, a cached labeled counter, a label-less counter and a label-less histogram. A gauge, not used by this policy, is generated like a counter, with `Set(value)` instead of `Increment()`.

Code 4.14: Generated telemetry block of the hook function (`_telemetry_body.cpp`) illustrating the co-existence of the seen telemetry features (metrics extension of the running example)

```
// --- Logging ---
if (__hook_denied) {
    fmt::memory_buffer ss_denied;
    fmt::format_to(std::back_inserter(ss_denied), FMT_COMPILE("denied open: name={} tid
    ={}"),
                   CopyObjectName(input_ObjectAttributes), GetCurrentThreadId());
    spdlog::warn(spdlog::string_view_t(ss_denied.data(), ss_denied.size()));
}

// --- Prometheus Metrics ---
// 2. Custom Counters
if (g_PromFamily_NtOpenFile_opens_total) {
    std::string cur_val = ToLabelString(__hook_denied ? "denied" : "allowed");
    uint64_t cur_hash = std::hash<std::string>{}(cur_val);
    thread_local uint64_t tl_last_hash_opens_total = 0;
    thread_local prometheus::Counter* tl_last_ptr_opens_total = nullptr;

    if (tl_last_ptr_opens_total == nullptr || tl_last_hash_opens_total != cur_hash) {
        tl_last_ptr_opens_total = &g_PromFamily_NtOpenFile_opens_total->Add({ {"verdict
    ", cur_val} });
        tl_last_hash_opens_total = cur_hash;
    }
    tl_last_ptr_opens_total->Increment();
}
if (g_PromMetric_NtOpenFile_failed_total) {
    if (!__hook_denied && __hook_result < 0) {
        g_PromMetric_NtOpenFile_failed_total->Increment();
```

```
    }
}

// 3. Custom Gauges

// 4. Custom Histograms
if (g_PromMetric_NtOpenFile_latency_ms) {
    if (!__hook_denied) {
        g_PromMetric_NtOpenFile_latency_ms->Observe((double)(elapsed_ms_NtOpenFile));
    }
}
```

4.5 Summary

This chapter presented how HookDesigner generates a hook DLL from an instrumentation DSL policy. The transpiler follows a pipeline of seven stages: the first four run offline on the developer's machine, and only the compiled DLL and the injector are needed on the target host. The user writes a YAML policy, which says which functions to hook and which behaviors to attach to them, and a C++ logic file, which holds the signatures of these functions and the code of their pre-logic and post-logic; the transpiler parses the logic file with Clang and checks the two files against each other. A pre-logic can also deny a call: the original function then does not run, and the caller receives the value declared in `on_deny`. An engine registry and a two-layer hierarchy of Jinja2 templates keep the engine-specific code apart from the code that the engines share, so the same policy generates hooks for Microsoft Detours and for Frida-Gum, together with the lifecycle functions `InitializeHooks` and `DeinitializeHooks`; MSVC then builds a self-contained DLL. The telemetry rules of the policy are translated into C++ when the code is generated, instead of being interpreted at runtime. Logs go through the asynchronous logger of spdlog, and metrics through a Prometheus HTTP endpoint inside the DLL, with a thread-local cache for labeled metrics. The generated DLL runs inside the target process, so a fault in its code, can crash the target: Chapter 5 shows how HookDesigner prevents this, and how the DLL is safely injected and ejected.

Chapter 5

Runtime Safety and DLL Lifecycle

Chapter 4 showed how HookDesigner turns a policy and a logic file into a self-contained hook DLL. This chapter follows that DLL into the target process, where it runs during the online stages 5–7 of the pipeline (Section 4.1.2). Section 5.1 shows how the generated code keeps the target running, following the *Crash Safety* design principle (Section 4.1.1), and ends with the complete structure of a generated hook function. Section 5.2 then shows how the injector loads the DLL and activates its hooks, and how the DLL is removed while threads may still be inside it, following the *Thread Safety* principle. As in Chapter 4, the examples come from the running example (Section 1.2).

5.1 DLL Runtime Safety

As established in Section 2.1.1, during DBI, the compiled payload runs directly inside the target application's virtual address space without memory isolation boundaries. While this shared-memory model enables DBI, it introduces several runtime safety issues: a fault or unhandled exception anywhere within the injected DLL—whether during lifecycle orchestration (`InitializeHooks`, `DeinitializeHooks`) or inside hook function execution (`_hook_body.cpp`)—immediately crashes the host process. In production environments, such unexpected terminations violate the *Crash Safety* design principle established in Section 4.1.1.

To satisfy this requirement, HookDesigner incorporates protection mechanisms in both the DLL lifecycle routines and the hook function body.

5.1.1 Asynchronous Exception Handling

In standard C++ development, Windows compilers typically use the synchronous exception model (`/EHsc`) [Mic26c]. Under `/EHsc`, C++ `try/catch` blocks are designed only to intercept exceptions thrown explicitly through the C++ `throw` statement. Under this standard configuration, a C++ `catch(...)` block does not catch OS-level faults—such as dereferencing a null pointer, reading an unmapped virtual address, or encountering a division by zero. In this way, the exception bypasses user code, triggers the default operating system exception filter, and terminates the host process immediately.

5.1.1.1 The /EHa Asynchronous Exception Model

HookDesigner wants a single `catch(...)` block to catch both kinds of errors: C++ exceptions raised by `throw` and OS-level faults. Without this, every protected region would also need the Microsoft-specific `__try`/`__except` construct of SEH. For this reason, the transpiler compiles the hook DLL with the asynchronous exception model [Mic26c], passing the `/EHa` flag to the MSVC compiler (Section 4.3.6). Under `/EHa`, the compiler generates structured unwinding tables that bridge Windows kernel SEH with C++ exception handling:

- Standard C++ `catch(...)` clauses are transformed to catch both synchronous C++ exceptions and asynchronous Win32 hardware faults.
- Proper stack unwinding and destructor execution are guaranteed even when an instruction triggers an OS fault.

5.1.1.2 Exception Boundaries

To prevent cascading failures and isolate runtime faults across the DLL, HookDesigner organizes exception boundaries across two tiers:

5.1.1.2.1 DLL Lifecycle Functions Tier: The exported lifecycle routines `InitializeHooks` and `DeinitializeHooks` are enclosed in `try/catch` blocks compiled under `/EHa`. While the template listings in (Section 4.3.5) focused on the primary attachment and detachment workflows, production safety requires explicit transaction rollback and resource cleanup inside these exception handlers, as shown in Code 5.1.

If an unhandled exception or hardware fault occurs during target address resolution, engine transaction commits, telemetry initialization the fault is safely intercepted, logged via `OutputDebugStringA`, and mapped to an explicit failure exit code (such as `0xE0000002`

for C++ exceptions or `0xE0000003` for SEH hardware faults) returned to the injector via `CallRemoteExport`. Crucially, if a fault occurs in `InitializeHooks` while an engine transaction is open (`in_transaction == true`), the exception handler must make sure that no staged hook is written to memory, so that the instrumentation is never partial:

- **Microsoft Detours:** Handled natively via `DetourTransactionAbort()`, which rolls back all staged hooks without modifying memory.
- **Frida-Gum:** Frida-Gum has no abort primitive, and `gum_interceptor_end_transaction()` writes all staged hooks to memory. For this reason, the `std::exception` handler (which also receives a failed replacement, Section 4.3.5.1) first disarms the staged hooks with `gum_interceptor_revert()` and only then closes the transaction with `gum_interceptor_end_transaction()`. It reverts only the hooks whose trampoline pointer `True<F>` is set, because Frida-Gum writes this pointer only when a replacement succeeds.

Finally, partially constructed telemetry structures and thread-local storage (TLS) allocations are destroyed (Section 5.1.3). This guarantees that a failure during initialization leaves zero dangling hooks and telemetry resources allowing the injector to safely unload the DLL without destabilizing the host application (Section 5.2).

The handlers of `DeinitializeHooks` are simpler. Under Detours, a fault inside the detach transaction aborts it and closes the thread handles, so the hooks stay attached. Under Frida-Gum, the handlers contain no revert loop: `gum_interceptor_revert()` on a hooked target only updates the interceptor's own tables and does not read the target code, so the detach procedure is not expected to fault in the first place. In both cases the handler then logs the error and returns a non-zero code, so the injector does not unload the DLL (Code 5.10).

Unlike the handlers of `InitializeHooks`, they do not destroy the telemetry structures (the logger and the metrics registry) or release the TLS slot of the reEntrancy guard; under Frida-Gum, they also keep the interceptor. After a fault, we cannot tell whether the hooks are still attached (the fault hit the detach transaction) or whether threads are still running inside the hook functions (Section 2.3.1). In both cases a thread may still run the hook code, which uses the TLS slot (Section 5.1.3) and writes to the logger and the metrics. If the telemetry structures were destroyed, that thread would use freed memory and crash the host process; if the TLS slot were released, the reEntrancy guard would work on an invalid TLS index. The only telemetry part that is already gone is the Prometheus HTTP exposer (Section 4.4.5), because `DeinitializeHooks` stops it before it detaches the hooks.

Code 5.1: Atomic transaction rollback and resource containment in lifecycle exception handlers.

```
// catch blocks inside InitializeHooks
catch (const std::exception& ex) {
    // Engine Transaction Rollback
    // Detours:
    if (in_transaction) {
        DetourTransactionAbort();
        for (HANDLE hThread : threads) {
            CloseHandle(hThread);
        }
    }

    // Frida-Gum (explicit revert loop before concluding transaction):
    if (in_transaction && interceptor) {
        {% for hook in hooks %}
        // True<F> is set only by a successful replace
        if (True{{ hook.function_name }}) {
            gum_interceptor_revert(interceptor, target_{{ hook.function_name }});
        }
        {% endfor %}
        gum_interceptor_end_transaction(interceptor);
    }
    if (interceptor) {
        g_object_unref(interceptor);
        interceptor = nullptr;
        gum_deinit_embedded();
    }

    // Teardown initialized telemetry & TLS slots
    g_PromExposer.reset();
    g_PromRegistry.reset();
    spdlog::shutdown();
    if (g_TlsIndex != TLS_OUT_OF_INDEXES) {
        TlsFree(g_TlsIndex);
        g_TlsIndex = TLS_OUT_OF_INDEXES;
    }

    // Stack buffer: a heap allocation could throw std::bad_alloc out of the catch
    block
    char msg[256];
    *fmt::format_to_n(msg, sizeof(msg) - 1, "[CRITICAL] Exception in InitializeHooks:
    {}\n", ex.what()).out = '\0';
    OutputDebugStringA(msg);
    return 0xE0000002; // Custom failure code for C++ exception returned to injector
}
catch (...) {
    // Detours: same abort as above.
    // Frida-Gum: no revert/end_transaction, gum_deinit_embedded discards the staged
    hooks.
    // Telemetry and TLS teardown identical to above...
```

```
    OutputDebugStringA("[CRITICAL] SEH/Hardware fault in InitializeHooks\n");
    return 0xE0000003; // Custom failure code for SEH/Hardware fault returned to
    injector
}
```

5.1.1.2.2 Hook Function Tier: Inside the body of each hook function (`_hook_body.cpp`), execution is segmented into decoupled, independent exception boundaries:

1. **Parameter Capture Boundaries:** Entry and exit argument captures are wrapped in dedicated exception boundaries. If an SEH hardware fault occurs, the error is logged with the corresponding parameter name, and execution safely continues without skipping the pre-logic or the post-logic.

2. **Pre-Logic and Post-Logic Boundaries:** The pre-logic and the post-logic execute inside separate, independent exception blocks. A fault within user code is contained, guaranteeing that the target function is still invoked via its trampoline (`TrueTargetFunction`), that the telemetry code is executed and that the return value is safely passed back to the caller. The only exception is a hook that declares `on_deny`: there, a fault in the pre-logic denies the call instead (fail-closed, Section 4.2.1.2.1).

3. **Telemetry Evaluation Boundary:** Telemetry evaluation is placed inside its own exception boundary, if any required parameter failed to capture, telemetry evaluation is automatically skipped. This prevents format strings and metric observations from computing over uninitialized or corrupted memory. Furthermore, any exception encountered during telemetry emission cannot destabilize the host application or corrupt the function return code.

Code 5.2 illustrates this decoupled pattern:

Code 5.2: Decoupled exception boundaries in `_hook_body.cpp` (schematic, on the `NtOpenFile` hook of the read-only extension of the running example)

```
// Entry Parameter Capture Boundary
bool telemetry_inputs_ready = false;
const char* capturing_input_param = nullptr;
try {
    capturing_input_param = "DesiredAccess";
    input_DesiredAccess = DesiredAccess;
    telemetry_inputs_ready = true;
} catch (const std::exception& ex) {
    char msg[256]; // stack buffer: no heap allocation inside the handler
    *fmt::format_to_n(msg, sizeof(msg) - 1, "[HOOK CAPTURE] Exception in {}: {}\n",
    capturing_input_param, ex.what()).out = '\0';
```

```
    OutputDebugStringA(msg);
} catch (...) {
    char msg[256];
    *fmt::format_to_n(msg, sizeof(msg) - 1, "[HOOK CAPTURE] SEH fault in {}\n",
    capturing_input_param).out = '\0';
    OutputDebugStringA(msg);
}

// Pre-Logic Boundary
try {
    ForceReadOnly(DesiredAccess, ObjectAttributes);
} catch (const std::exception& ex) {
    OutputDebugStringA("[HOOK EXCEPTION] Exception in pre-hook\n");
} catch (...) {
    OutputDebugStringA("[HOOK EXCEPTION] Hardware/SEH fault in pre-hook\n");
}

NTSTATUS __hook_result = TrueNtOpenFile(FileHandle, DesiredAccess, ObjectAttributes,
                                        IoStatusBlock, ShareAccess, OpenOptions);

// Post-Logic Boundary (generated only when a post-logic is declared)
try {
    UserPostLogic(__hook_result);  // placeholder: this hook declares no post-logic
} catch (const std::exception& ex) {
    OutputDebugStringA("[HOOK EXCEPTION] Exception in post-hook\n");
} catch (...) {
    OutputDebugStringA("[HOOK EXCEPTION] Hardware/SEH fault in post-hook\n");
}

//Exit Parameter Capture Boundary
bool telemetry_outputs_ready = false;
const char* capturing_output_param = nullptr;
try {
    capturing_output_param = "DesiredAccess";
    output_DesiredAccess = DesiredAccess;
    telemetry_outputs_ready = true;
} catch (const std::exception& ex) {
    char msg[256];
    *fmt::format_to_n(msg, sizeof(msg) - 1, "[HOOK CAPTURE] Exception in {}: {}\n",
    capturing_output_param, ex.what()).out = '\0';
    OutputDebugStringA(msg);
} catch (...) {
    char msg[256];
    *fmt::format_to_n(msg, sizeof(msg) - 1, "[HOOK CAPTURE] SEH fault in {}\n",
    capturing_output_param).out = '\0';
    OutputDebugStringA(msg);
}

// Telemetry Evaluation Boundary
```

```
if (telemetry_inputs_ready && telemetry_outputs_ready) {
    try {
        EvaluateTelemetry();   // the log rule of the hook
    } catch (const std::exception& ex) {
        OutputDebugStringA("[TELEMETRY EXCEPTION] Exception in telemetry\n");
    } catch (...) {
        OutputDebugStringA("[TELEMETRY EXCEPTION] Unknown exception in telemetry\n");
    }
}

return __hook_result;
```

5.1.2 Transpiler Type Analysis and Memory Access Helpers

As already explained in Section 4.4.2.2, DBI authors often need to inspect function parameters on function entry and exit. However, manually writing C++ code to capture parameters' value is error-prone: forgetting a null check, dereferencing an invalid pointer, or traversing an unbounded string can easily cause memory access violations. To help the work of policy authors, HookDesigner *automates* access to primitive variables and strings in a safe way. By analyzing parameter types at transpile time, the framework automatically generates the appropriate access solution, preventing the author from writing repetitive and error-prone C++ boilerplate.

5.1.2.1 Type Analysis

As introduced in Section 4.4.2.2, every parameter required for telemetry must be safely captured on function entry or exit. To automate this without requiring manual boilerplate from the user, the transpiler performs static type analysis on each parameter's Clang type representation (preserved from Stage 2, Section 4.2.2.1).

The analysis classifies each parameter through three conceptual phases:

- **Canonical Alias Resolution:** In C and Win32 development, types are frequently masked behind typedefs (such as `DWORD`, `LPSTR`, or `size_t`). The transpiler traverses `get_canonical()` to resolve these aliases to their underlying primitive representation.
- **Pointer Indirection Resolution:** By reentrantly unwinding pointee types, the transpiler calculates the pointer depth d. Types with $d = 1$ pointing to character types (`char`, `wchar_t`) are classified as string candidates, and types with $d = 1$ pointing to the native `UNICODE_STRING` structure as counted strings, whereas non-pointer primitives have $d = 0$.

- **Safe Access Expression:** For string pointers, the transpiler generates safe copy calls (`CopyString`), and for counted strings `CopyUnicodeString`. For dereferenceable primitive pointers (excluding opaque handles such as `void*` and complex structures), it nests d dereference helpers (`ReadPointer`). Opaque handles and direct values are passed unchanged.

This automated classification enables the transpiler to emit uniform assignments of the form `input_<p> = safe_read_expr` without requiring template branching in the hook body. Table 5.1 formalizes the complete type mapping matrix.

Clang Type	**Depth (d)**	**Category**	**Generated Read Expression**	**Storage Type**
`int`, `double`, `DWORD`	0	Primitive Value	`param`	`T` (original type)
`const char*`, `char*`	1	C-String	`CopyString(param)`	`std::string`
`const wchar_t*`, `wchar_t*`	1	Wide C-String	`CopyString(param)`	`std::string` (UTF-8)
`PUNICODE_STRING`	1	Counted NT String	`CopyUnicodeString(param)`	`std::string` (UTF-8)
`int*`, `double*`	1	Primitive Pointer	`ReadPointer(param)`	Underlying type (`int`)
`int**`	2	Nested Pointer	`ReadPointer( ReadPointer(param))`	Underlying type (`int`)
`void*`, `HANDLE`	1	Opaque Handle	`param`	Raw pointer type

Table 5.1: Mapping clang type and generated expression

5.1.2.2 Memory-Safety Helper Functions

Most of the generated expressions leverage helper functions defined in `_user_logic_helpers.cpp`:

- **Pointer Dereferencing (`ReadPointer<T>`):** When reading data through a pointer, standard C++ dereferencing (`*ptr`) crashes the process if the pointer is null or unmapped. HookDesigner defines a templated helper:

```
template <typename T>
static T ReadPointer(const T* ptr, T fallback = T()) {
    if (!ptr) {
        return fallback;
    }
    return *ptr;
}
```

If `ptr` contains a non-null but unmapped address (e.g., a dangling pointer pointing to deallocated memory), the resulting read violation is trapped by the outer `/EHa` exception boundary.

- **Bounded String Copying (`CopyString`):** Passing raw string pointers (`char*` or `wchar_t*`) directly to logging formatters is dangerous: if the target buffer lacks a null terminator or points to invalid memory, formatting routines read past the buffer boundary until they fault. HookDesigner generates explicit calls to `CopyString`:

```
static std::string CopyString(const char* source, size_t maxChars = 512) {
    if (!source) return "null";
    size_t len = strnlen(source, maxChars);
    return std::string(source, len);
}
```

The bound of 512 characters covers Windows file paths (`MAX_PATH` is 260 characters) and registry key paths, while it still limits the scan of a corrupted or non-terminated buffer; longer strings are truncated. For wide strings (`wchar_t*`), an overloaded implementation uses `wcsnlen` and converts the result to UTF-8 using `WideCharToMultiByte(CP_UTF8)`, ensuring consistent log formatting.

- **Counted String Copying (`CopyUnicodeString`):** Functions of the Windows native API (`ntdll.dll`) pass names as `UNICODE_STRING` structures: a length in bytes (`Length`), a capacity, and a pointer to a wide-character buffer that is not guaranteed to end with a NUL character [Mic26o]. `CopyString` cannot be used here, because it searches for a terminator that may not exist. `CopyUnicodeString` takes the length from `Length`, caps it at 512 characters, and converts the buffer to UTF-8 as the wide overload of `CopyString` does; a null structure pointer yields `"null"`.

5.1.2.3 Member Access Contract

As introduced in Section 4.4.2.2, when a policy rule references an internal struct member using the arrow or dot operator (e.g., `{input_ObjectAttributes->RootDirectory}`), the

transpiler detects the operator and *bypasses automated Clang type analysis* on the accessed field. Instead, it emits the member access expression defined in the YAML policy directly into the generated C++ code.

This design establishes an explicit contract:

- **Author Flexibility:** The policy author can inspect nested member field, HookDesigner will not refuse the policy.
- **Author Responsibility:** The author is responsible for ensuring pointer or string member safety from the YAML policy, exploiting the helpers provided in `logic_helpers.cpp`. In the running example, the `ObjectName` field of the `OBJECT_ATTRIBUTES` structure (Section B.2) is a counted string reached through a structure, so the author wraps it as `CopyUnicodeString(input_ObjectAttributes->ObjectName)`. The logic file of the running example packs this check in two small helpers, `CopyObjectName` and `ObjectRoot`, which also return a safe value when `ObjectAttributes` is `NULL`; the log rules call them instead of the raw member access (Code 4.10).
- **Exception Boundary Safety:** However, any possible error of the policy author is intercepted by the outer `/EHa` SEH containment boundary, preventing any possible host crash.

5.1.2.4 Array-to-Pointer Decay

A fundamental characteristic of the C and C++ type systems is *array-to-pointer decay*: function parameters declared as arrays decay into raw pointer types within the Clang type analysis without preserving runtime size metadata. Consequently, HookDesigner's automated analysis (Table 5.1) cannot deduce the allocated extent of an arbitrary pointer argument at transpile time without external semantic annotations.

To uphold strict memory safety and avoid out-of-bounds reads, HookDesigner deliberately enforces dereferencing: automated parameter capture for non-character pointer types (`T*`) strictly dereferences the element at index zero via the `ReadPointer<T>` helper. If an instrumentation policy requires inspecting across array elements, this logic cannot be expressed through automated YAML telemetry rules. Instead, buffer inspection is explicitly delegated to the pre-logic or the post-logic implemented within the companion C++ logic file (Section 4.2.2). Inside these functions, the developer operates with full C++ expressiveness, enabling array traversal.

5.1.3 ReEntrancy Guard

A hook function that calls a target function, directly or through another target function, enters a hook function again on the same thread, and can loop until the stack overflows (Section 2.3.1). Detours does nothing against these re-entrant calls, and Frida-Gum stops only the direct form, and only in standard replacement mode (Section 3.3.1). This section explains why HookDesigner adds its own guard to the hook functions of both engines, and how the guard works.

5.1.3.1 A guard for Both Engines

Suppose that the pre-logic `CheckFileAllowed` reads the deny list from a configuration file the first time it runs. The pre-logic opens this file, and the open is itself a call to `NtCreateFile` (Section B.1). Without a guard in the hook functions, the two engines behave differently. With Detours, this call enters the hook function of `NtCreateFile` again, and the loop presented in Section 2.3.1 begins:

HookedNtCreateFile → CheckFileAllowed → NtCreateFile → HookedNtCreateFile → ...

With Frida-Gum, the result depends on the function that the program called first:

- **Direct form.** If the program opened its file through `NtCreateFile`, the open of the configuration file reaches the same function from its own pre-logic. The newest entry of the dispatcher belongs to `NtCreateFile`, so the dispatcher runs the original function directly, and the loop never starts.
- **Indirect form.** If the program opened its file through `NtOpenFile`, which has the same pre-logic, the newest entry belongs to `NtOpenFile`. The dispatcher therefore runs the hook function of `NtCreateFile` in full: the hook function treats a file open made by the instrumentation as if the program had made it, and its pre-logic starts loading the deny list a second time, while the first load is still in progress. If the logic of the second hook function also called the first function, the two hook functions would call each other until the stack overflows.

In addition, for functions that never return, HookDesigner installs the Frida-Gum hook with `gum_interceptor_replace_fast` (Section 5.1.5.2); this mode does not use the dispatcher (Section 3.2.1), so no check runs at all.

For these reasons, HookDesigner generates the same guard in the hook functions of both engines: a flag for each thread, shared by all the hooks of the DLL. While any hook function runs, the flag is set, and a call to any target function on the same thread goes

straight to the original function. The flag is set at every entry of a hook function, with both engines, because a hook function cannot know in advance whether its logic will call a target function. What changes between the engines is the mechanism that stops each re-entrant call:

- With Detours, and with Frida-Gum's fast replacement, every re-entrant call reaches the hook function, and the guard of HookDesigner stops both forms.
- With Frida-Gum's standard replacement, a call in the direct form never reaches the hook function, because the dispatcher sends it to the original function first: Frida-Gum's built-in mechanism stops it, and the guard of HookDesigner never sees it. Only the calls in the indirect form reach the hook function, where the guard of HookDesigner stops them.

5.1.3.2 The ReEntrancyHookGuard Definition

HookDesigner stores the flag of the guard in a TLS slot. During the initialization of the payload (`InitializeHooks`), the framework allocates a process-wide TLS slot index:

```
static DWORD g_TlsIndex = TLS_OUT_OF_INDEXES;
// Inside InitializeHooks():
g_TlsIndex = TlsAlloc();
```

Using this slot, HookDesigner implements a resource acquisition is initialization (RAII) helper structure named `ReEntrancyHookGuard` in the helper file `_user_logic_helpers.cpp`, shown in Code 5.3:

Code 5.3: `ReEntrancyHookGuard` RAII implementation in `_user_logic_helpers.cpp`.

```
struct ReEntrancyHookGuard {
    bool active;
    ReEntrancyHookGuard() : active(false) { Acquire(); }
    ~ReEntrancyHookGuard() { Release(); }

    void Acquire() {
        if (!active) {
            TlsSetValue(g_TlsIndex, (LPVOID)1);
            active = true;
        }
    }

    void Release() {
        if (active) {
            TlsSetValue(g_TlsIndex, (LPVOID)0);
            active = false;
```

```
        }
    }
};
```

5.1.3.3 The New Hook Function Control Flow

At the entry of each generated hook function (`Hooked<F>`), before executing any code, the function inspects the thread's reEntrancy state:

```
if (TlsGetValue(g_TlsIndex) != 0) {
    return True<FunctionName>(param1, param2, ...);
}
```

If the TLS value is non-zero, the thread is already executing inside an active hook function. The function immediately redirects execution to the trampoline pointer (`True<FunctionName>`), without running the pre-logic, the post-logic or the telemetry code.

If the thread is not in a reentrant call (`TlsGetValue == 0`), the function instantiates a local `ReEntrancyHookGuard` on the stack, which sets the TLS flag to `1` (`Acquire()`). Crucially, right before invoking the trampoline to execute the original target function, the function calls:

```
hook_reEntrancy_guard.Release();
```

This pre-trampoline release is a deliberate choice:

- While the pre-trampoline and post-trampoline code run, the guard is *active* (`1`), blocking any re-entrant hook loops triggered by the pre-logic, the post-logic or the telemetry code.
- While the original target function executes, the guard is *released* (`0`). If the target function legitimately calls *another* target function as part of its normal operation, that nested call is permitted to enter its own hook function and be monitored.

Under Frida-Gum's standard replacement, this release has no effect on a call to the *same* function made by the original function: the dispatcher skips it before the hook function runs (Section 3.3.1), so a recursive taget function can be correctly monitored with Detours but not with Frida-Gum.

After the trampoline returns, the function re-acquires the guard (`hook_reEntrancy_guard.Acquire()`) to protect the rest of the post-trampoline code: the post-logic and the telemetry code.

Finally, when the hook function completes and returns, `hook_reEntrancy_guard` goes out of scope on the stack. Its RAII destructor automatically invokes `Release()`, restoring the TLS slot to `0`. By relying on RAII, even if an unexpected exception occurs at any point of the pre-trampoline or post-trampoline code, the stack unwinding mechanism automatically disarms the guard, ensuring the thread is always returned to a fair, non-reentrant state. Figure 5.1 in Section 5.1.5 shows these steps within the complete control flow of a generated hook function.

5.1.4 System Error State Preservation

As discussed in Section 2.2, the hook function must leave the machine state of the thread as the target function would leave it. A thread, however, also carries a *software* error state that the hook function must preserve: the Win32 *last-error code*. Windows keeps this value per thread, so all modules of the process read and write the same value [Mic26h]. A caller typically reads it with `GetLastError()` right after a failed call, to learn the reason of the failure.

Inside a hook function, a lot of code runs between the entry and the return to the caller, and much of it changes the last-error code as a side effect. Without protection, the caller would read the error left by the hook function instead of the one produced by the original function, and could take a different execution path: the hook function would change the behavior of the program it observes.

HookDesigner prevents this with `SystemErrorGuard`, a RAII helper defined in `_user_logic_helpers.cpp` and shown in Code 5.4.

Code 5.4: `SystemErrorGuard` in `_user_logic_helpers.cpp` (simplified).

```
struct SystemErrorGuard {
    DWORD win32_error;   // errno is saved and restored in the same way

    SystemErrorGuard() : win32_error(GetLastError()) {}
    ~SystemErrorGuard() { SetLastError(win32_error); }

    void Capture() { win32_error = GetLastError(); }
    void Restore() const { SetLastError(win32_error); }
};
```

Each generated hook function uses the guard at four points:

1. **Entry:** the guard is the first object constructed in the hook function, so it saves the caller's last-error code before any other code runs. In particular, it is constructed *before* the re-entrancy check, whose `TlsGetValue` call would otherwise clear the value.

2. **Before the original function:** `Restore()` is called right before every call to `True<F>`, both on the main path and on the re-entrant bypass path. The original function therefore starts from the same error state it would see without the hook.

3. **After the original function:** `Capture()` is called once `True<F>` returns, and saves the error state produced by the original function. When execution time is measured, `Capture()` follows the end timestamp, just as `Restore()` precedes the start timestamp, so the timed window contains only the original function. `QueryPerformanceCounter` modifies neither the last-error code nor `errno`, so the saved state is the same.

4. **Return:** the destructor writes the saved value back on every return path.

As a consequence, the post-logic, the output capture and the telemetry code can change the last-error code freely: the caller always observes the value produced by the original function. The call-denial branch (Section 4.2.1.2.1) relies on the same mechanism: when `last_error` is declared, it writes that value into the guard, and the destructor returns it to the caller.

5.1.5 The Complete Hook Function Structure

All the presented safety features converge in the body of the generated hook function (`_hook_body.cpp`). This body varies on two levels:

- **At code generation:** the transpiler emits either the standard body or the body for non-returning functions (Section 5.1.5.2), and it emits the call-denial branch (Section 5.1.5.3) only if the hook declares `on_deny`. A given hook function therefore contains only one of these variants.

- **At runtime:** the generated code takes only two decisions: the re-entrancy check (Section 5.1.3) and, if present, the deny-call verdict.

Figure 5.1 shows all the resulting paths in a single flowchart. The listings that follow show the generated code: in full for the standard body, and only the part that differs for each variant.

5.1.5.1 Standard Hook Function Body

To illustrate how these defenses integrate in practice for standard returning targets, Code 5.5 presents a complete generated C++ Hook function; it follows the allowed-call

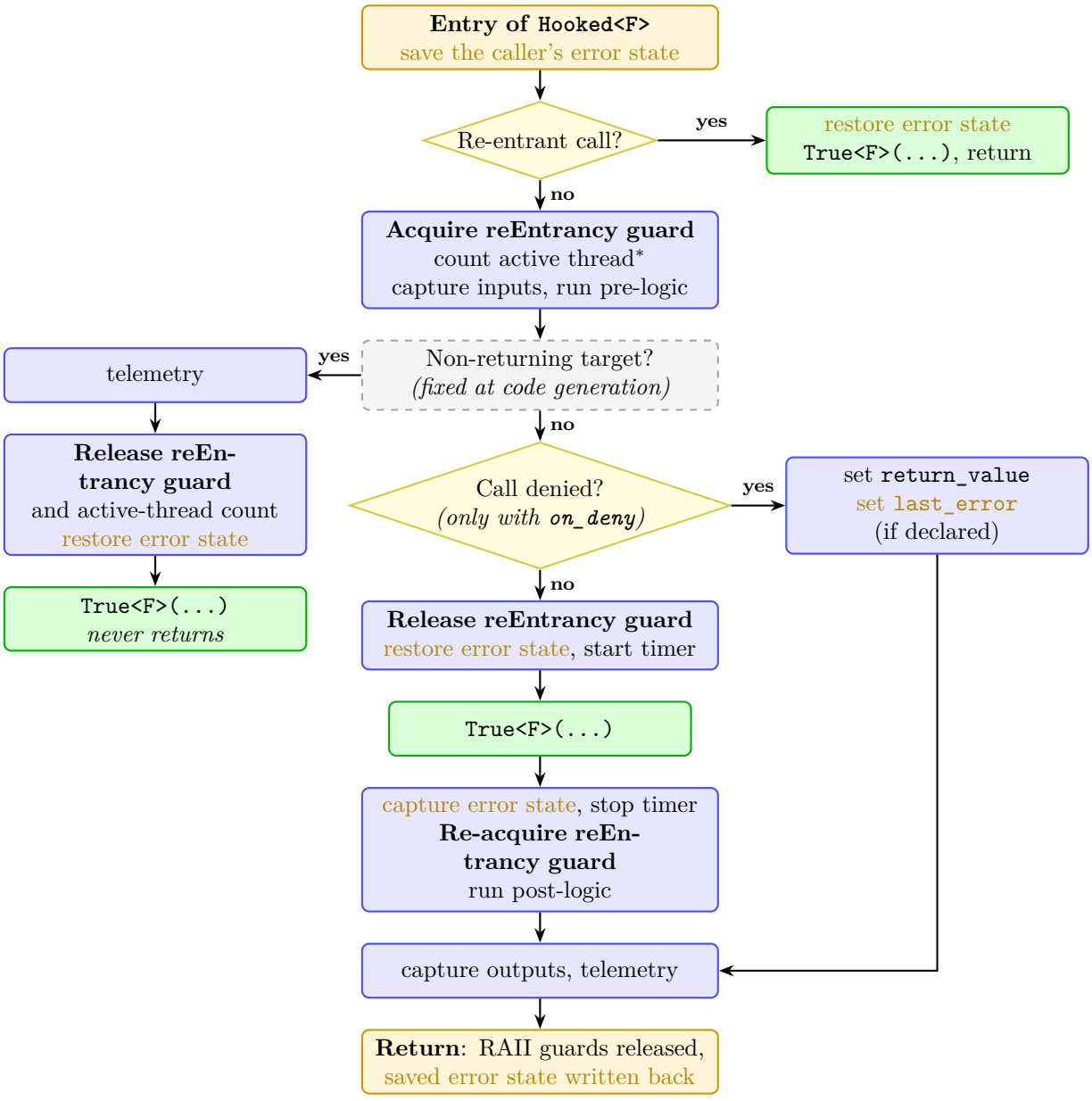


Figure 5.1: Control flow of a generated hook function. Diamonds are runtime decisions; the dashed box is a choice fixed at code generation, so a hook function contains only one of the two bodies. Green nodes call the original function, and amber text marks the steps of the error guard (Section 5.1.4). Exception boundaries are omitted.

path in the middle of Figure 5.1. It is the hook function for `NtOpenFile` of the read-only extension (Code 4.2), exactly as the transpiler generates it for Detours. This hook declares a pre-logic but no post-logic; when a post-logic is declared, its call is placed after `Acquire()`, in its own exception boundary (Code 5.2). Right after the re-entrancy guard, the function also declares `hook_active_threads`, a second guard that counts the threads running inside the hook functions, so that the DLL is not unloaded while one of them is still running (Section 2.3.1); Section 5.2.3.1 describes this guard and the cases where Frida-Gum does not need it.

Code 5.5: Complete generated hook function demonstrating integrated safety mechanisms and decoupled exception boundaries (`NtOpenFile`, read-only extension of the running example).

```
NTSTATUS NTAPI HookedNtOpenFile(PHANDLE FileHandle, ACCESS_MASK DesiredAccess,
        POBJECT_ATTRIBUTES ObjectAttributes, PIO_STATUS_BLOCK IoStatusBlock,
        ULONG ShareAccess, ULONG OpenOptions) {
    SystemErrorGuard system_error_guard;

    if (TlsGetValue(g_TlsIndex) != 0) {
        system_error_guard.Restore();
        return TrueNtOpenFile(FileHandle,DesiredAccess,ObjectAttributes,IoStatusBlock,
    ShareAccess,OpenOptions);
    }

    ReEntrancyHookGuard hook_reEntrancy_guard;
    HookActiveThreads hook_active_threads;

    bool telemetry_inputs_ready = false;
    const char* capturing_input_param = nullptr;
    ACCESS_MASK input_DesiredAccess = {};
    POBJECT_ATTRIBUTES input_ObjectAttributes = {};

  try {
        capturing_input_param = "DesiredAccess";
        input_DesiredAccess = DesiredAccess;
        capturing_input_param = "ObjectAttributes";
        input_ObjectAttributes = ObjectAttributes;
        telemetry_inputs_ready = true;
    } catch (const std::exception& ex) {
        // Handler messages use a stack buffer: a heap allocation could throw std::
    bad_alloc out of the catch block
        char msg[256];
        *fmt::format_to_n(msg, sizeof(msg) - 1, "[HOOK CAPTURE] Exception capturing
    parameter '{}' in NtOpenFile: {}\n",
                          capturing_input_param ? capturing_input_param : "unknown", ex
    .what()).out = '\0';
        OutputDebugStringA(msg);
    } catch (...) {
```

```
        char msg[256];
        *fmt::format_to_n(msg, sizeof(msg) - 1, "[HOOK CAPTURE] Hardware/SEH fault
    capturing parameter '{}' in NtOpenFile\n",
                          capturing_input_param ? capturing_input_param : "unknown").
    out = '\0';
        OutputDebugStringA(msg);
    }

    // Pre-logic execution
    try {
        ForceReadOnly(DesiredAccess,ObjectAttributes);
    } catch (const std::exception& ex) {
        char msg[256];
        *fmt::format_to_n(msg, sizeof(msg) - 1, "[HOOK EXCEPTION] Exception in pre-hook
     NtOpenFile: {}\n", ex.what()).out = '\0';
        OutputDebugStringA(msg);
    } catch (...) {
        OutputDebugStringA("[HOOK EXCEPTION] Hardware/SEH fault in pre-hook NtOpenFile\
    n");
    }

    hook_reEntrancy_guard.Release();
    system_error_guard.Restore();

    LARGE_INTEGER start_time_NtOpenFile;
    LARGE_INTEGER end_time_NtOpenFile;
    QueryPerformanceCounter(&start_time_NtOpenFile);
    NTSTATUS __hook_result = TrueNtOpenFile(FileHandle,DesiredAccess,ObjectAttributes,
    IoStatusBlock,ShareAccess,OpenOptions);
    QueryPerformanceCounter(&end_time_NtOpenFile);
    double elapsed_ms_NtOpenFile = (g_PerfFrequency.QuadPart > 0) ? ((double)(
    end_time_NtOpenFile.QuadPart - start_time_NtOpenFile.QuadPart) * 1000.0 / (double)
    g_PerfFrequency.QuadPart) : 0.0;

    system_error_guard.Capture();
    hook_reEntrancy_guard.Acquire();

    bool telemetry_outputs_ready = false;
    const char* capturing_output_param = nullptr;
    ACCESS_MASK output_DesiredAccess = {};

  try {
        capturing_output_param = "DesiredAccess";
        output_DesiredAccess = DesiredAccess;
        telemetry_outputs_ready = true;
    } catch (const std::exception& ex) {
        char msg[256];
        *fmt::format_to_n(msg, sizeof(msg) - 1, "[HOOK CAPTURE] Exception capturing
    output parameter '{}' in NtOpenFile: {}\n",
```

```
                        capturing_output_param ? capturing_output_param : "unknown",
    ex.what()).out = '\0';
        OutputDebugStringA(msg);
    } catch (...) {
        char msg[256];
        *fmt::format_to_n(msg, sizeof(msg) - 1, "[HOOK CAPTURE] Hardware/SEH fault
    capturing output parameter '{}' in NtOpenFile\n",
                        capturing_output_param ? capturing_output_param : "unknown").
    out = '\0';
        OutputDebugStringA(msg);
    }

    if (telemetry_inputs_ready && telemetry_outputs_ready) {
        try {
        // --- Logging ---
        if (input_DesiredAccess != output_DesiredAccess) {
            fmt::memory_buffer ss_readonly;
            fmt::format_to(std::back_inserter(ss_readonly), FMT_COMPILE("read-only open
    : name={} access=0x{:x} to 0x{:x} status=0x{:x} lat={:.3f}ms"), CopyObjectName(
    input_ObjectAttributes), input_DesiredAccess, output_DesiredAccess, (ULONG)
    __hook_result, elapsed_ms_NtOpenFile);
            spdlog::info(spdlog::string_view_t(ss_readonly.data(), ss_readonly.size()))
    ;
        }
        } catch (const std::exception& ex) {
            char msg[256];
            *fmt::format_to_n(msg, sizeof(msg) - 1, "[TELEMETRY EXCEPTION] Exception in
     telemetry NtOpenFile: {}\n", ex.what()).out = '\0';
            OutputDebugStringA(msg);
        } catch (...) {
            OutputDebugStringA("[TELEMETRY EXCEPTION] Hardware/SEH fault in telemetry
    NtOpenFile\n");
        }
    }

    return __hook_result;
}
```

5.1.5.2 Adaptation for Non-Returning Targets

The standard hook function body operates on the premise that the intercepted target function eventually returns after execution. However, several system APIs are *non-returning functions*: they terminate the thread or process, or transfer control directly to an external exception dispatcher, never returning control to the caller.

If a non-returning function is instrumented using the standard returning function structure,

the instrumentation payload encounters the following architectural failures:

1. **Post-Trampoline Telemetry dead-code:** In `_hook_body.cpp`, telemetry evaluation is part of the post-trampoline code, so it is never reached with this type of function.

2. **Bypassed C++ Stack Unwinding:** When a process or thread terminates, the operating system immediately tears down the user-mode execution context. Standard C++ stack unwinding is skipped, meaning that local RAII destructors on the stack are *never executed*. Consequently, the destructor of `HookActiveThreads` (Code 5.9) never runs, causing the global `activeThreads` counter to remain permanently greater than zero. During payload ejection, this causes the injector to believe threads are permanently trapped in the hook function, triggering an unnecessary DLL leak.

HookDesigner resolves these issues through automated adaptations applied to the standard hook function body:

1. **Pre-Trampoline Telemetry:** In `_hook_body.cpp`, telemetry evaluation is moved into the pre-trampoline code, so it runs *before* the trampoline call (Code 5.6).

2. **Targeted Thread Tracking and Explicit Guard Destruction:** Since process termination skips local destructors, the function explicitly calls the `hook_active_threads.Release()` method of the guard (Section 5.2.3.1) immediately before invoking the terminal trampoline (Code 5.6). This decrements the global `activeThreads` counter in advance, ensuring DLL ejection synchronization safety (Section 5.2.3). Under Frida-Gum, the dispatcher's own tracking (Section 3.3.1) cannot follow a function that never returns, so the framework installs these hooks with `gum_interceptor_replace_fast` (Section 3.2.1) and uses the `HookActiveThreads` guard for them; Section 5.2.3.1 explains why, and when each engine uses the guard.

These adaptations form the left branch of Figure 5.1. Before the terminal call, the function also restores the caller's error state (Section 5.1.4); no `Capture()` follows, since the original function never returns and the guard's destructor never runs. Code 5.6 shows the generated code for the `ExitProcess` hook of Code 4.4, whose log rule records the exit code of the process, limited to the part that differs from the standard body. The log rule runs before the process ends, but the asynchronous logger (Section 4.4.4) writes the message later, on its own thread; if `ExitProcess` ends the process first, the message can be lost. For the same reason, the test program of the running example waits for a short time before it exits, so that the logger can write its last messages.

Code 5.6: Generated hook function structure for non-returning functions (`ExitProcess` of the extension in Code 4.4; simplified, only the part that differs from Code 5.5).

```
void WINAPI HookedExitProcess(UINT uExitCode) {
    // Entry, guards and input capture: as in the standard body (this hook has no pre-
    logic)

    // Guarded Inverted Telemetry (BEFORE Trampoline)
    if (telemetry_inputs_ready) {
        try {
            fmt::memory_buffer ss_exit;
            fmt::format_to(std::back_inserter(ss_exit), FMT_COMPILE("process exit: pid
    ={} code={}"),
                           GetCurrentProcessId(), input_uExitCode);
            spdlog::info(spdlog::string_view_t(ss_exit.data(), ss_exit.size()));
        } catch (...) {
            OutputDebugStringA("[TELEMETRY EXCEPTION] Hardware/SEH fault in telemetry
    ExitProcess\n");
        }
    }

    // EXPLICIT GUARD RELEASE: Decrements activeThreads before trampoline
    hook_reEntrancy_guard.Release();
    hook_active_threads.Release();
    system_error_guard.Restore();   // no Capture(): the destructor never runs

    // Trampoline Invocation (Never returns)
    TrueExitProcess(uExitCode);
}
```

5.1.5.3 Adaptation for Call Denial

When a hook declares `on_deny` (Section 4.2.1.2.1), the transpiler adds a runtime branch to the standard body, shown on the right of Figure 5.1. Code 5.7 shows the generated code for the native `NtOpenFile` function of Code 4.4, which declares both `on_deny` and a post-logic, limited to the part that differs from the standard body.

Code 5.7: Generated call-denial branch for `NtOpenFile`, with the post-logic of the extension in Code 4.4 (simplified, only the part that differs from Code 5.5).

```
NTSTATUS NTAPI HookedNtOpenFile(PHANDLE FileHandle, ACCESS_MASK DesiredAccess,
        POBJECT_ATTRIBUTES ObjectAttributes, PIO_STATUS_BLOCK IoStatusBlock,
        ULONG ShareAccess, ULONG OpenOptions)
{
    // Entry, guards and input capture: as in the standard body

    bool __hook_denied = true;              // fail-closed default
```

```
    try {
        __hook_denied = !CheckFileAllowed(ObjectAttributes);
    } catch (...) {                          // C++ exceptions and SEH faults
        OutputDebugStringA("[HOOK EXCEPTION] Hardware/SEH fault in pre-hook NtOpenFile
    (call denied)\n");
    }

    NTSTATUS __hook_result{};                // read by telemetry on both paths
    if (__hook_denied) {
        // TrueNtOpenFile is not called and the post-logic is skipped
        __hook_result = STATUS_ACCESS_DENIED;   // on_deny.return_value
    } else {
        hook_reEntrancy_guard.Release();
        system_error_guard.Restore();
        __hook_result = TrueNtOpenFile(FileHandle, DesiredAccess, ObjectAttributes,
                                       IoStatusBlock, ShareAccess, OpenOptions);
        system_error_guard.Capture();
        hook_reEntrancy_guard.Acquire();

        // Post-logic execution: only for allowed calls
        try {
            CheckOpenedFile(FileHandle, __hook_result);
        } catch (...) {
            OutputDebugStringA("[HOOK EXCEPTION] Hardware/SEH fault in post-hook
    NtOpenFile\n");
        }
    }

    // Output capture, telemetry and return: as in the standard body
}
```

The branch relies on the following design choices:

- **Fail-closed default:** `__hook_denied` starts as `true` and is set only by a pre-logic that returns normally, so a pre-logic that throws or faults leaves the call denied.
- **Denied path:** it only sets `return_value` and, when `last_error` is declared, writes it into the error guard (`system_error_guard.win32_error`), whose destructor returns it to the caller (Section 5.1.4). It skips the trampoline call and the post-logic, and then joins the rest of the post-trampoline code: the output capture and the telemetry code.
- **ReEntrancy guard:** on the denied path the guard is never released, because there is no original call to release it around, so the whole hook function runs protected from re-entrancy.

- **Post-logic:** its call is generated inside the allowed branch, since it processes the result of an original call.

5.2 DLL Lifecycle and Deployment

After the offline compilation stage (Stage 4, Section 4.1.2) completes, the instrumentation artifact exists as a self-contained Windows DLL. To monitor and manipulate the host application, this payload must be moved to the online context: it must be mapped into the target process's virtual address space and activate its hooks (Stage 5), execute safely alongside the application (Stage 6), and cleanly detach when monitoring terminates (Stage 7).

This section examines the deployment and runtime lifecycle of HookDesigner payloads, managed by the standalone injector utility (`injector.exe`, implemented in `injector.cpp`).

5.2.1 DLL Injection

In all the modern operating systems, address space isolation (Section 2.1.1) prevents one process from directly manipulating the execution state of another. To introduce the compiled payload into a running target, HookDesigner implements dynamic library injection via the standard Windows debugging and threading APIs.

5.2.1.1 The DLL Injection Process

To attach to a running target process identified by its process identifier (PID), the standalone injector utility (`injector.exe`) leverages the classical remote thread injection technique introduced in Section 2.1.2. After obtaining a process handle via `OpenProcess`, the injector writes the absolute path of the hook DLL into the target's virtual address space using `VirtualAllocEx` and `WriteProcessMemory`.

To execute `LoadLibraryA` remotely, the injector exploits an architectural property of the Windows subsystem: core system libraries (specifically `Kernel32.dll`) are mapped to identical virtual base addresses across all processes on a given system boot. Consequently, the address of `LoadLibraryA` in the injector's address space is identical in the target process. The injector spawns a remote thread pointing to `LoadLibraryA` with the path buffer as its parameter via `CreateRemoteThread`, and waits for thread completion. Once the remote thread finishes and the memory buffer is released, the DLL is mapped into the target process.

5.2.1.2 Relative Virtual Address Offset Invariance

Once `LoadLibraryA` returns, the hook DLL is loaded in the target process, but its hooks are still disabled. The injector must now remotely invoke the module's exported initialization function, `InitializeHooks`. However, under ASLR, the operating system loader maps the hook DLL to an arbitrary base address in the target process that differs from where it might be loaded in the injector.

To resolve the address of exported remote functions across ASLR boundaries, `injector.cpp` exploits the property of *RVA offset invariance* within PE binaries via the `CallRemoteExport()` helper (Code 5.8):

1. **Local DLL Loading:** The injector loads the hook DLL locally with `LoadLibraryExA` specifying `DONT_RESOLVE_DLL_REFERENCES`. This flag instructs the Windows loader to map the binary and its PE headers into memory without invoking `DllMain` and without resolving external dependencies, avoiding useless side effects.

2. **RVA Offset Calculation:** The injector queries the exported function's local virtual address and subtracts the local base address of the module, which gives the RVA of the function. The RVA depends only on the module file and not on the base address (Section 2.1.3), so it is the same in every address space.

3. **Remote Base Address Discovery:** The injector enumerates the target process's loaded modules using the Toolhelp32 snapshot API (`CreateToolhelp32Snapshot`, locating the target's remote base address `hRemoteModule` matching the DLL filename.

4. **Remote Function Execution:** The remote function address is computed by adding the offset relative to the remote base. After freeing the temporary local module mapping with `FreeLibrary`, the injector launches a remote thread targeting $pRemoteFunc$ and waits for it to complete.

Code 5.8: Cross-process export resolution via RVA offset invariance in `CallRemoteExport`.

```
// 1. Map DLL locally without side-effects
HMODULE hLocalMod = LoadLibraryExA(localDllPath.c_str(), NULL,
    DONT_RESOLVE_DLL_REFERENCES);
FARPROC pLocalFunc = GetProcAddress(hLocalMod, exportName.c_str());

// 2. Compute invariant RVA offset and resolve target address under ASLR
DWORD_PTR offset = (DWORD_PTR)pLocalFunc - (DWORD_PTR)hLocalMod;
LPVOID pRemoteFunc = (LPVOID)((DWORD_PTR)hRemoteModule + offset);
FreeLibrary(hLocalMod);

// 3. Trigger remote execution in target process
```

```
HANDLE hThread = CreateRemoteThread(hProcess, NULL, 0,
    (LPTHREAD_START_ROUTINE)pRemoteFunc, NULL, 0, NULL);
WaitForSingleObject(hThread, INFINITE);
```

5.2.1.3 The Loader Lock Issue

A common anti-pattern in DBI is to perform complex tasks directly inside the DLL entry point, `DllMain`, as introduced in Section 2.1.2. While convenient for trivial payloads, executing complex instrumentation routines inside `DllMain` introduces severe concurrency issues on Windows due to the OS *Loader Lock*.

Whenever the Windows operating system maps or unmaps a DLL, or when a process creates or terminates a thread, the OS loader must acquire an internal, process-wide synchronization primitive known as the Loader Lock. While this lock is held, the loader sequentially executes the `DllMain` routines of all loaded modules. Consequently, binary engineering guidelines explicitly prohibit thread synchronization, waiting on external events, and socket initialization within `DllMain` [Mic26f].

HookDesigner's telemetry (whose initialization would be in `DLL_PROCESS_ATTACH`:) (Section 4.4) relies on two background mechanisms:

- **Asynchronous Logging** (`spdlog`): Spawns a background consumer thread to drain buffers and flush log messages.
- **Metrics Server** (`prometheus-cpp`): Spawns CivetWeb HTTP listener threads to serve metric scrapers.

A circular deadlock occurs whenever the calling thread inside `DllMain` blocks waiting upon those background threads. In HookDesigner's architecture, executing lifecycle logic under Loader Lock exposes two possible deadlock modes:

1. **The Initialization Rollback Deadlock:** During normal startup, telemetry initialization requires asynchronous thread creation without an immediate startup join. However, if any subsequent initialization step fails, the exception handler immediately invokes the rollback sequence (Code 5.1) to prevent partial instrumentation. This cleanup triggers a blocking call (`WaitForSingleObject`) on the newly spawned worker threads. Because those workers are permanently trapped waiting for the Loader Lock held by `DllMain` and the thread in `DllMain` waits on those worker threads, the result is a circular deadlock.
2. **The Deinitialization Deadlock:** Similarly, when an instrumentation payload is dynamically unloaded, the operating system acquires the Loader Lock before invoking

`DllMain(DLL_PROCESS_DETACH)`. Executing `DeinitializeHooks` inside this handler requires tearing down and joining all background telemetry threads before the module is unmapped from memory. As in the rollback scenario, joining these exiting threads while holding the Loader Lock triggers the same circular deadlock seen in the rollback mechanism.

5.2.1.4 The Two-Phase Protocol

To eliminate this issue, HookDesigner separates DLL mapping from hooks activation using a *Two-Phase Protocol*:

- **Phase 1 (Passive Mapping):** The injector invokes `CreateRemoteThread` targeting `LoadLibraryA`. Inside the generated payload (`_dllmain.cpp`), `DllMain` executes under Loader Lock but performs only minimal operations:

```
BOOL APIENTRY DllMain(HMODULE hModule, DWORD ul_reason_for_call, LPVOID lpReserved
    ) {
    if (ul_reason_for_call == DLL_PROCESS_ATTACH) {
        DisableThreadLibraryCalls(hModule);
    }
    return TRUE;
}
```

  Calling `DisableThreadLibraryCalls` informs the Windows Loader that the DLL does not require notifications when application worker threads are created or terminated (`DLL_THREAD_ATTACH` or `DLL_THREAD_DETACH`). This eliminates thread-notification overhead during application runtime. Crucially, `DllMain` creates no threads, makes no network calls. It returns `TRUE` immediately, allowing the OS loader to release Loader Lock cleanly.

- **Phase 2 (Active Initialization):** Once the Phase 1 remote thread exits, the injector computes the remote address of `InitializeHooks` via RVA offset invariance (Code 5.8) and launches a second remote thread. Because this second thread runs entirely outside `DllMain` and so outside Loader Lock, it can safely initialize the `spdlog` background thread pool, bind the Prometheus HTTP metrics server, and install transactional inline hooks, bypassing the Loader Lock deadlock risk.

Furthermore, the injector inspects the remote exit code returned by `InitializeHooks`. In injection mode, if initialization fails (returning a non-zero code), the injector automatically executes a rollback by remotely invoking `FreeLibrary` to unload the DLL, leaving the host process running uncorrupted.

5.2.1.5 Engine-Specific Initialization Mechanics

While Section 4.3.5 introduced the simplified template structure used to emit the `InitializeHooks` skeleton (Codes 4.8 and 4.9), HookDesigner generates a more complex initialization process during Phase 2. This process incorporates all the features we have seen so far throughout Sections 4.4 and 5.1, adapting the attachment strategy to the underlying engine as summarized in Table 5.2.

Initialization Step	**Microsoft Detours**	**Frida-Gum**
ReEntrancy Safety	Allocates TLS index (`TlsAlloc`)	Allocates TLS index (`TlsAlloc`)
Latency Timer	`QueryPerformanceFrequency()`, only if `has_execution_time`	Identical
Telemetry Engine	Initialized via `_telemetry_init.cpp` (`spdlog` thread pool, Prometheus Exposer)	Identical initialization via shared partial `_telemetry_init.cpp`
Runtime Core	None (Direct binary patching)	Calls `gum_init_embedded()` and `gum_interceptor_obtain()`
Address Resolution	Evaluates `GetProcAddress` or base + RVA offset	Evaluates `GetProcAddress` or base + RVA offset
Thread Enumeration	Calls `EnlistAllThreads()` via Toolhelp32 snapshot	None
Transactional Scope	`DetourTransactionBegin()`	`gum_interceptor_begin_transaction()`
Thread Enlistment	Calls `DetourUpdateThread()` for all worker threads	None
Hook Attachment	`DetourAttach( &(PVOID&)True<F>, Hooked<F>)`	Calls `gum_interceptor_replace()` (standard) or `replace_fast()` (non-returning)
Transaction Commit	`DetourTransactionCommit()` (suspends threads, relocates IP, applies patch)	`gum_interceptor_end_transaction()` (writes the patch while threads run, no IP relocation)

Table 5.2: Comparison of Hooks initialization between Microsoft Detours and Frida-Gum.

5.2.2 Injector Launch Mode

Dynamic injection into an active process (`injector.exe inject <PID> <DLL>`) is appropriate when instrumenting long-running processes. However, this approach introduces a problem: any operations performed by the target before the injector attaches are missed. To solve this problem, `injector.cpp` provides a *launch mode* designed for launching the process already containing the instrumentation payload; it can be invoked as:

```
injector.exe launch <Executable_Path> <DLL_Path> [optional_target_args...]
```

The running example needs this mode: its test program opens all its files right after it starts and then exits, so a hook injected into the running process would miss the opens. It is therefore started with its hook DLL already loaded, for example:

```
injector.exe launch filedeny_target.exe hook_detours.dll <target arguments>
```

5.2.2.1 Suspended-Process Mechanism

The execution sequence for launch mode proceeds through four steps:

1. **Process Instantiation:** The utility calls `CreateProcessA` specifying the process creation flag `CREATE_SUSPENDED`. When launching a process with this flag, the primary thread is created and suspended before reaching the application entry point [Yos+17].

   Consequently, when `InitializeHooks` is invoked via a remote thread prior to resuming the primary thread, `GetModuleHandleA` can only resolve modules present in the static import address table (IAT) or those imported by the injected DLL itself. Any library that the target application loads dynamically at runtime via `LoadLibrary` or through delay-loading (`/DELAYLOAD`) will not yet be present in the process environment block (PEB) loader database; resolving symbols within such modules yields `NULL`, causing the hook initialization routine to log a warning and skip attachment for those targets.

2. **Hook Activation:** While the main thread remains frozen, the injector applies the Two-Phase Protocol (Section 5.2.1.4):

   - It injects the instrumentation DLL via a remote thread pointing to `LoadLibraryA`.
   - It computes the remote address of `InitializeHooks` and activates all hooks via a second remote thread.

Upon completion of `InitializeHooks`, the injector validates the remote thread's exit code: if it is non-zero (indicating Hooks initialization failure), the injector aborts and terminates the suspended process via `TerminateProcess` to avoid executing an uninstrumented or partially configured target.

3. **Primary Thread Resumption:** When `InitializeHooks` returns `0` (success), the injector calls `ResumeThread`, decrementing the suspend count to zero. The primary thread awakens and begins executing its startup sequence. Because hooks are already active, any function intercepted by the policy that executes during application startup is captured from its very first call.

4. **Process Termination:** The injector observes process execution, queries its exit code via `GetExitCodeProcess`, and cleanly terminates.

Figure 5.2 compares the dynamic injection pipeline with the suspended launch pipeline, illustrating how both share the Two-Phase protocol while targeting different process states and lifecycle requirements.

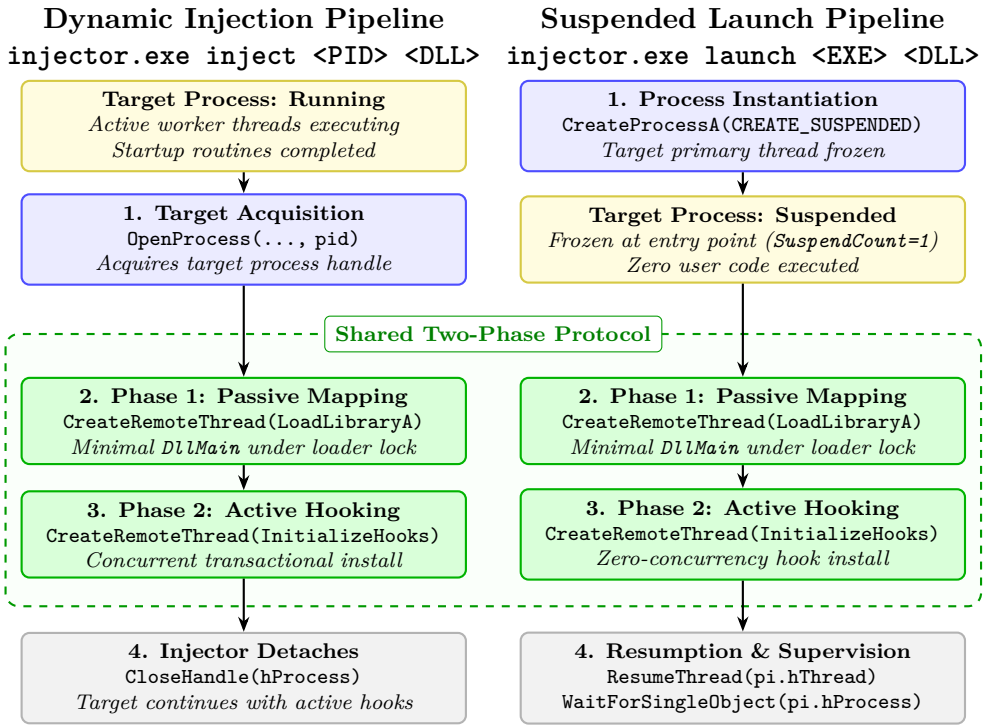


Figure 5.2: Architectural comparison between Dynamic Injection mode and Suspended Launch mode

5.2.3 DLL Ejection

When instrumentation is no longer required, the payload must detach cleanly so that the target process can continue running without interruption. As analyzed in Section 2.3.1, reversing prologue bytes is necessary but not sufficient: in a multi-threaded application, worker threads may already be executing within hook functions when detachment begins. If the host unmaps the DLL via `FreeLibrary` while a worker thread's instruction pointer points to unmapped memory, the process crashes immediately with an Access Violation.

To prevent this, HookDesigner uses a thread-tracking mechanism during hook detachment.

5.2.3.1 Hook Active Thread Tracking

HookDesigner counts the threads that are running inside its hook functions with a RAII helper, `HookActiveThreads`, defined in `_user_logic_helpers.cpp` and shown in Code 5.9. It is the guard of the manual tracking described in Section 3.3.1.

Code 5.9: HookDesigner implementation of active thread tracking in `_user_logic_helpers.cpp`.

```
static std::atomic<int> activeThreads{0};

// RAII helper to track active hook execution count
struct HookActiveThreads {
    bool active;
    HookActiveThreads() : active(true) {
        activeThreads++;
    }
    ~HookActiveThreads() {
        if (active) activeThreads--;
    }
    void Release() {
        if (active) {
            activeThreads--;
            active = false;
        }
    }
};
```

- **Atomic Counter:** The counter `activeThreads` is a global atomic variable (`std::atomic<int>`). Its constructor uses atomic increment (`activeThreads++`) on entry, and its destructor executes atomic decrement (`activeThreads-`) on exit.
- **Explicit Guard Release for Terminating APIs:** As detailed in Section 5.1.5.2, functions like `ExitProcess` (Code 5.6) never return, so stack unwinding and local

destructors are bypassed. For this reason the guard also has a `Release()` method: the hook function calls `hook_active_threads.Release()` just before it calls the original function, so the counter goes down before the thread or the process ends.

5.2.3.1.1 The Guard in Each Engine Not every generated hook function declares this guard. As for the re-entrancy guard (Section 5.1.3.1), the choice depends on what the engine already does (Section 3.3.1):

- **Detours.** Detours does not count threads, so every hook function declares the guard, right after the re-entrancy guard (Code 5.5). Unlike the re-entrancy guard, the counter is not released before the call to the original function: while the original function runs, its return address still points into the hook function, so the thread is still inside the DLL (Section 2.3.1).
- **Frida-Gum, returning functions.** The hook is installed with `gum_interceptor_replace`, and the dispatcher counts the thread from its entry until it returns, the original call included (Section 3.3.1). The hook function does not declare the guard, and the drain relies on `gum_interceptor_flush()`.
- **Frida-Gum, non-returning functions.** The dispatcher lowers its count only when the function returns, which never happens here: after a call to `ExitThread`, for example, the count stays above zero, and `gum_interceptor_flush()` never reports that the threads have left. For this reason HookDesigner installs these hooks with `gum_interceptor_replace_fast`, which does not use the dispatcher (Section 3.2.1), and the hook function declares the guard, as under Detours.

With both engines the guard is declared after the re-entrancy check, so a re-entrant call that goes straight to the original function does not change the counter. Such a call happens only on a thread that is already running the logic of another hook function, and that thread is already counted, by the guard or by the dispatcher. Table 5.3 shows how the ejection combines the two counts.

5.2.3.2 The DLL Ejection Mechanism

When `injector.exe eject <PID> <DLL>` is executed, it remotely invokes the exported function `DeinitializeHooks` via `CallRemoteExport`, using the same method of the injection, (Code 5.8), for preventing Loader Lock deadlock (Section 5.2.1.3) possible also during ejection. The payload then performs the following five-step ejection sequence:

1. **Terminate Background Listener Threads:** The payload resets the Prometheus HTTP exposer. The underlying CivetWeb server runs internal threads blocked in

`select()` or `accept()` calls; destroying the exposer closes the listening socket and terminates these web worker threads cleanly. This step must occur first so that external HTTP scraping requests do not arrive while metric families and registries are being destroyed.

2. **Revert Hook Patches:** The payload detaches all inline hooks in one transaction of the engine (Section 3.5); how safe this step is depends on the engine. Under Detours, the payload enlists every thread of the process (`EnlistAllThreads` and `DetourUpdateThread`), so `DetourTransactionCommit` writes the original prologue bytes back while these threads are suspended, and moves a thread whose IP is inside the trampoline back to the matching instruction of the restored prologue. Under Frida-Gum, `gum_interceptor_revert` followed by `gum_interceptor_end_transaction` copies the original bytes back while the other threads keep running, so, as during attachment, a thread that runs the prologue at that exact moment may still crash. Once the transaction commits, original prologue bytes are restored in memory. New function calls execute the unhooked target code directly; no new threads can enter the hook functions.

3. **Bounded Drain Wait:** Although new calls cannot enter, worker threads that were already inside the hook functions before Step 2 may still be executing. To synchronize safely, the payload polls a drain condition for at most $T_{\text{drain}} = 2\,\text{s}$ (Algorithm 2). The wait has a limit because a thread blocked in the original function can stay inside the hook function for an unlimited time (Section 2.3.1); Section 5.2.3.3 describes what happens when the limit expires. What "drained" means depends on the engine and, for Frida-Gum, on the policy (Table 5.3); the template fixes this choice once, at code-generation time.

   - Under Detours, every hook function increments the `activeThreads` counter, so the drain condition is that the counter reaches zero.
   - Under Frida-Gum with only returning hooks, the interceptor tracks in-flight invocations itself, and `gum_interceptor_flush()` returning `TRUE` (no thread is still inside the interceptor's trampolines) is sufficient.
   - Under Frida-Gum with at least one non-returning hook, those hooks bypass the interceptor's tracking (Section 5.1.5.2), so both conditions are required.

   Whenever the counter takes part in the condition, a *confirmation interval* T_{confirm} follows before the module may be considered free. The `HookActiveThreads` destructor decrements the counter just before the hook function returns, so a thread can read zero while still executing the epilogue (register restoration and the final `ret`) inside the DLL's pages. Unmapping at that instant would fault a preempted thread, so the counter is re-read after T_{confirm}. This is a heuristic, not a proof: a thread preempted for longer than T_{confirm} at exactly that point would go undetected. A second call after

a failed drain skips the revert (the hooks are already detached) and re-runs only the drain phase, which is what makes the repeated ejection of Section 5.2.3.3 possible.

4. **Destruction of Telemetry Subsystems:** Only after Step 3 confirms that all worker threads have drained (`g_Ejected == true`) the payload destroy the logging and metrics registries. This ordering is critical: if loggers or registries were destroyed while a worker thread was still inside `_telemetry_body.cpp` formatting a log or incrementing a counter, the thread would encounter destroyed pointers, triggering a use-after-free crash.

5. **Runtime and Resource Deallocation:** The payload releases the TLS slot. Under Frida-Gum, it unreferences the interceptor and shuts down embedded Frida runtime engine. Finally, `DeinitializeHooks` returns `0` to signal a clean drain and safe detachment to the injector, or `1` if active threads remain trapped.

Generated for	**Drained Condition**	***needConfirm***
Detours	`activeThreads` $= 0$	yes
Frida-Gum, returning hooks only	`gum_interceptor_flush()`	no
Frida-Gum, ≥ 1 non-returning hook	`gum_interceptor_flush()` $\wedge$ `activeThreads` $= 0$	yes

Table 5.3: Drain condition for each engine and policy.

5.2.3.3 Ejection-Failure DLL Leak Strategy

The bounded wait of Step 3 can expire while a thread is still inside a hook function. This happens when the thread is blocked in the original function (Section 2.3.1), for example in a call to `recv` that waits for network data.

5.2.3.3.1 The Controlled Leak Mechanism To prevent this event, HookDesigner implements a defense coordinated between the injected DLL and the injector utility, illustrated in Figure 5.3:

1. **DLL – Data Structures Preservation (`DeinitializeHooks`):** When the 2-second spin-wait loop expires and threads remain trapped `DeinitializeHooks` recognizes that clean ejection is impossible. Instead of proceeding with standard cleanup, it:

Algorithm 2 `DeinitializeHooks`. `Drained` and `needConfirm` are fixed per engine and policy when the code is generated (Table 5.3); they are not evaluated per call.

Require: $T_{\text{drain}} = 2000\,\text{ms}$, $T_{\text{confirm}} = 1000\,\text{ms}$, $T_{\text{poll}} = 50\,\text{ms}$

if *ejected* **then**
    return TRUE
end if
if *hooksAttached* **then**
    stop the metrics exposer (if configured)
    revert all hooks in a single engine transaction
    if the transaction failed **then**
        return FALSE
    end if
    hooksAttached $\leftarrow$ FALSE
end if
$t_0 \leftarrow$ NOW()
while NOW() $- t_0 < T_{\text{drain}}$ **do**
    if DRAINED **then**
        if *needConfirm* **then**
            sleep T_{confirm}
        end if
        if not *needConfirm* **or** *activeThreads* $= 0$ **then**
            ejected $\leftarrow$ TRUE
            break
        end if
    end if
    sleep T_{poll}
end while
if *ejected* **then**
    destroy the metrics registry and shut down the logger (if configured)
    release the engine runtime (Frida-Gum only) and free the TLS slot
end if
return *ejected*

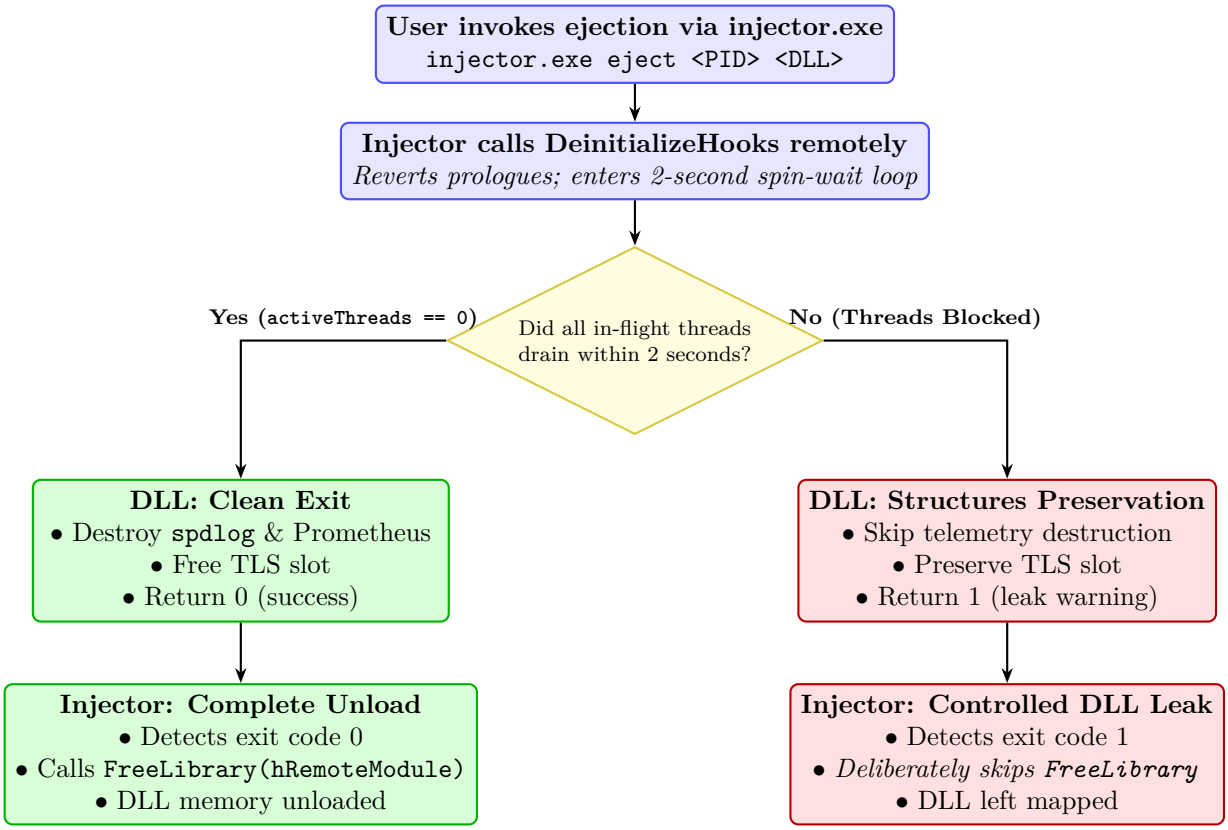


Figure 5.3: The Ejection-Failure DLL Leak Strategy

- **Preserves Telemetry Registries:** It deliberately skips telemetry data destruction. Leaving these structures intact ensures that when the blocked worker thread eventually wakes up and runs the post-trampoline code, the telemetry code will not crash due to null pointer dereferences.
- **Preserves TLS Index:** It deliberately skips releasing the allocated TLS slot: when a blocked worker thread unblocks and returns from the target API, it runs the post-trampoline code of the hook function, which calls `hook_reEntrancy_guard.Acquire()` and subsequently `Release()` inside its RAII destructor (Section 5.1.3). Both routines call `TlsSetValue` using `g_TlsIndex`, if `TlsFree` had been invoked, `g_TlsIndex` would become invalid. Preserving `g_TlsIndex` guarantees that the re-entrancy tracking of the post-trampoline code and the RAII destructors complete safely without process corruption.
- **Returns Status Code:** It returns `1` (thread leak warning code) to the injector, signaling that hooks have been detached but threads remain trapped in hook functions.

2. **Injector – Safety Fallback:** The injector inspects the remote thread's exit code, as shown in Code 5.10:

Code 5.10: Injector fallback logic in `injector.cpp`.

```
DWORD deinitExitCode = (DWORD)-1;

CallRemoteExport(hProcess, hRemoteModule, dllPath, "DeinitializeHooks", &
    deinitExitCode);

if (deinitExitCode == 1) { // trapped threads warning: DLL leak  fallback
    std::cout << "[WARNING] Hooks successfully detached, but some threads remain
    blocked inside them.\n";
    std::cout << "[WARNING] Skipping FreeLibrary (leaking DLL in target memory) to
     prevent target crash.\n";
} else if (deinitExitCode != 0) { // Detachment error
    std::cerr << "[ERROR] DeinitializeHooks reported error code: 0x"
              << std::hex << deinitExitCode << ". Skipping FreeLibrary.\n";
} else { // Exit code 0: Clean drain confirmed
    std::cout << "Ejecting DLL...\n";
    HANDLE hThread = CreateRemoteThread(hProcess, NULL, 0,
        (LPTHREAD_START_ROUTINE)pFreeLibrary, (LPVOID)hRemoteModule, 0, NULL);
    WaitForSingleObject(hThread, INFINITE);
    CloseHandle(hThread);
}
```

   If `deinitExitCode` is `1`, `injector.exe` outputs a warning and *deliberately skips* calling `FreeLibrary`. If `deinitExitCode` is any other non-zero value, an error occurred during detachment, and `FreeLibrary` is similarly bypassed to prevent unmapping unpredictable state. As in the trapped-thread case, the exception handlers of `DeinitializeHooks` skip Steps 4 and 5 of the ejection sequence, so the telemetry structures and the TLS slot are still valid (Section 5.1.1.2). Only when `deinitExitCode == 0` (clean thread drain confirmed) does the injector safely invoke `FreeLibrary` to unmap the DLL.

From a software engineering perspective, leaking memory is typically considered a defect. However, in DBI—particularly in critical industrial software—leaking a DLL under ejection failure is a safety-oriented design decision.

Considering the two alternatives when ejection encounters blocked threads:

1. **Strict Memory Reclamation (`FreeLibrary`):** The injector unmaps the module. When the blocked thread wakes up, its return address points to deallocated memory (`MEM_FREE`), triggering an instantaneous Access Violation, that makes crash the host process.

2. **Controlled Payload Leak (HookDesigner Fallback):** The module remains mapped in virtual memory. Because hooks were already reverted in Step 2, no *new* calls can ever enter the DLL. The memory overhead of the leaked payload is ususally negligible. When the blocked thread eventually unblocks, it unwinds its stack through the still-valid code pages, returns to the host application, the ejection mechanism can be repeated and the DLL can be freed.

In critical environments, often *process stability supersedes complete memory reclamation.* By combining the reversion of the prologues (Step 2) with the bounded drain wait (Step 3) and the controlled DLL leak fallback, HookDesigner never unmaps the DLL while a thread may still be running inside it. The only remaining risk is the one of Step 2 under Frida-Gum: a thread that runs a prologue at the exact moment its original bytes are written back.

5.3 Summary

This chapter described how the hook DLL generated by HookDesigner protects the target process while it runs inside it. The DLL is compiled with `/EHa`, so a `catch(...)` block also catches hardware faults. The lifecycle functions undo a failed initialization, so that the hooks are never installed only in part, and each hook function runs the parameter capture, the pre-logic, the post-logic and the telemetry in separate exception boundaries, so that a fault in one of them is contained and the caller still receives a result. At transpile time, a type analysis of the parameters chooses safe read helpers for the values that the telemetry captures. A per-thread ReEntrancy guard sends the calls that the hook code makes to target functions straight to the original function. The generated hook function combines these mechanisms in one body, with a variant for functions that never return and one for denied calls. The injector loads the DLL with a remote thread, and then calls `InitializeHooks`, in a second remote thread that runs outside the Loader Lock, so that the telemetry threads can start without a deadlock. In launch mode the target is created suspended, so the hooks are active before the program reaches its entry point. At ejection, `DeinitializeHooks` reverts the hooks, waits for the threads that are still inside the hook functions, and destroys the telemetry only after they have left. If some threads stay blocked, the DLL is left loaded on purpose, because unloading it would crash the target, and the ejection can be repeated later.

Appendix A

Code-Cache Based DBI Frameworks

This appendix extends the summary given in Section 2.4.

A.1 Core Architectural Components

Modern code-cache-based DBI frameworks rely on a unified architectural pattern comprising the following components:

1. **The Dispatcher (Control Flow Manager):** The dispatcher is the central component of the code-cache DBI engine, responsible for orchestrating the execution state. Because the application and the code-cache DBI framework share the same process but operate under different scopes, they must safely transit between "framework mode" and "application mode". The dispatcher triggers a context-switch mechanism that carefully saves and restores the application's state whenever execution exits or enters the code cache.

2. **JIT Compiler:** Instead of translating the entire binary before execution, these architectures translate the application's instructions dynamically in small fragments. During this translation phase, the framework decodes and encodes (into the framework's abstraction) the original instructions and has the opportunity to insert custom instrumentation code or patches before emitting the new sequence.

3. **The Code Cache (The Execution Sandbox):** To mitigate the massive performance penalty of repeatedly translating the same code, code-cache-based DBI frameworks employ a software-based code cache. The translated and potentially instrumented fragments are stored in this sandboxed memory region. When the

application re-executes a previously translated section of code, it runs directly from the code cache, which significantly optimizes execution speed.

4. **The Client Interface (Instrumentation API):** This is the abstraction layer exposed to the developer. It provides the necessary functions to inspect the program state, traverse instruction sequences, and specify exactly where and how instrumentation should be injected.

A.2 The Execution Lifecycle: The JIT Loop

To formally describe how these components interact at runtime, we can outline the execution lifecycle of a standard block of code, illustrated in Figure A.1:

1. *Interception:* The target application attempts to execute a block of original code. The code-cache DBI engine intercepts this attempt and yields control to the Dispatcher.

2. *Cache Lookup:* The Dispatcher queries a fast-lookup hash table to check if the target memory address has already been translated. If it results in a cache hit, the CPU is redirected to the Code Cache.

3. *Translation (On Cache Miss):* If the code has not been encountered before, the JIT Compiler decodes original binary instructions one by one to form a straight-line sequence that terminates at a control transfer, such as a branch, call, or return.

4. *Instrumentation:* The engine queries the Client Interface to determine if the user wishes to instrument the block or a single instruction within it. If so, the instrumentation instructions are injected.

5. *Emission and Linking:* Most code-cache-based DBI frameworks use this phase to apply patches to the control-flow instructions at the end of the sequence, guaranteeing that when the block finishes running, it will redirect back to the framework rather than jumping to a native memory address. If the block ends in a direct branch to another previously translated block, the engine "links" them together directly in the cache (if it was an indirect branch, it cannot know the target before executing it), allowing the CPU to bypass the Dispatcher in future iterations. The modified block is emitted into the Code Cache.

6. *Execution:* The Dispatcher performs a context switch, restoring the application state, and commands the CPU to execute the newly cached block natively.

7. *Code Cache Exit:* Execution remains safely inside the code cache — often jumping directly from one translated block to another through "linking"—until it hits a boundary. This boundary might be an unlinked branch or an indirect branch whose target has not been translated yet. When this happens, the block executes an exit stub which saves the CPU's state and transfers control back to the Dispatcher, restarting the loop.

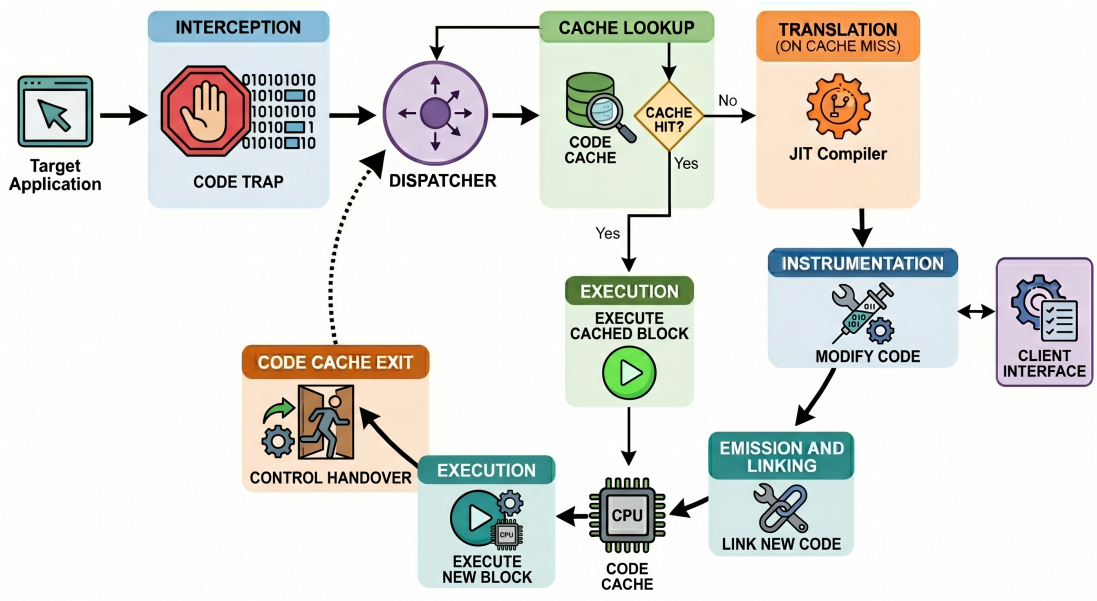


Figure A.1: The Execution Lifecycle of a Code-Cache-Based DBI framework

A.3 Comparison with Inline Hooking

Table A.1 summarizes the differences between the two DBI paradigms: inline hooking (Section 2.3) and the code-cache based DBI described in this appendix.

A.4 The Client Interface: Instrumentation vs. Analysis Functions

Code-cache-based DBI frameworks require the developer to interact with the engine using a *two-tiered callback paradigm*. The developer must supply two distinct types of functions to the engine, each operating at fundamentally different times during the execution lifecycle:

Dimension	Inline hooking	Code-cache based DBI
Mechanism	A jump written at the start of the target function sends each call to the hook function; the trampoline runs the moved instructions and jumps back to the rest of the function.	A dispatcher passes each new block of code to a JIT compiler, which translates it and inserts the instrumentation code; the CPU runs only the translated copies stored in the code cache.
Granularity	Function level: the hook function runs when a target function is called, and can also run code after the original function returns.	Instruction level: instrumentation code can be inserted at any instruction, for example to log every conditional branch.
Runtime overhead	Lower: only the calls to target functions pay an extra cost (jump, hook function and trampoline); the rest of the program runs as before.	Higher: every instruction passes through the JIT compiler, even when it is not instrumented.
Machine state	Preserved by the compiler, because the hook function is an ordinary function that follows the ABI; the hook function must also declare exactly the parameter list and return type of the target.	Saved and restored by the engine, because engine-generated code runs between the program and the instrumentation code.
Original code	Changed in memory: the first bytes of each target function are overwritten, and written back when the hooks are removed.	Never changed, and never executed directly by the CPU.
Engine complexity	Lower: the engine changes only the prologue of each target function, so it must find where the prologue instructions end and write the jump safely while other threads run.	Higher: the engine needs a dispatcher and a JIT compiler that can translate every instruction that the program runs.

Table A.1: Comparison of the two main DBI paradigms.

- *The Instrumentation Function (The "Where" and "What"):* This callback is invoked by the code-cache DBI engine *only once* per newly discovered sequence of instructions during the JIT compilation phase. Its purpose is to inspect the static properties of the instructions (e.g., opcodes, operands) and decide *if* and *where* custom code should be injected. If injection is required, it uses the API to queue the insertion of the analysis function.
- *The Analysis Function (The "Action"):* This is the actual custom payload injected into the Code Cache. Unlike the instrumentation function (executed only at compile time), it is executed *every time* the CPU runs the corresponding instrumented block of code. For this reason, the analysis function must be heavily optimized and efficient.

The level of abstraction provided by the Client Interface significantly alters how developers interact with the engine.

A.5 High-Level Abstraction API: Intel PIN

Intel PIN [Luk+05] is a standard academic code-cache-based DBI framework with a proprietary (free for non-commercial use) license. It provides a heavily abstracted C++ API.

To utilize Intel PIN, developers do not modify the core engine itself. Instead, they write a custom C++ program known as a *PINtool*. A PINtool represents the physical implementation of the Client Interface discussed previously; it is the user-defined module that contains both the Instrumentation Functions and the Analysis Functions. Architecturally, a PINtool is compiled as a standard shared library (a DLL on Windows). When the user launches a target application under PIN, the engine automatically injects the PINtool DLL into the target process's memory space. Once loaded, the engine invokes the PINtool's `main` function, allowing the tool to register its specific callbacks. In Code A.1, we can observe the granularity achievable using code-cache-based DBI: in this case, we can log conditional branches to determine which paths are taken during code execution.

Code A.1: An example of a PINtool for branch logging

```
// Analysis function (Runs at execution time in the Code Cache)
void record_branch(ADDRINT instruction_address, ADDRINT target_address, BOOL taken) {

  if (taken) { /* Log Taken */ }
  else       { /* Log Not Taken */ }
}

// Instrumentation function (Runs at JIT compilation time)
void trace(INS instruction, void *v) {
```

```
  // PIN abstracts away module resolution: we want to instrument
  // only target application instructions, filter possible external libraries
  RTN rtn = INS_Rtn(instruction);
  if (RTN_Valid(rtn) && IMG_IsMainExecutable(SEC_Img(RTN_Sec(rtn)))) {
    // If it is inside the target executable, check if it is an IF branch
    if (INS_IsBranch(instruction) && INS_HasFallThrough(instruction)) {
      // Inject analysis function right before the jump happens.
      INS_InsertCall(instruction, IPOINT_BEFORE, (AFUNPTR)record_branch,
      IARG_INST_PTR,
      IARG_BRANCH_TARGET_ADDR,
      IARG_BRANCH_TAKEN,
      IARG_END);
    }
  }
}
```

A.6 Low-Level API: DynamoRIO

DynamoRIO [BGA03] is the open-source alternative that exposes a lower-level C API. The equivalent branch-logging example seen previously for PIN is implemented in Code A.2.

Code A.2: DynamoRIO branch instrumentation example

```
// Analysis function
static void record_branch(app_pc instr_addr, app_pc target_addr, int taken) {

  if (taken) { /* Log Taken */ }
  else       { /* Log Not Taken */ }
}

// Instrumentation function
static dr_emit_flags_t event_bb_analysis(void *drcontext, void *tag, instrlist_t *bb,
   instr_t *inst, bool for_trace, bool translating,
void *user_data) {

  // Manual module tracking to filter out OS libraries
  app_pc instr_addr = instr_get_app_pc(inst);
  module_data_t *mod = dr_lookup_module(instr_addr);
  bool is_main_exec = false;

  if (mod != NULL) {
    module_data_t *main_mod = dr_get_main_module();
    if (main_mod != NULL) {
      if (mod->start == main_mod->start) { is_main_exec = true;
      }
      dr_free_module_data(main_mod);
```

```
    }
    dr_free_module_data(mod);
  }

  if (!is_main_exec) return DR_EMIT_DEFAULT;
  // Check if it is a conditional branch (CBR)
  if (instr_is_cbr(inst)) {
    // Inject analysis function into the basic block (bb)
    dr_insert_cbr_instrumentation(drcontext, bb, inst, (void *)record_branch);
  }
  return DR_EMIT_DEFAULT;
}
```

Using DynamoRIO, developers can directly manipulate the instruction flow. In PIN, developers generally "wrap" existing instructions (adding a call before or after an instruction) but cannot easily modify the instruction itself. In DynamoRIO, the developer is handed a linked list of instructions (`instrlist_t`) and can easily manipulate or modify them (e.g., reordering them or moving a load instruction earlier to test for race condition).

Furthermore, because PIN is designed to be highly opaque, using `INS_InsertCall` prompts PIN's JIT compiler to automatically analyze the custom analysis function for deciding which registers need to be saved and restored based on what the C++ code actually uses. With DynamoRIO, developers can manage the state and flags manually for specialized analysis.

In conclusion, while DynamoRIO may present a steeper learning curve than Intel PIN, it affords developers greater transparency and raw power, providing fine-grained control over the translated code and the code cache.

A.7 Summary of Code-Cache-Based DBI Frameworks

To summarize the key differences between the analyzed code-cache-based DBI engines, Table A.2 compares the features and capabilities of Intel PIN and DynamoRIO.

Feature	Intel PIN	DynamoRIO
API Style	High-level C++ API	Low-level C API
Code Modification	High-level wrapping (instrumentation only before/after instructions)	Direct instruction manipulation
State/Flags Handling	Fully automatic (always analyzed and optimized by JIT compiler)	Manual or automatic (developer can choose)
Platform Support	Windows, Linux	Windows, Linux, macOS, Android (x86, x64, ARM, ARM64)
License	Proprietary (Free for non-commercial use)	BSD (Open Source)

Table A.2: Comparison of the analyzed code-cache-based DBI frameworks.

Appendix B

How a Program Opens a File on Windows

The running example of this thesis (Section 1.2) stops a program from opening some files. Its hooks intercept two functions of the Windows *native API*, `NtCreateFile` and `NtOpenFile`. This appendix explains what these functions are, why the example hooks them instead of a better-known function such as `CreateFileW`, and the types of their parameters and of their result that the listings of this thesis use.

B.1 The Layers of a File Open

A program rarely talks to the operating system kernel directly. It calls functions of the Win32 API, such as `CreateFileW`, or functions of a library built on top of it, such as `_wfopen` of the C runtime or the `std::ifstream` class of the C++ standard library. The Win32 functions in turn call the functions of the native API, which are exported by `ntdll.dll`; these functions are the last step in user mode, and they enter the kernel through a system call [Yos+17].

For files, the two native functions are:

- `NtCreateFile`, which "creates a new file or directory, or opens an existing file, device, directory, or volume" [Mic26k];
- `NtOpenFile`, which "opens an existing file, device, directory, or volume, and returns a handle for the file object" [Mic26l].

This layering decides where a hook must be placed. A hook on `CreateFileW` sees only the

programs that call CreateFileW: a program that calls NtOpenFile directly passes below it. A hook on NtCreateFile and NtOpenFile, instead, sees the file opens of every layer above them, whatever function the program used. For this reason the running example hooks the two native functions. A program can still skip ntdll.dll by executing the system call instruction itself; no user-mode hook can see such a call.

B.2 Parameters and Result

Code B.1 shows the declaration of NtOpenFile, the function used by most listings of this thesis. NtCreateFile has the same first four parameters, followed by seven more that describe how the file is created [Mic26k].

Code B.1: Declaration of NtOpenFile [Mic26l].

```
NTSTATUS NTAPI NtOpenFile(
    PHANDLE            FileHandle,       // out: the handle of the opened file
    ACCESS_MASK        DesiredAccess,    // in:  the rights requested
    POBJECT_ATTRIBUTES ObjectAttributes, // in:  the name of the file
    PIO_STATUS_BLOCK   IoStatusBlock,    // out: details on the result
    ULONG              ShareAccess,      // in:  what other opens may do
    ULONG              OpenOptions);     // in:  options of the open
```

DesiredAccess is a bit mask of the rights that the caller asks for, for example FILE_READ_DATA to read the data of the file and FILE_WRITE_DATA to write it [Mic26k].

ObjectAttributes is a pointer to an OBJECT_ATTRIBUTES structure, which holds the name of the file in two members [Mic26m]:

- ObjectName points to a UNICODE_STRING, a string with an explicit length that is not guaranteed to end with a NUL character [Mic26o];
- RootDirectory is an optional handle to a directory. If it is NULL, ObjectName must contain the full path of the file, for example \??\C:\data\a.txt; otherwise, ObjectName is a name relative to that directory, for example a.txt [Mic26k; Mic26m].

So the full path of a file is not always in the name that a hook function receives: for a relative open, part of it is hidden behind the RootDirectory handle.

NTSTATUS: Native functions do not use GetLastError: they return a 32-bit NTSTATUS value. Since NTSTATUS is a signed type, a negative value is a warning or an error. The running example uses the error STATUS_ACCESS_DENIED (0xC0000022): "a process has requested access to an object but has not been granted those access rights" [Mic26a].

Appendix C

Preliminary Experiments

To empirically validate the concepts discussed in Chapters 2 and 3 and to fully understand the operational constraints, security boundaries, and low-level deployment challenges of each framework, a propedeutic suite of testing applications was developed. All source code and custom injection utilities have been organized and made available within the project's repository.[1]

The experimental activity was structured into three different scenarios, each designed to evaluate specific architectural properties of the instrumentation engines.

C.1 Scenario 1: Single-Threaded Hooking and Process Injection/Ejection Mechanics

- **Test Rationale:** This preliminary scenario aimed to verify the fundamental lifecycle management of DBI within the Windows ecosystem.

  This involved implementing a custom, dual-mode utility (`injector.cpp`) to handle both the insertion and the ejection of the instrumentation payload.

  Concurrently, the test evaluated the syntactic and structural paradigms of the three inline hooking frameworks (Microsoft Detours, MinHook, and Frida-Gum) when intercepting a standard system API—`MessageBoxW` in `user32.dll`.

- **Empirical Results:** Every analyzed framework successfully intercepted the execution flow in the pre-logic. This capability allowed for the modification of string

[1] https://github.com/davde342/MasterThesis/tree/main/code/first_examples

pointers targeting the dialog box text and caption, the reassignment of the configuration flags to `MB_YESNO`, and the runtime mutation of the return value in the post-logic, effectively forcing an `IDYES` outcome regardless of the user's explicit interaction.

Furthermore, forcing the programmatic module ejection successfully triggered the `DLL_PROCESS_DETACH` branch within each hook's entry point, safely restoring the original prologue instruction bytes and confirming a flawless ejection in single-threaded environments.

- **Encountered Obstacles:** In this idealized single-threaded scenario, no big challenges occurred. Setting up Microsoft Detours was straightforward due to its thoroughly documented APIs, the abundance of reference examples, and a standard paradigm that is conceptually simpler than that of Frida-Gum.

  MinHook demonstrated a similar workflow to Detours but simplified the implementation further by automatically managing transactional patching internally.

  Conversely, approaching Frida-Gum for the first time presented a steeper learning curve; given its more complex architecture, distinct paradigm, and less exhaustive documentation; comprehensively understanding its API behaviors and mapping them to my use case required a direct analysis of the huge and dense codebase within the official Frida repository.

C.2 Scenario 2: Multi-Threaded Hooking and Concurrency Management

- **Test Rationale:** To simulate a realistic production workload and test the instrumentation frameworks under thread concurrency, a multithreaded target engine was used (`target_multithread.cpp`). This target utilizes a synchronized task queue where a producer thread continuously queues activities with varying workloads, while three concurrent consumer threads process these tasks in parallel by invoking the exported routines `ProcessTask` and `LogTaskResult`.

- **Empirical Results:** Real-time monitoring of thread behaviors was achieved, allowing a dynamic optimization policy: the injected hook intercepted high-latency tasks (>500ms) and programmatically constrained their complexity metrics to 100ms prior to native routing, illustrating the possibility of inline hooking for runtime policy enforcement and load adaptation.

- **Encountered Obstacles:** Two major obstacles I faced in this scenario. First, performing hook detachment in a concurrent context introduced a possible race condition: active worker threads could be descheduled while executing inside the hook

callback, causing an immediate Access Violation (`0xC0000005`) once the library was unmapped.

Second, in Frida-Gum, intercepting routines that process floating-point arguments (such as the `double` parameter in `LogTaskResult`) encountered an ABI mismatch because these parameters are passed via `XMM0`–`XMM3` registers under the Windows x64 ABI, while Frida expects them in other registries.

- **Implemented Mitigations:** To address the detachment race condition in concurrent environments, framework-specific synchronization protocols were deployed (as formalized in Sections 2.3.1 and 3.3.1).

  As detailed in Section 3.2, to bypass the floating-point register limitation inherent in Frida-Gum's Invocation Listener mode, the high-level callback attachment was discarded in favor of Frida-Gum's Function Replacement mode via `gum_interceptor_replace`, allowing the C/C++ compiler to handle exact register allocation natively.

C.3 Scenario 3: Instruction-Level Assembly Tracking via Code-Cache-Based DBI Engines

- **Test Rationale:** To achieve low-level instruction granularity, a target program containing conditional branches was compiled (`PIN_dynamo_target.cpp`), alongside an Intel PIN tool (`pintool.cpp`) and a DynamoRIO client (`dynamo.c`). The rationale was to program the underlying JIT compilers to isolate all conditional branch instructions (`INS_IsBranch` / `instr_is_cbr`) and dynamically inject an analysis callback (`record_branch`) used to log the target address (taken branch). The APIs of both engines are described in Appendix A.

- **Empirical Results:** Both code-cache-based DBI engines yielded execution trace graphs. Using IntelPIN it was also possible to write the source file code line number correspondent to the taken branch

- **Encountered Obstacles:** An initial operational challenge involved navigating the different APIs of the two frameworks; DynamoRIO exposes lower-level, low-level utilities compared to the highly abstracted API of Intel PIN, resulting in a steeper learning curve. Beyond API familiarity, the primary technical obstacle across both engines was attempting to log not only the target runtime address of each taken branch but also its source code line number from `PIN_dynamo_target.cpp` (compiled with debug symbols) to maximize execution trace readability.

- **Implemented Mitigations:** After several unsuccessful attempts, I have found that for solving the source line number obstacle under Intel Pin, I had to utilize the

`PIN_GetSourceLocation` API. However, direct invocation of this routine inside the analysis function triggers a critical runtime error due to missing client locks. To mitigate this, calls to `PIN_GetSourceLocation` were wrapped inside a thread-safe synchronization pair using `PIN_LockClient()` and `PIN_UnlockClient()`, ensuring proper access to Pin's symbol database.

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